Temperature Shocks, Worker Age, and Sectoral Reallocation into Informality: Evidence from Vietnam

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This version: March 2026

Abstract

Do climate shocks push agricultural workers into formal or informal employment? Using three decades of Vietnamese survey and weather data, I show that extreme heat drives workers out of agriculture and into informal employment across all age groups, but only younger workers also enter formal employment. A three-sector Roy model links these patterns to temperature-driven declines in agricultural returns, with agricultural attachment and age-varying formal-sector barriers determining who exits and where they go. Supporting evidence confirms both channels: heat reduces yields and farm revenue without price adjustments, while job postings and a vignette experiment document age-based barriers in formal hiring.

Keywords: temperature, labor allocation, informality, worker age, Vietnam

JEL Codes: O12, Q54, J46

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1 INTRODUCTION

Climate change is reshaping agricultural livelihoods across the developing world. A growing body of evidence documents that temperature shocks reduce agricultural employment and increase non-agricultural work (e.g., Albert et al., 2021; Colmer, 2021; Emerick, 2018; Jessoe et al., 2018; M. Liu et al., 2023). Yet the non-agricultural sector in developing countries is not homogeneous: it encompasses both formal employment offering stable wages and social protections, and a large informal sector characterized by self-employment, casual labor, and limited earnings growth (Donovan et al., 2023; La Porta & Shleifer, 2014; Ulyssea, 2020). Whether climate-displaced agricultural workers end up in formal or informal jobs has very different implications for their welfare and for the economy's long-run development trajectory, yet existing work treats the non-agricultural sector as a single destination and cannot distinguish between the two.

Who ends up in formal versus informal employment depends critically on who is being displaced. In practice, the agricultural workforce in much of the developing world is aging (HelpAge International, 2014; Ren et al., 2023; Rigg et al., 2020). Older farmers have accumulated sector-specific skills and assets that may raise the cost of leaving agriculture (Foster & Rosenzweig, 1995), and older workers more broadly face steeper barriers to formal hiring (Neumark et al., 2019). Climate shocks are thus displacing a workforce whose age composition may shape both the extent of reallocation and whether it leads predominantly to formal or informal employment. This paper examines how climate shocks reallocate workers between formal and informal employment, and how this reallocation varies with worker age.

I study this question using Vietnam, a lower-middle-income country that shares key labor market features with much of the developing world: a large agricultural workforce, a segmented non-agricultural sector where informality remains prevalent, and a farm population that skews notably older than the non-farm labor force (Y. Liu et al., 2020; McCaig & Pavcnik, 2015, 2017). The country has experienced substantial warming over recent decades, with considerable sub-national climatic variation. Its nationally representative household surveys span nearly three decades and contain detailed information on employment sector, formality status, and individual demographics—a combination that is rare in developing-country data and essential for decomposing reallocation along both dimensions.

To guide the empirical analysis, I develop a conceptual framework describing a three-sector economy (agriculture, informal non-agriculture, and formal non-agriculture) with heterogeneous workers who choose sectors based on comparative advantage, following

Roy (1951). The framework integrates three components: (i) a temperature-driven "push" out of agriculture operating through relative productivity loss under conditions where local agricultural prices do not absorb the productivity shock (e.g., Kongsamut et al., 2001; Matsuyama, 1992), (ii) within-cohort variation in access to formal employment, with the share of workers facing high formal barriers increasing with age, and (iii) agricultural attachment (such as accumulated crop knowledge) that raises the effective payoff of remaining in agriculture for older workers. The model predicts that temperature shocks push workers out of agriculture, with the magnitude of exit declining in age due to accumulated agricultural attachment. Formal entry also declines in age, driven by both attachment and the rising share of workers who lack direct access to formal employment. The predicted effect on informal entry is ambiguous: fewer older workers leave agriculture, but a larger share of older cohorts faces barriers that channel agricultural exit toward informality. Whether informal entry varies with age therefore depends on the relative strength of these two opposing forces, making the informal response a useful diagnostic for assessing whether both mechanisms are at work. The framework also predicts that older workers who do enter formal employment should exhibit larger earnings gains than younger entrants, reflecting stronger selection on skill—a mechanism analogous to immigrant self-selection under higher migration barriers (Borjas, 1987).

I test these predictions using nearly three decades of province-age group panel data from Vietnam's household surveys (1993–2018), linked to detailed daily weather records from the ERA5 reanalysis, and supplemented with data on migration, crop yields, land use, and formal firm censuses. Following the climate impacts literature (Dell et al., 2014), I exploit year-to-year variation in weather while controlling for province-by-age group fixed effects and region-by-age group-by-year fixed effects. This approach relies on the assumption that, conditional on these fixed effects, local weather variation is orthogonal to unobserved determinants of sectoral employment. These age-group-specific time effects absorb any region-level trends that differentially affect age groups, such as cohort-specific gains in educational attainment or age-targeted labor market policies, which could otherwise confound the temperature-age interaction. Several checks support the validity of this approach. Future temperature does not predict current outcomes, ruling out pre-existing differential trends. Neither province-level migration rates nor individual-level measures of geographic mobility respond to temperature shocks, confirming that the observed reallocation occurs within local labor markets rather than through geographic sorting. Results are consistent when estimated on the longitudinal panel component (available from 2004), which tracks the same individuals across waves and thus rules out compositional turnover within province-age group cells, and when weather is measured at the finer

district level. The estimates are also robust to alternative functional forms for temperature, spatial interpolation methods, leave-one-out province exclusions, and Monte Carlo permutation tests.

Two main findings emerge. First, temperature shocks significantly affect sectoral labor allocation on average. Hot temperatures, measured as cumulative wet-bulb degree days above 27°C (the 97th percentile of the daily temperature distribution), reduce the agricultural employment share and increase both formal and informal non-agricultural shares. A one-standard-deviation increase in heat exposure within provinces (approximately 6.7 degree days) reduces the agricultural employment share by approximately 4.3 percentage points and increases the formal and informal non-agricultural shares by roughly 1.8 and 2.5 percentage points, respectively ($p < 0.01$). These effects represent about 10% of the respective sector mean shares. The response is driven by lagged rather than contemporaneous temperature exposure: heat experienced 12–24 months before the interview drives the sectoral labor changes, while exposure in the most recent 12 months has no significant effect. This is consistent with a gradual process of job search and transition rather than an immediate switch.[1] There is no effect on the share of inactive or unemployed individuals, indicating that the margin of adjustment is sectoral reallocation rather than movement into non-employment (either inactive or involuntarily unemployed).

Second, these average effects mask substantial heterogeneity by age group, and the pattern varies across destination sectors. Younger workers, those in their twenties and thirties, exhibit the largest decline in agricultural employment and enter both formal and informal non-agriculture. Middle-aged workers show a more moderate agricultural decline, with gains concentrated in informal employment and a weaker formal response. Older workers show the smallest agricultural decline, with their adjustment occurring almost entirely through informal non-agriculture. I reject equality of heat effects across age groups for agriculture ($p < 0.01$) and formal non-agriculture ($p < 0.01$) but not for informal non-agriculture ($p = 0.75$), consistent with the framework's prediction that the informal response is ambiguous across age groups.

To understand these patterns, I examine supporting evidence on mechanisms. On the "push" side, I show that extreme heat disproportionately reduces agricultural revenue per worker relative to non-agriculture, reduces hours worked in agriculture, and lowers crop yields, while prices of agricultural commodities such as rice, the main staple crop, do not respond to local temperature shocks. With prices unaffected, these losses translate directly into lower agricultural returns, creating incentives for exit.

[1] Cold temperatures do not affect sectoral employment shares. Other weather extremes like rainfall show qualitatively similar patterns but are imprecisely estimated and less robust.

On the barrier side, I provide three complementary pieces of evidence on age-based labor market frictions: sector earnings gaps consistent with positive selection among older formal entrants, widespread age restrictions in Vietnamese job postings, and experimental evidence of age penalties in hiring evaluations from a vignette study with manufacturing firms. I also rule out alternative explanations including differential physiological vulnerability to heat, educational differences across cohorts, and age as a proxy for health.

Finally, I use these age-specific estimates to project how the economy's aggregate reallocation capacity evolves under demographic change. Combining the estimated responses with population projections from the United Nations World Population Prospects, I show that population aging alone weakens the aggregate reallocation response by roughly 13% by mid-century, suggesting that the economy's capacity to absorb climate shocks through labor mobility is gradually eroding.

This paper makes three contributions. First, it decomposes climate-driven labor reallocation in a developing country by the formality status of destination employment. A growing literature documents that temperature shocks push workers out of agriculture and into non-agricultural employment (Albert et al., 2021; Colmer, 2021; Emerick, 2018; M. Liu et al., 2023), and recent work frames this sectoral reallocation as a key channel for climate adaptation, both domestically (Colmer, 2021) and through international trade (Nath, 2025), alongside within-agriculture responses such as input adjustments (Aragón et al., 2021; Jagnani et al., 2021). However, the literature generally treats non-agricultural employment as a single destination. In practice, non-agricultural labor markets in developing countries are sharply segmented between a formal sector offering stable wages and social protections and a large informal sector characterized by self-employment and limited earnings growth (La Porta & Shleifer, 2014; Ulyssea, 2020). This paper shows that extreme heat pushes workers into informal employment across all age groups, but only younger workers also enter formal employment—a distinction the existing literature does not make.

Second, the paper develops a three-sector Roy model that explains both the temperature-driven push out of agriculture and the age-dependent sorting across formal and informal destinations, and provides direct evidence on the underlying mechanisms. The framework integrates agricultural attachment and age-varying formal-sector barriers to generate a diagnostic prediction: if both forces operate simultaneously, the informal response should be approximately age-invariant even as agricultural exit and formal entry decline sharply with age, a pattern the data confirm. I confirm the push channel by showing that extreme heat reduces agricultural returns without offsetting price adjustments, consistent with evidence on the agricultural productivity costs of extreme heat (Deschênes

& Greenstone, 2007; Schlenker & Roberts, 2009). On the barrier side, I provide original evidence on employer stated preferences: job postings data reveal widespread age restrictions in formal hiring, and a vignette experiment with manufacturing firms documents substantial age penalties in hiring evaluations, extending experimental evidence on age discrimination (Carlsson & Eriksson, 2019; Neumark et al., 2019) to a developing-country context where such evidence is scarce. Earnings patterns among sector-switchers further show that older formal entrants exhibit larger gains, consistent with stronger positive selection under higher barriers (Borjas, 1987).

Third, the paper shows that the economy's capacity to reallocate labor in response to climate shocks depends on the demographic composition of the agricultural workforce — and that this capacity is eroding. A large body of work on labor market adjustment demonstrates that reallocation is shaped by sector-specific experience and switching costs (Dix-Carneiro, 2014), and that structural shocks disproportionately reallocate younger workers while older workers remain in declining sectors (Autor & Dorn, 2009), yet the climate-labor literature implicitly treats the agricultural workforce as demographically homogeneous (e.g., Albert et al., 2021; Colmer, 2021; M. Liu et al., 2023).[2] Combining the estimated age-specific responses with population projections, I show that population aging alone weakens the aggregate reallocation response by roughly 13% by mid-century. As agricultural workforces across the developing world age rapidly (HelpAge International, 2014; Ren et al., 2023; Rigg et al., 2020), the labor reallocation channel that existing research identifies as a key margin of adaptation may be weakening precisely when it is most needed.

The remainder of the paper proceeds as follows. Section 2 introduces a simple conceptual framework that guides the empirical analysis. Section 3 describes the data and provides descriptive patterns of the temperature-employment relationship. Section 4 details the empirical strategy and discusses identification assumptions. Section 5 presents the main results and robustness checks. Section 6 examines mechanisms, Section 7 presents a projection exercise emphasizing the role of labor force composition, and Section 8 concludes.

2 CONCEPTUAL FRAMEWORK AND TESTABLE PREDICTIONS

This section presents a simple framework to understand how temperature shocks affect labor allocation across sectors in a developing economy with a segmented non-agricultural

[2] A notable exception is Emerick (2018), who documents larger reallocation effects among more educated households. Education heterogeneity carries favorable demographic implications, as rising educational attainment increases the pool of workers who can reallocate; the age heterogeneity documented here works in the opposite direction, as agricultural workforces are aging.

labor market. The goal is twofold: to clarify the conditions under which heat drives workers out of agriculture, and to generate testable predictions about which non-agricultural jobs absorb them—predictions that depend on worker characteristics in ways that are not obvious *a priori*.

The framework highlights three forces. First, extreme heat reduces the return to agricultural work through both physiological constraints on labor performance and an agronomic channel, creating a "push" out of agriculture. Whether this push actually reallocates labor depends on whether agricultural prices absorb the shock: if reduced local supply raises commodity prices enough to offset the productivity loss, workers may remain in agriculture. The reallocation mechanism therefore requires that local agricultural prices are approximately unresponsive to local productivity shocks—a condition that can be satisfied under several institutional arrangements, including international price-taking, government price stabilization, or sufficient domestic market integration. This condition is testable, and the distinction matters: in a setting with locally determined prices, reduced agricultural output could raise food prices enough to attenuate or even reverse the reallocation.

Second, access to formal non-agricultural jobs is limited by entry barriers (e.g., hiring restrictions, screening practices, age cutoffs) that vary both across and within worker cohorts. This creates a segmented labor market in which workers pushed out of agriculture may end up in very different destinations depending on who they are, not just on wages.

Third, older workers accumulate agricultural attachment (e.g., land tenure, crop-specific knowledge, local trade relationships) that raises the opportunity cost of leaving agriculture at any age. This generates age-graded exit from agriculture on top of the destination-sorting driven by formal barriers.

The framework yields several predictions that jointly guide the empirical analysis. Some of them are intuitive: heat exposure is expected to push workers out of agriculture, and age-based formal barriers would channel older exiters into informality. The remaining are less obvious. First, the predicted effect of heat on informal non-agricultural entry is ambiguous in age: agricultural attachment reduces the number of older exiters, but a larger share of older cohorts faces high formal barriers, channeling their exit toward informality. Whether informal entry varies with age therefore depends on the relative strength of these two opposing forces, making the informal response a useful diagnostic for assessing whether both mechanisms are jointly at work. Second, the model predicts that among the subset of older workers who do enter formal employment, observed earnings gains should be larger than for younger formal entrants—a pattern that reflects selection rather than differential returns.

Appendix Section A presents the full model, assumptions, and proofs. The remainder of this section outlines the key elements and predictions.

2.1 Setup

Consider an economy with three sectors: agriculture (g), informal non-agriculture (i), and formal non-agriculture (f). Workers differ in non-agricultural skill ε but are equally productive in agriculture. Employment in period 1 is predetermined. At the start of period 2, each worker chooses a sector to maximize net earnings. This structure follows the Roy selection framework (Roy, 1951), in which workers sort across sectors based on comparative advantage, as applied to developing-country labor markets by Lagakos and Waugh (2013).

A worker who remains in agriculture earns w_g, which depends on temperature through both labor performance and crop productivity. A worker who enters informal non-agriculture earns $w_i\tilde{\varepsilon} - \kappa_i$, where $\tilde{\varepsilon} \equiv (1-b) + b\varepsilon$ with $b \in (0,1)$ captures the lower skill intensity of informal work. The entry cost κ_i is additive and (as a benchmark) does not vary with age. A worker who enters formal non-agriculture earns $w_f\varepsilon - \kappa_f$, where κ_f varies across workers and the probability of facing high barriers increases with age, capturing hiring restrictions, screening, and other barriers documented in the labor market. The segmentation of non-agricultural employment into formal and informal sectors, with barriers governing access to formal employment, is a central feature of developing-country labor markets (La Porta & Shleifer, 2014; Ulyssea, 2020).

An important component is *agricultural attachment*: older workers have accumulated sector-specific capital that raises the effective value of staying in agriculture by an amount $\delta(a)$, which is increasing in age. This captures the well-documented stickiness of agricultural labor in developing economies, for example, learning by doing accumulates over time (Foster & Rosenzweig, 1995). Agricultural attachment is not a cost of entering non-agriculture; it is a benefit of remaining in agriculture. The distinction matters: it makes older workers less likely to exit agriculture, but it does not make informality harder to access for those who do exit.

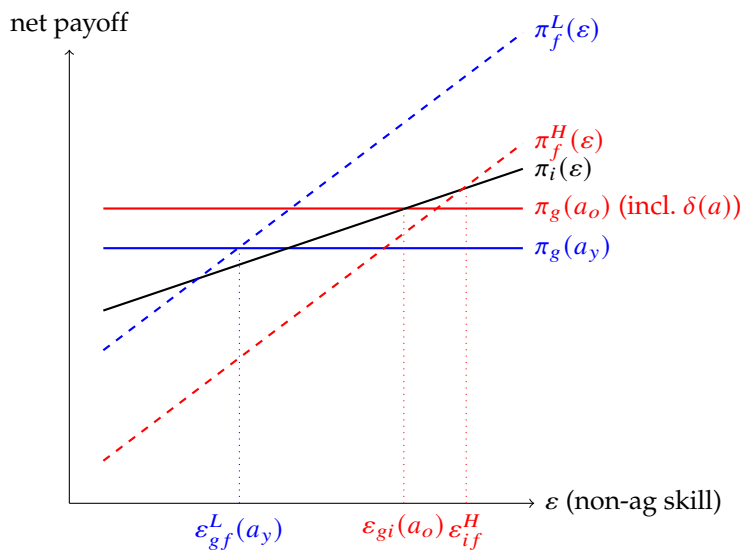
The worker's destination upon exiting agriculture depends on her formal entry cost. For workers with *low* formal barriers ($\kappa_f = \kappa_f^L$), a single threshold governs the allocation: the exit cutoff $\varepsilon_{gf}^L(a)$ at which formal employment just dominates agriculture. Workers below this cutoff stay in agriculture; workers above it exit directly into formal jobs, bypassing informality entirely.

For workers with *high* formal barriers ($\kappa_f = \kappa_f^H$), two thresholds govern the allocation. The lower threshold $\varepsilon_{gi}(a)$ is the skill level at which informal non-agriculture just dom-

inates agriculture, determining whether the worker exits agriculture at all. The upper threshold $\varepsilon_{if}^{H} \equiv \varepsilon_{if}(\kappa_f^H)$ is the skill level at which formal dominates informal. Conditional on facing κ_f^H, this informal–formal cutoff does not depend on age or agricultural attachment. Workers below $\varepsilon_{gi}(a)$ remain in agriculture, those between $\varepsilon_{gi}(a)$ and ε_{if}^{H} enter informal work, and those above ε_{if}^{H} enter formal work.

Since the share of workers facing high barriers rises with age, older cohorts are more likely to be in the intermediated regime, where informality serves as the absorbing margin for agricultural exit. Figure 1 illustrates this logic, separating payoffs that shift with age (through $\delta(a)$) from payoffs that shift with barrier type (through κ_f^L versus κ_f^H).

Figure 1: Payoff Structure



Notes: Agriculture pays a flat return $\pi_g(a) = w_g + \delta(a)$, where $\delta(a)$ captures age-varying attachment to agriculture (higher for older cohorts, shifting $\pi_g(a)$ up). Informal non-agriculture has net payoff $\pi_i(\varepsilon) = w_i[(1-b)+b\varepsilon]-\kappa_i$. Formal non-agriculture has net payoff $\pi_f(\varepsilon) = w_f\varepsilon-\kappa_f$; the figure plots a low-barrier case $\pi_f^L(\varepsilon)$ with κ_f^L and a high-barrier case $\pi_f^H(\varepsilon)$ with $\kappa_f^H > \kappa_f^L$. In the configuration shown, low-barrier workers in a young cohort exit agriculture directly into formal employment at $\varepsilon_{gf}^L(a_y)$ (bypassing informality). In this case, the low-barrier informal–formal cutoff $\varepsilon_{if}^L \equiv \varepsilon_{if}(\kappa_f^L)$ lies to the left of $\varepsilon_{gf}^L(a_y)$ and is omitted. High-barrier workers in an older cohort exit into informal work at $\varepsilon_{gi}(a_o)$ and enter formal only when $\varepsilon \geq \varepsilon_{if}^H$, the informal–formal indifference cutoff for high-barrier workers. A heat shock lowers w_g, reducing $\pi_g(a)$ and expanding the set of workers who prefer to exit agriculture.

The key comparative statics follow from how temperature and age shift these thresholds, and from how the composition of barrier types within a cohort changes with age.

2.2 Predictions

Prediction 1: Heat pushes workers out of agriculture. When temperature rises, agricultural returns w_g fall, either through some combination of reduced labor productivity,

fewer workable hours, and lower crop yields, while non-agricultural returns are less affected. This shifts the agriculture payoff line downward in Figure 1, reducing the relevant exit threshold (either $\varepsilon^L_{gf}(a)$ for low-barrier workers or $\varepsilon_{gi}(a)$ for high-barrier workers), inducing workers who were previously indifferent to exit. This is the "push" out of agriculture, a mechanism central to the structural transformation literature (Kongsamut et al., 2001; Matsuyama, 1992). Nath (2025) shows that in the absence of trade barriers, temperature-driven agricultural productivity losses generate precisely this pattern of sectoral reallocation at the global level.

Importantly, the push requires that agricultural commodity prices do not rise to offset the productivity loss. This condition holds when local prices are approximately insensitive to local supply shocks, as would be expected under several institutional arrangements present in Vietnam. As one of the world's largest rice exporters, Vietnam's integration into international rice markets implies that province-level productivity shocks are too small to move world prices. Additionally, throughout much of the sample period, the Vietnamese government directly regulated rice prices through state marketing agencies that set producer prices administratively and controlled export volumes (Ghosh & Whalley, 2004), and subsequently through floor price guarantees mandating that farm-gate prices remain above production costs (Government of Vietnam, 2009). Whether prices are set by world markets as in the case of small open economy (Matsuyama, 1992) or by policy, the implication remains the same: a local productivity shock from extreme heat reduces agricultural returns rather than being offset by higher output prices. I test this condition directly in Section 6 and confirm that local rice prices do not respond to local temperature shocks. In settings with locally determined prices, in contrast, reduced agricultural output could raise food prices enough to keep workers in agriculture, attenuating or even reversing the reallocation (e.g., M. Liu et al., 2023).

Prediction 2: Older workers exit agriculture less. Because older workers have accumulated agricultural attachment $\delta(a)$, their effective agricultural payoff is higher. In Figure 1, their agriculture line sits above the young workers' line. The same heat shock shifts both lines down by the same amount, but the older workers' line started higher, so fewer of them cross the exit threshold. Formally, both exit cutoffs, $\varepsilon^L_{gf}(a)$ and $\varepsilon_{gi}(a)$, are increasing in $\delta(a)$. Conditional on barrier type, the cutoff moves with w_g at an age-invariant rate, but because older cohorts start from a higher cutoff and the skill distribution has a thinning right tail, a given decline in w_g implies a smaller change in the *mass* of workers who cross the cutoff (i.e., a smaller $|dN_g(a)/dT|$). This prediction is consistent with evidence that declining occupations age in place: the same shock induces larger employment exits among younger than older workers, as accumulated skill specificity raises the cost of switching

(Autor & Dorn, 2009).

Prediction 3: Formal employment gains are concentrated among younger workers. The formal employment response to heat depends on how many workers within a cohort are in a position where a decline in agricultural returns directly expands their formal employment. Workers with low formal barriers are in this position: their exit threshold $\varepsilon^L_{gf}(a)$ is pinned by the agriculture–formal comparison, which moves with w_g. Workers with high formal barriers are not: their entry into formal employment is governed by the informal–formal cutoff ε^H_{if}, which is unaffected by changes in agricultural returns. Since the share of workers facing low formal barriers is larger among younger cohorts, the aggregate formal employment response to heat is concentrated among the young.

This prediction is consistent with extensive evidence that age-based hiring barriers limit older workers' access to formal employment, documented primarily in developed economies (Carlsson & Eriksson, 2019; Neumark et al., 2019). The present paper examines whether a similar age pattern in formal entry appears in a developing-country context where the informal sector provides an alternative absorbing margin.

Prediction 4: Informal non-agricultural entry is ambiguous in age. Informal employment is generated by workers who face high formal barriers: when they exit agriculture, they enter informality because the bar for formal entry is too high. The informal response to heat within an age group therefore depends on two quantities: how many workers face high barriers, and how many of those are at the margin of exiting agriculture (i.e., near $\varepsilon_{gi}(a)$).

Agricultural attachment reduces the second quantity. Among workers facing high barriers, older cohorts have fewer workers near the exit threshold (because $\delta(a)$ raises the effective agricultural payoff), so the same heat shock produces less informal entry per high-barrier worker. But the share of workers facing high barriers increases with age, pulling in the opposite direction: even though each high-barrier older worker is less likely to exit, there are more of them.

Whether the informal response rises or falls with age therefore depends on the relative strength of these forces. If only agricultural attachment mattered, the informal response would decline with age. If only the rising share of high-barrier workers mattered, it would increase. If the informal responses are similar across age groups, that would be consistent with both mechanisms operating simultaneously at roughly offsetting magnitudes.

Prediction 5: Older formal switchers exhibit larger earnings gains (selection). A final prediction concerns the earnings of workers who switch into formal non-agriculture.

Older formal entrants are more selected on non-agricultural skill, for two reinforcing reasons. First, among workers with low formal barriers, agricultural attachment $\delta(a)$ raises the exit threshold, so the older workers who still enter formal are drawn from further in the right tail of the skill distribution. Second, a larger share of older formal entrants come from the high-barrier pool, where the formal entry threshold is higher still. Both channels imply that observed earnings among older formal switchers are higher on average, even though the formal wage schedule itself is unchanged. This logic parallels the classic analysis of immigrant self-selection by Borjas (1987), where higher migration barriers produce more positively selected entrants with higher observed outcomes. The model thus predicts larger earnings gains for older formal entrants, a pattern that could otherwise be mistakenly interpreted as higher returns to experience.

For informal switchers, the selection effect is attenuated because informal work is less skill-intensive: earnings depend less on ε, compressing the observable differences across age groups. This provides a testable contrast between the two non-agricultural sectors.

3 DATA AND SUMMARY STATISTICS

3.1 Data and Measurement

In this section, I briefly discuss the main data sources and variables of interest. For detailed variable definition and data construction, see Appendix Section B. The main sources of data include the Vietnamese household surveys, the population and formal firm census, and the global climate and weather reanalysis ERA5 database.

Employment Data. I use the Vietnam Living Standards Surveys 1993-1998, the Vietnam Household Living Standards Surveys 2002-2018 to construct measures of the sectoral composition of employment and sectoral hours of work. The surveys are nationally and provincially representative.[3] Although VHLSS is a repeated cross-sectional survey, it contains a rotating panel sub-component that tracks individuals over a period of up to three consecutive waves (e.g., 2004–2006–2008), which allows me to analyze individual transition across sectors over a longer time than is usually feasible. The analysis sample includes workers with available information on industry of employment, as well as types of employers for household members age 24–64. I focus on this age range to capture working-age individuals with completed education.

The key variables of interest are the sector in which an individual was working during the reference period and their working hours. These variables are constructed using data

[3]The earliest household surveys in 1990s are not representative at the province level, I re-visit this point in the robustness checks section.

from the employment modules of the household survey, which covers hours worked, industries, and types of employer of the two most time-consuming jobs. For each job, an individual is asked whether he or she works for his or her own household or for other households, collectives, state-owned enterprises, private domestic enterprises, or foreign-invested enterprises. Following McCaig and Pavcnik (2018), I consider an individual as working in the informal sector if he or she is self-employed or works as an employee in household businesses. I also consider working in collectives or cooperatives as informal in order to make the definition consistent over a longer time period, although this should not affect the analysis much.[4] This employer-type classification is based on the same survey question across all waves, ensuring that the definition of (in)formality does not change over the sample period. Worker-side measures of informality (labor contracts and social insurance) are only available from 2014, precluding their use in the main analysis. A cross-check using the worker-side definition of GSO and ILO (2018) for the 2014–2018 waves confirms that the two measures are highly correlated, with a Pearson correlation coefficient of approximately 0.9 (Appendix Section B.2).

Historical Weather Data. The main weather data are from ERA5 reanalysis, which combines model data with global observations into a consistent dataset on a 0.25°× 0.25°grid (approximately 28 km × 28 km at the equator) (Hersbach et al., 2020). Reanalysis data provide consistent estimates of atmospheric parameters over time and space (Auffhammer et al., 2013) and are increasingly used in developing countries where station data are limited (Ortiz-Bobea, 2021). I focus on grid-level daily mean wet-bulb temperature and daily total precipitation.[5]

I use wet-bulb temperature (WBT) rather than dry-bulb (ambient) temperature because it more accurately captures thermal strain during physical labor. The human body dissipates heat through sweating, the effectiveness of which falls sharply as humidity rises, meaning high humidity can severely limit cooling even at moderate ambient temperatures (Parsons, 2014). WBT combines temperature and humidity into a single physiologically relevant index (Raymond et al., 2020), and has been increasingly adopted in economics to study heat effects on productivity in settings with manual or outdoor labor

[4]Employment in cooperatives and collectives contributes less than 1% of total employment of adults aged 24–64 since 2000. Their classification as informal follows the Cooperative Law of 2012, which does not consider cooperatives a type of enterprise, making them analogous to the household businesses that McCaig and Pavcnik (2018) use as the primary source of informal employment.

[5]The decision to use daily mean rather than maximum values stems from the fact that reanalysis outputs are generally sensitive to extreme values. While most reanalysis datasets agree on mean values across space, they are not in full agreement about the timing or magnitude of deviations from this mean (Auffhammer et al., 2013). In settings where station-level data are of good quality and available at high spatio-temporal density, maximum or minimum values are commonly used, e.g., Graff Zivin and Neidell (2014).

(Adhvaryu et al., 2020; LoPalo, 2023; Somanathan et al., 2021). The distinction is especially important in Vietnam, where diverse topography and monsoon exposure generate substantial cross-provincial variation in humidity. At dry-bulb temperatures of 27–30°C, WBT in the data ranges from roughly 17°C to 28°C, spanning conditions from comfortable to severely impairing for physical work (Appendix Figure B2).

Grid-level data are aggregated to the province level using inverse-distance weights from the four nearest grid points to each province's geographic centroid. Because administrative boundaries changed during the study period, I use the original 1993 boundaries, yielding a consistent sample of 52 provinces.[6]

To summarize thermal exposure, I construct wet-bulb degree-day measures from daily WBT, denoted WBT_{dpt} for day d in province p and reference year t:

$$\text{DD9}_{pt} \equiv \sum_d \max\left\{9 - \text{WBT}_{dpt}, 0\right\}$$
$$\text{DD27}_{pt} \equiv \sum_d \max\left\{\text{WBT}_{dpt} - 27, 0\right\}$$

DD9_{pt} and DD27_{pt} capture cumulative exposure to cold and extreme heat, respectively. For example, two days with WBT = 28°C and 30°C contribute 1 + 3 = 4 degree-days to DD27_{pt}. The 9–27°C interval captures the middle 95% of the WBT distribution in the sample (Appendix Figure B2) and is motivated by evidence that performance losses are highly nonlinear, with substantial declines once WBT exceeds the 26–30°C range (Kjellstrom et al., 2009; Somanathan et al., 2021). Precipitation is measured as cumulative rainfall over the same reference period, entered as a second-degree polynomial to capture potential nonlinear effects.

Other Data. In addition to employment and weather data, I use the Enterprise Census, which covers all formally registered firms in Vietnam during the period 2002–2016, to construct firm-level and province-level longitudinal datasets of labor productivity, proxied by revenue per worker, for the formal non-agricultural sector. I also compile province-level migration rates, agricultural cultivated area, and crop yields from statistical yearbooks.

Merging Employment Data with Weather Data. Employment variables, including sector of employment and hours of work, refer to the 12-month period preceding each individual's interview date. Because interviews within a province occur in different months, individuals from different households in the same province may not share the same

[6]For reference, Vietnamese provinces average roughly 5,000 km^2, comparable to western US counties. The economic regions average approximately 55,000 km^2, comparable to small US states.

weather exposure. However, the majority of interviews take place during June–August, limiting variation in reference periods within province-years. Given that more than 96% of household members remain in the same province over the full reference period, I assume that individual j interviewed in month m of year t in province p has been exposed to the weather distribution of province p during the 24 full months prior to m.[7] The 24-month window reflects the gradual nature of sectoral reallocation—involving job search, skill acquisition, and sometimes relocation—and aligns with the two-year interval between the post-2000 VHLSS survey waves. To examine the temporal structure of the reallocation response, I also construct four non-overlapping 6-month exposure windows (18–24, 12–18, 6–12, and 0–6 months before the interview), as discussed in Section 5.1. Individual-level data are then collapsed to the province-year or province-age group-year level by computing weighted means using survey sampling weights.

The main analysis is conducted at the province-year and province-age group-year levels, which permits a consistent panel spanning the full 1993–2018 study period. District boundaries underwent substantial changes during the 1990s, precluding a district-level panel over the full period. From 2004 onward, the VHLSS includes stable district identifiers and a rotating panel component that tracks the same individuals across consecutive waves, which I exploit to test robustness to finer geographic resolution and fixed individual composition using ERA5-Land reanalysis data at $0.1° \times 0.1°$resolution (Muñoz-Sabater et al., 2021).

3.2 Summary Statistics and Descriptive Evidence

The final sample is an unbalanced panel covering 52 provinces across seven regions over 1993–2018, yielding 569 province-year observations or 1,707 province-age group-year observations depending on the level of analysis. All summary statistics and analyses incorporate sampling weights.

Table 1 summarizes the weighted province-year panel. The labor market is heavily agricultural: on average, 46% of workers are employed in agriculture. Non-agricultural employment is substantial and largely informal—28% of workers are in informal non-agriculture compared with 17% in formal non-agriculture, highlighting the importance of informality as a major absorber of labor outside agriculture. Sectoral composition varies meaningfully both across and within provinces over time: the agricultural share ranges from 0.01 to 0.98, with a within-province standard deviation of 0.11, indicating sizable shifts in sectoral allocation over the sample period.

[7] For example, an individual interviewed in January 2010 in province p is assumed to be exposed to the weather of province p from January 2008 to December 2009.

Table 1: Summary Statistics

	Mean	Overall SD	Within SD	Min	Max	N
	(1)	(2)	(3)	(4)	(5)	(6)
Outcomes (0–1 shares)						
Agriculture	0.457	0.211	0.114	0.012	0.981	569
Formal Non-Agriculture	0.174	0.102	0.061	0.000	0.467	569
Informal Non-Agriculture	0.281	0.099	0.065	0.000	0.744	569
Inactive/Unemployed	0.088	0.050	0.019	0.000	0.245	569
Weather Variables						
WBT Degree Days Below 9°C	8.928	19.071	8.324	0.000	195.973	569
WBT Degree Days Above 27°C	8.847	16.405	6.748	0.000	83.790	569
Total Precipitation (m)	3.901	0.750	0.388	2.034	7.535	569
Demographics						
Share of Male Individuals	0.481	0.015	0.012	0.398	0.525	569
Share of Kinh Ethnictity Individuals	0.859	0.210	0.065	0.000	1.000	569
Average Age	41.5	1.6	1.4	34.8	47.1	569
Share of Individuals Completing High School	0.234	0.119	0.069	0.000	0.630	569

Notes: Unit of analysis is province-year. Sample weights applied throughout. Outcome variables are weighted employment shares in each sector summing to one. Overall SD is the standard deviation of the raw variable; Within SD is the standard deviation after removing province fixed effects. Weather variables are cumulative measures over the 24-month period preceding survey interviews. Degree days below 9°C (DD9) and above 27°C (DD27) are constructed from daily mean wet-bulb temperature. Total precipitation is measured in meters. Sources: Employment and demographic data from VHLSS 1993–2018. Weather data from ERA5 reanalysis.

Weather exposure exhibits substantial variation. Wet-bulb degree days above 27°C (DD27) average 9 but range up to 84, with a within-province standard deviation of 6.7, implying meaningful year-to-year fluctuations in extreme heat exposure. Cold exposure (DD9) is episodic, with a long right tail (maximum 196). Precipitation averages 3.9 meters with relatively less within-province variation (within SD 0.39).

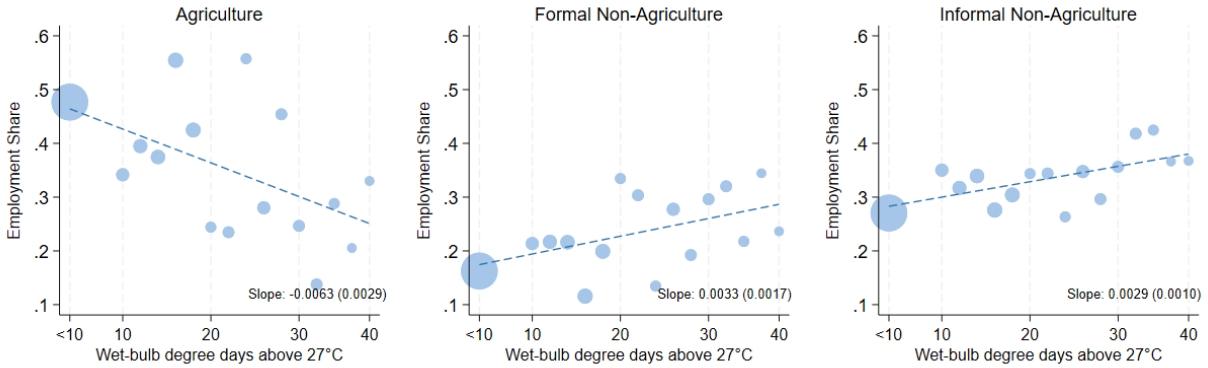
Demographic composition is broadly stable within provinces over time. The male share averages 0.48 with little within-province variation, and mean age is 41 years. The share of the majority ethnic group varies substantially across provinces. Educational attainment is relatively low during this period—only 23% complete high school on average—and shows limited change over time within provinces (within SD 0.07).

Figure 2 presents the raw relationship between wet-bulb degree days above 27°C and sectoral employment shares. Observations are grouped into bins based on temperature exposure: one bin for values below 10, and 2-degree-day increments from 10 to 40. It appears that provinces experiencing higher heat exposure tend to have lower agricultural employment shares and higher non-agricultural shares (Panel A). Furthermore, the rela-

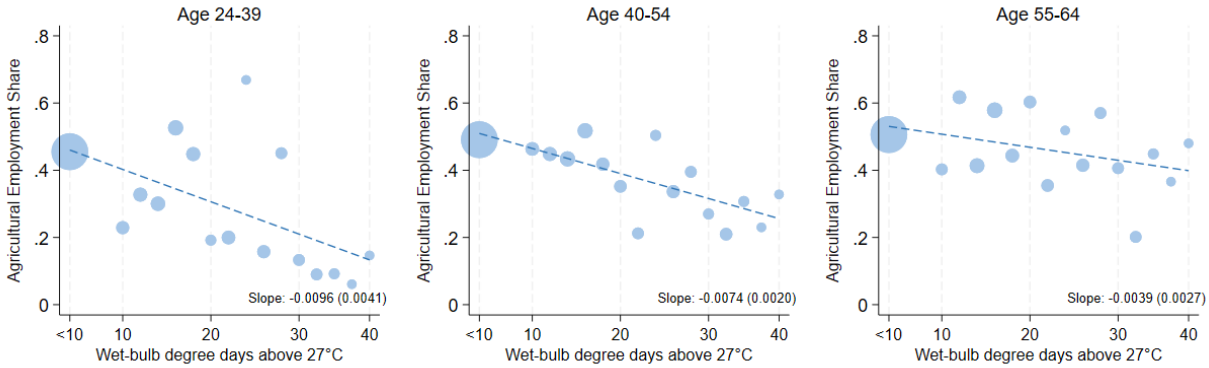
tionships appear stronger for younger workers than for older workers (Panel B).

Figure 2: Descriptive Evidence: Extreme Temperatures and Sectoral Employment Shares

Panel A: Heat Exposure and Sectoral Employment Shares



Panel B: Heat Exposure and Agricultural Employment Shares: By Age Group



Notes: Each panel plots the raw relationship between wet-bulb degree days above 27°C and sectoral employment shares. Observations are grouped into bins based on temperature exposure: one bin for values below 10, and 2-degree-day increments from 10 to 40. Marker size is proportional to the square root of the number of province-year observations in each bin. The dashed line shows the linear fit. Data are weighted by sampling weights.

These cross-sectional correlations, however, need not reflect causal impacts. Provinces that differ in temperature exposure may also differ in many other dimensions such as geography, soil quality, market access, historical development trajectories that could independently affect labor market structure. For example, hotter provinces may be located in lowland coastal areas with better access to urban labor markets, or may have climatic conditions less suitable for certain agricultural activities regardless of heat stress on workers. More generally, cross-sectional comparisons conflate the effects of temperature with all time-invariant characteristics correlated with climate (Auffhammer et al., 2013; Dell et al., 2014). In addition, secular changes in cohort composition, education, or region-specific development policies could also be correlated with both temperature exposure

and sectoral employment, further complicating the interpretation of age-heterogeneity in temperature-labor share relationship.[8] The next section presents empirical approaches to identify plausibly causal effects of temperature on sectoral labor allocation.

4 EMPIRICAL STRATEGY

4.1 Baseline Specification: Average Effects

To estimate the average temperature effects, I follow the standard panel fixed effects approach in the climate-economy literature (e.g., Auffhammer et al., 2013; Dell et al., 2012; Deschênes & Greenstone, 2007), exploiting year-to-year variation in weather within provinces. Province fixed effects remove all time-invariant confounders, ensuring that identification comes from plausibly exogenous inter-annual weather variation rather than potentially endogenous differences across locations. Specifically, I estimate the following equation:

$$y_{prt} = f(\mathrm{WBT}_{pt}) + g(\mathrm{R}_{pt}) + \gamma_p + \gamma_{rt} + \varepsilon_{prt} \tag{1}$$

where y_{prt} is an outcome in province p, economic region r, and year t. Outcomes include the employment shares (main job) in agriculture, informal non-agriculture, and formal non-agriculture. The term R_{pt} is a vector of non-temperature weather controls measured over the same reference period as WBT_{pt}, including second-degree polynomials in rainfall.

Province fixed effects γ_p absorb time-invariant differences across provinces, and region-by-year fixed effects γ_{rt} absorb shocks common to provinces within an economic region in a given year, including region-specific growth episodes, policy changes, and discrete climate events such as the 2015–2016 El Niño drought that severely affected the Mekong Delta and Central Highlands (Reliefweb, 2016). Specifications relying on year fixed effects alone would fail to account for such regional shocks.

A concern with demanding fixed effects is that they may absorb too much weather variance, leaving insufficient power to detect effects. Fisher et al. (2012) demonstrate this in their comment on Deschênes and Greenstone (2007): with corrected data, moving from county-plus-year to county-plus-state-by-year fixed effects leaves only 1.3% of observations with residuals exceeding 1°F. In this setting, residual variation is more substantial. After removing province and region-by-year fixed effects, residual standard deviations are 6.92 degree days for cold (36% of raw variation) and 1.88 degree days for heat (12%). Importantly, 37% and 31% of observations have cold and heat residuals exceeding 1 de-

[8] For example, Porzio et al. (2022) emphasize that cohort-driven exits from agriculture can be tightly linked to education expansion, which may mechanically coincide with changes in climate exposure over time.

gree day, respectively—well above the 1.3% documented in Fisher et al. (2012), suggesting sufficient identifying variation remains under this demanding specification (Appendix Table D1).

All regressions are weighted using survey sampling weights so that estimates correspond to effects for the average person.[9] In addition to standard errors clustered at the province level, I report Conley standard errors allowing spatial correlation up to 150 km and serial correlation up to five lags (Conley, 1999) using the weighted Stata module developed by Colella et al. (2019).

The main object of interest in equation (1) is the response function $f(\mathrm{WBT}_{pt})$. In the baseline specification, I summarize the temperature distribution using degree days constructed from daily wet-bulb temperature, WBT_{dpt}, and model $f(\cdot)$ as a piecewise-linear function:

$$f(\mathrm{WBT}_{pt}) \equiv \beta_{DD9}\,\mathrm{DD9}_{pt} + \beta_{DD27}\,\mathrm{DD27}_{pt}$$

With this function form, β_{DD9} and β_{DD27} represent the effect of one additional degree day below 9°C or above 27°C, respectively, accumulated over the 24-month reference period. For robustness checks, I also consider more flexible representations of $f(\cdot)$, including temperature bins, degree-day bins, and fourth-order polynomials. These specifications allow for richer non-linearities while remaining parsimonious.

4.2 Heterogeneous Effects By Age Group

To examine whether the temperature effects vary across age groups, I estimate an age-group extension of the baseline specification, following the fully saturated structure used to estimate age-specific climate impacts in panel settings (e.g., Carleton et al., 2022):

$$y_{aprt} = f_a(\mathrm{WBT}_{pt}) + g_a(\mathrm{R}_{pt}) + \gamma_{ap} + \gamma_{art} + \varepsilon_{aprt} \tag{2}$$

where y_{aprt} is the outcome for age group $a \in \{24\text{–}39,\ 40\text{–}54,\ 55\text{–}64\}$ in province p, region r, and year t. The age-group-by-province fixed effects γ_{ap} absorb time-invariant differences in sectoral allocation across age groups within a province. The age-group-by-region-by-year fixed effects γ_{art} absorb age-specific shocks common within a region-year. The identifying variation thus comes from within-province deviations in temperature exposure interacted with age group, net of broad regional age-specific trends. In practice,

[9] Weights are constructed from the survey sampling weights and normalized to sum to one within each survey year. Specifically, the weight for province p in year t is $w_{pt} = \frac{\sum_{j\in p} sw_{jpt}}{\sum_p \sum_{j\in p} sw_{jpt}}$, where sw_{jpt} is the sampling weight for individual j in province p and year t.

estimating equation (2) is equivalent to estimating equation (1) while allowing the coefficients on weather variables to differ flexibly by age group.

4.3 Identification

For the average temperature effects, the identifying assumption is that, conditional on the included fixed effects and controls, province-level year-to-year deviations in temperature are orthogonal to unobserved determinants of sectoral employment:

$$\mathbb{E}\left[\varepsilon_{prt} \mid \mathrm{WBT}_{pt}, \mathrm{R}_{pt}, \gamma_p, \gamma_{rt}\right] = 0$$

Intuitively, the coefficients are identified from within-province fluctuations in heat exposure around province-specific averages, netting out region-year shocks that could jointly affect labor demand and sectoral allocation. This structure directly addresses concerns that slow-moving changes such as education expansion, demographic transitions, or region-level development strategies could otherwise confound the relationship between temperature and labor reallocation.

For the heterogeneity analysis, the identifying assumption extends naturally:

$$\mathbb{E}\left[\varepsilon_{aprt} \mid \mathrm{WBT}_{pt}, \mathrm{R}_{pt}, \gamma_{ap}, \gamma_{art}\right] = 0$$

Threats and interpretation. A potential concern with estimating temperature effects on province-level labor market outcomes is that temperature exposure may proxy for other time-varying province-level changes correlated with sectoral employment. The baseline analysis mitigates this in several ways. Province fixed effects absorb all time-invariant confounders, and region-by-year fixed effects net out regional shocks—including discrete climate events such as the 2015–16 El Niño drought that severely affected the Mekong Delta and Central Highlands (Reliefweb, 2016). Second-order polynomials of precipitation control for non-temperature weather shocks. To further assess the validity of the design, I include leads of temperature exposure as a falsification test: if future weather, which has not yet been realized, predicts current employment outcomes, it would suggest that the specification captures spurious trends rather than causal effects. I also conduct Monte Carlo permutation tests that randomly reassign weather series across provinces and re-estimate the baseline specification. If the estimated effects reflect a genuine relationship between temperature and employment, the actual estimates should fall in the tails of the distribution generated under random assignment. Finally, leave-one-out exercises are used to verify if results are not driven by any single province.

Equation (2) identifies differential reduced-form responses by age group under the

same exogeneity assumption as the baseline. The age-group-by-province fixed effects absorb time-invariant differences in sectoral allocation across age groups within a province, including persistent cohort-level characteristics such as average education or historical exposure to market institutions. The age-group-by-region-by-year fixed effects absorb age-specific shocks common within a region-year, such as age-targeted hiring, region-specific policies, or sectoral demand shifts that affect age groups differentially.

A separate concern is that the age gradient in sectoral responses could reflect cohort-specific attributes rather than age-specific barriers. Although the fixed effects control for average differences across cells and broad cohort-specific trends, they may not fully absorb within-province changes in cohort composition. If the age gradient were driven by cohort-level differences in human capital, then conditioning on education—the dimension along which Vietnamese cohorts differ most markedly—should attenuate it. I assess this by controlling for demographic composition within province-age group cells and estimating effects separately by education level. As a further test, I exploit the rotating panel component of the VHLSS (2004–2018), which tracks the same individuals across up to three consecutive survey waves. Because the main analysis uses repeated cross-sections, different birth cohorts could populate each province-age group cell across waves, and the observed age gradient could in principle reflect compositional turnover rather than age-specific behavioral responses. Estimating the specification on this panel sub-sample directly addresses this concern.

5 EMPIRICAL RESULTS

5.1 Average Temperature Effects

Table 2 reports estimates of equation (1). Each column relates sectoral employment shares (coded 0–1) to wet-bulb degree days below 9°C (DD9) and above 27°C (DD27), controlling for province fixed effects and region-year fixed effects. Cold exposure has no economically or statistically meaningful effect on sectoral labor shares.

In contrast, heat exposure is associated with a substantial reallocation of employment across sectors: one additional degree-day above 27°C reduces the agricultural employment share by roughly 0.64 percentage points, and increases the formal and informal non-agricultural shares by roughly 0.26 and 0.37 percentage points, respectively. The inactive and unemployed share is not significantly affected. Inference is robust to both province-clustered standard errors (in parentheses) and Conley spatial HAC standard errors that allow for spatial correlation up to 150 km and serial correlation up to five lags (in brackets).

To put these estimates into perspective, Appendix Table D2 presents the same spec-

Table 2: Wet-bulb Temperature and Sectoral Labor Share

	Agriculture	Formal Non-Agriculture	Informal Non-Agriculture	Inactive and Unemployed
	(1)	(2)	(3)	(4)
DD9	0.0007	-0.0001	-0.0005	-0.0001
	(0.0004)	(0.0001)	(0.0003)	(0.0001)
	[0.0003]	[0.0001]	[0.0003]	[0.0000]
DD27	-0.0064	0.0026	0.0037	0.0000
	(0.0016)	(0.0012)	(0.0011)	(0.0005)
	[0.0015]	[0.0008]	[0.0012]	[0.0004]
Mean Outcome	0.457	0.174	0.281	0.088
Observations	569	569	569	569
Province FE	Yes	Yes	Yes	Yes
Region-Year FE	Yes	Yes	Yes	Yes

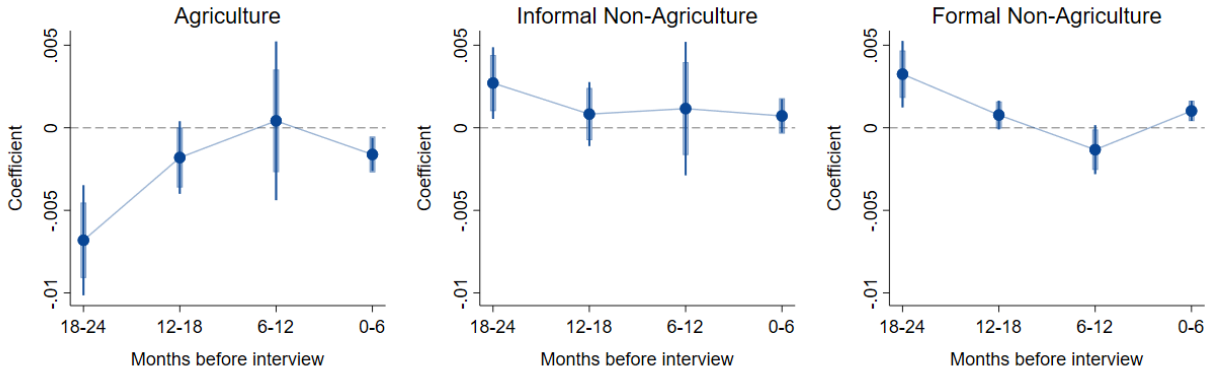
Notes: Each column presents the effects of one additional degree day below 9°C (DD9) or above 27°C (DD27) on sectoral labor shares. Estimates obtained from equation (1). Unit of analysis is province-year. All regressions control for the second-order polynomials of precipitation. Robust standard errors clustered at the province level and Conley standard errors that allow for spatial correlation up to 150 km and serial correlation up to five lags are reported in parentheses and brackets, respectively. Province distances are computed from province geographic centroids. All regressions use sampling weights.

ification but reports effects in within-province standard deviation units. A one within-province standard deviation increase in degree days above 27°C (approximately 6.7 degree days) is associated with a 4.3 percentage point decline in the agricultural share—about 10% of the mean agricultural share (0.45), which is offset by increases of 1.8 and 2.5 percentage points in the formal and informal non-agricultural shares, respectively. The precipitation terms indicate a non-linear relationship with sectoral allocation (with the linear and squared terms entering with opposite signs), but these precipitation-related estimates are generally less precisely estimated than the temperature effects. In what follows, I focus the discussion on temperature effects.

These estimates pool temperature exposure over a 24-month window. To examine how the reallocation unfolds over time, I estimate a distributed-lag specification that replaces the pooled measure with four non-overlapping 6-month exposure windows (18–24, 12–18, 6–12, and 0–6 months before the interview), each entering simultaneously alongside corresponding controls for cold exposure and precipitation. Coefficients are scaled by a factor of four so that their sum is comparable to the pooled estimate in Table 2. The results in Figure 3 illustrate that the reallocation effect is concentrated in the most distant exposure window: heat experienced approximately 12–24 months before the interview drives the largest sectoral shifts, with the effect attenuating at shorter lags. This pattern is

robust to both clustering and spatial inference.

Figure 3: Dynamic Effects of Heat Exposure on Sectoral Labor Shares



Notes: Each panel presents the distributed-lag effects of one additional degree day above 27°C (DD27) on sectoral labor shares, estimated separately for four non-overlapping 6-month exposure windows (18–24, 12–18, 6–12, and 0–6 months before the interview). Coefficients are scaled so that their sum is comparable to the pooled 24-month estimate in Table 2. Unit of analysis is province-year. Estimates obtained from equation (1). All regressions control for cold exposure (DD9) and second-order polynomials of precipitation, each entering in the same four 6-month windows. Thin dark bars show 95% confidence intervals using robust standard errors clustered at the province level. Thick light bars show 95% confidence intervals using Conley standard errors that allow for spatial correlation up to 150 km and serial correlation up to five lags. Province distances are computed from province geographic centroids.

To have a better understanding of which industry climate-induced migrants move into, I classify non-agricultural employment into three main groups: less, medium, and high skill-intensive industries.[10] This further breakdown of the destination sector suggests that roughly 66% of the labor reallocation induced by heat exposure is into less skill-intensive industries (Appendix Table D3), including low-tech manufacturing (e.g., manufacture of food products and beverage, textile and apparel), less knowledge-intensive services (e.g., wholesale and retail, transport, accommodation and food services), construction and mining.

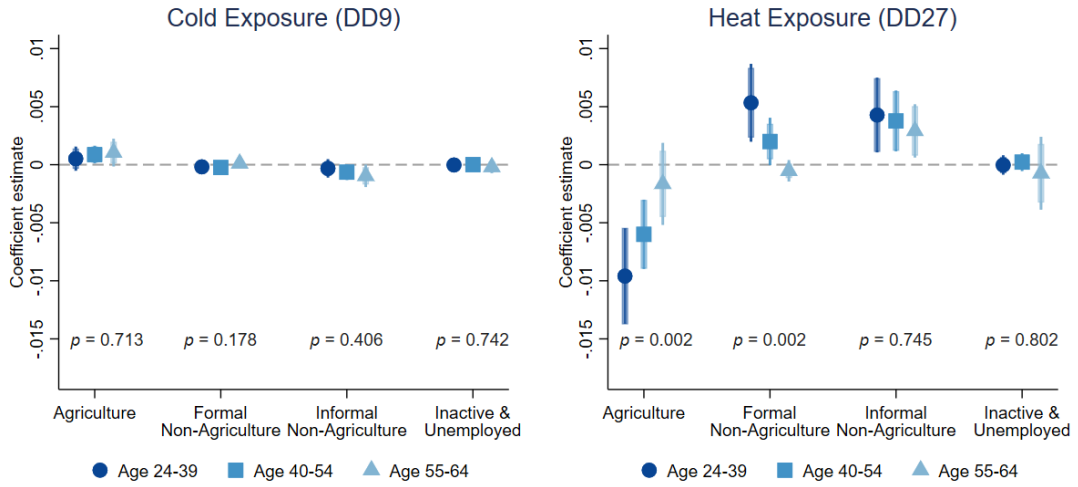
5.2 Heterogeneous Effects by Age Group

Figure 4 reports heterogeneous temperature effects by age group (24–39, 40–54, and 55–64), estimated from equation (2) with province-by-age-group fixed effects and region-by-

[10]Industries are ranked following the Statistical Classification of Economic Activities in the European Community. Less skill-intensive industries include those classified as low-tech manufacturing, less knowledge-intensive services, construction and mining; medium skill-intensive industries include those classified as medium-tech manufacturing and public utilities; high skill-intensive industries include those classified as high-tech manufacturing and knowledge-intensive services according to the Statistical Classification of Economic Activities in the European Community. The mean educational attainment of workers in each industry is presented in Appendix Table B3. For details, see Annex 3 – High-tech Aggregation by Statistical Classification of Economic Activities in the European Community (NACE Rev.2)

age-group-by-year fixed effects. Cold exposure (DD9) has small and statistically insignificant effects on sectoral labor shares for all age groups, and I cannot reject equality of cold temperature effects across cohorts (joint p-values are 0.71 for agriculture, 0.18 for formal non-agriculture, 0.41 for informal non-agriculture, and 0.74 for inactivity/unemployment).[11]

Figure 4: Wet-bulb Temperature and Sectoral Labor Share: Heterogeneity by Age Group



Notes: Each panel presents the effects of one additional degree day below 9 °C (DD9) or above 27°C (DD27) on sectoral labor shares of different age groups. Unit of analysis is province-age group-year. Estimates obtained from equation (2). All regressions control for cold exposure (DD9) and the second-order polynomials of precipitation and their interactions with group dummies. Thin dark bars show 95% confidence intervals using robust standard errors clustered at the province level. Thick light bars show 95% confidence intervals using Conley standard errors that allow for spatial correlation up to 150 km and serial correlation up to five lags. Province distances are computed from province geographic centroids. p-values from the F-test of significant age cohort differences using standard errors clustered at the province level are reported. The results are qualitatively similar when using Conley standard errors.

In contrast, heat exposure (DD27) induces pronounced reallocation that varies by age. For workers aged 24–39, one additional degree day above 27°C reduces the agricultural labor share by 1 percentage points and increases formal and informal non-agricultural shares by 0.53 and 0.43 percentage points, respectively ($p < 0.01$). For workers aged 40–54, the agricultural share falls by 0.6 percentage points ($p < 0.01$), with corresponding increases of 0.2 (formal, $p < 0.1$) and 0.40 (informal, $p < 0.01$) percentage points. For workers aged 55–64, the agricultural and formal responses are smaller and imprecisely estimated, but informal non-agricultural employment increases by roughly 0.30 percentage points ($p < 0.05$). Consistent with these patterns, I reject equality of heat effects across age groups for agriculture (joint $p < 0.01$) and formal non-agriculture (joint $p < 0.01$), but not for informal non-agriculture (joint $p = 0.75$) or inactivity/unemployment (joint p =

[11]The corresponding full regression results are presented in Appendix Table D4.

0.80). Pairwise tests show that the DD27-induced decline in agricultural employment is larger for ages 24–39 than middle and older groups, while differences in informal non-agricultural responses across cohorts are not statistically significant. I find no evidence that heat exposure affects the inactivity/unemployment share for any age group, suggesting that the main margin of adjustment is sectoral reallocation rather than movement into non-employment.

One concern with interpreting the age gradient is that it may reflect cohort-specific attributes rather than age-specific barriers. The sample covers a period of rapid structural transformation and cohort upgrading: older workers observed in the 1990s grew up under central planning with no private formal sector, while older workers observed in the 2010s had substantially more exposure to market institutions and formal education. Although the province-by-age-group fixed effects and region-by-age-group-by-year fixed effects control for average differences across cells and broad cohort-specific trends, they may not fully absorb within-province cohort composition changes. If the age gradient were substantially driven by cohort-level differences in human capital, conditioning on education (the dimension along which Vietnamese cohorts differ most markedly) should attenuate it. However, estimating the specification separately for workers with and without a high school diploma, the age gradient in formal non-agricultural employment persists within both education groups, whereas the informal non-agricultural response remains statistically indistinguishable across age groups in both samples (Appendix Table D5). This suggests that the barrier is specific to formal sector access rather than a general cohort-level difference in willingness or ability to leave agriculture.

To further rule out that different workers entering and exiting each cell drive the differential effects across age groups, I exploit the rotating panel component of the VHLSS (2004–2018), which tracks the same individuals across up to three consecutive survey waves, with weather measured at the finer district level. The age-specific patterns are consistent: transitions into formal non-agriculture remain concentrated among younger workers ($p = 0.016$ for the joint test of equality across age groups), while the informal response remains statistically indistinguishable across age groups ($p = 0.51$, Appendix Table D6).

Together, these results suggest that heat exposure shifts workers out of agriculture across all age groups, but transitions into formal non-agriculture are concentrated among younger workers, while older workers adjust primarily through informal channels. This pattern likely reflects barriers associated with age itself rather than cohort-level differences in skills or institutional exposure.[12]

[12]This age gradient is also not driven by the employer-type classification of formality. A cross-tabulation

5.3 Robustness Checks and Additional Results

The main results presented so far are robust to a range of specification checks detailed in Appendix Section C. These include adding time-varying demographic controls (education, gender, ethnicity), dropping province-year cells with few underlying observations, leave-one-out province exclusions, alternative spatial interpolation of gridded weather data (averaging all grid points within a provincial geographic boundary), and alternative functional forms of temperature exposure including cumulative bins, degree day bins, and polynomial specifications (Appendix Table D7; Appendix Figures D1–D5). Across all specifications, the qualitative pattern is unchanged: heat exposure reduces the agricultural employment share and increases non-agricultural employment, with transitions into formal non-agriculture concentrated among younger workers and informal absorption broad-based across age groups. A Monte Carlo permutation test confirms that the estimates fall far outside the distribution of 10,000 placebo coefficients for both the average and age-specific results (Appendix Figures D6–D7).

Decomposing temperature exposure into 12-month windows (a lead capturing the 12 months following the interview, contemporaneous exposure 0–12 months before, and lagged exposure 12–24 months before) corroborates the finer 6-month dynamic profile in Figure 3: the lagged component drives the reallocation effects while contemporaneous exposure has no independent effect (Appendix Table D8). In addition, lead coefficients, capturing future temperature not yet realized, are small and statistically insignificant across all sectors and age groups, supporting the causal interpretation of the estimates (Appendix Tables D8–D9).

Finally, I examine whether inter-provincial migration could mechanically generate the observed sectoral shifts. Three exercises address this concern. A decomposition following McCaig and Pavcnik (2018) confirms that most structural change occurs within provinces (Appendix Table D10). Province-level migration rates from statistical yearbooks do not respond to temperature exposure (Appendix Table D11). And an individual-level measure—the number of months each household member stayed at home during the reference period, which unlike the province-level measure captures whether individuals remained in their own household—shows no reduction for any age group in response to heat (Appendix Table D12). Results are also consistent when estimated on the longitudinal panel component, which holds individual composition fixed (as already discussed in Section 5.2, Appendix Table D6). Together, these results indicate that the observed reallo-

of employer-type and worker-side informality measures shows that the share of unprotected workers within formal firms is itself increasing in age (Appendix Table B2), suggesting that an alternative worker-side definition would, if anything, produce a steeper age gradient.

cation occurs among workers who remain in place rather than through migration-driven compositional change.

6 POTENTIAL MECHANISMS

Guided by the theoretical framework, this section presents empirical evidence on two potential mechanisms underlying the main results. The first addresses the "push" from agriculture: heat reduces agricultural returns relative to other sectors, shifting comparative advantage and driving workers out. The second addresses why formal employment is accessible primarily to younger workers: switching costs vary by age, with older workers facing barriers that limit their options to the informal sector.

6.1 The Push: Relative Returns

Heat exposure may reduce agricultural returns through multiple channels: lower productivity, fewer workable hours, or both. If non-agricultural returns are less affected, comparative advantage shifts away from agriculture, pushing workers out. I examine each channel in turn.

Labor Productivity. I directly test whether hot temperatures have differential effects on sectoral labor productivity by assembling a longitudinal dataset of province-level production for agriculture, formal non-agriculture, and informal non-agriculture from 2002 to 2016 using both VHLSS and enterprise census (See Appendix Section B for details). I estimate the baseline equation (1) where the outcome is log of revenue per worker in each sector (agriculture, informal non-agriculture, and formal non-agriculture) in province p of region r in year t.[13]

Table 3 presents the results. Cold exposure has no significant effect on sectoral labor productivity. Hot exposure, however, has a significantly larger impact on agriculture than other sectors. One additional degree day above 27°C over the 24-month reference period reduces revenue per worker in agriculture by approximately 0.56% (p-value < 0.01). Given a within-province standard deviation of roughly 7 degree days, a one SD heat exposure implies a 3.5% decline in agricultural productivity. The corresponding effects on formal and informal non-agricultural productivity are statistically insignificant, though the formal sector estimate is imprecisely measured.

[13]Revenue in agriculture is not restricted to crop production but also includes revenues from livestock, aquaculture, farm service, and forestry. Ideally, one should estimate the marginal product of labor in each sector and then examine the effect of hot temperatures on that outcome. Data limitations, however, do not allow me to estimate the marginal product of labor.

Table 3: Wet-bulb Temperature and Sectoral Labor Productivity

	Agriculture	Formal Non-Agriculture	Informal Non-Agriculture
	(1)	(2)	(3)
DD9	0.0008	-0.0032	0.0016
	(0.0005)	(0.0023)	(0.0016)
	[0.0004]	[0.0026]	[0.0016]
DD27	-0.0056	0.0051	-0.0008
	(0.0027)	(0.0074)	(0.0033)
	[0.0021]	[0.0054]	[0.0035]
Mean Outcome	2.279	5.331	3.488
Observations	416	416	416
Province FE	Yes	Yes	Yes
Region-Year FE	Yes	Yes	Yes

Notes: Each column presents the effects of one additional degree day below 9 or above 27°C (DD27) on labor productivity, proxied by revenues per worker (log), at the province level. Estimates obtained from equation (1). Unit of analysis is province-year. All regressions control for second-order polynomials of precipitation. Robust standard errors clustered at the province level and Conley standard errors that allow for spatial correlation up to 150 km and serial correlation up to five lags are reported in parentheses and brackets, respectively. Province distances are computed from province geographic centroids.

Although non-farm labor productivity is unaffected on average, firm-level analysis of the mining and quarrying sector—where workers face thermal conditions similar to, or even more demanding than, agriculture—reveals substantial productivity losses, particularly among small and older establishments (Appendix Table D13). This supports the interpretation that heat affects labor productivity through physiological constraints on human performance in climate-exposed work environments.

Labor Supply. Heat may also reduce agricultural returns by lowering labor supply. If hot temperatures increase labor disutility through health effects and impaired task performance (Rode et al., 2022), workers may reduce hours worked. Table 4 tests this channel. Panel A reports effects on hours worked conditional on working in a sector; Panel B reports unconditional hours, treating non-participants as zero.

One additional degree day above 27°C reduces conditional hours in agriculture by approximately 20 hours per year (2% of sample mean, Panel A, Column 1). Hours in formal non-agriculture are unaffected among existing workers, while informal non-agriculture shows a modest increase, possibly reflecting reallocation of effort toward non-farm activities. Panel B shows unconditional hours, which capture both intensive and extensive margin responses. The increase in unconditional hours in formal and informal non-agriculture (Columns 2–3) reflects new workers entering these sectors rather than existing workers increasing hours, consistent with the extensive margin reallocation documented

Table 4: Wet-bulb Temperature and Labor Supply

	Agriculture	Formal Non-Agriculture	Informal Non-Agriculture	Total Hours
	(1)	(2)	(3)	(4)
Panel A: Conditional Hours				
DD27	-19.62	0.62	10.05	
	(7.10)	(3.18)	(4.57)	
	[7.03]	[2.46]	[3.86]	
Mean Outcome	1281	1790	1825	
Observations	465	462	465	
Panel B: Unconditional Hours				
DD27	-15.83	6.00	6.48	-3.12
	(4.60)	(1.79)	(2.92)	(5.66)
	[4.77]	[1.68]	[2.13]	[4.01]
Mean Outcome	778	686	438	1910
Observations	517	517	517	517
Province FE	Yes	Yes	Yes	Yes
Region-Year FE	Yes	Yes	Yes	Yes

Notes: Each column presents the effects of one additional degree day above 27°C (DD27) on hours of work. Dependent variables are average number of hours worked, winsorized at the top 1% of the individual distribution by year. In Panel A, hours are conditional on working in a sector (intensive margin). In Panel B, hours are unconditional, with individuals not working in a sector coded as zero hours (intensive and extensive margin). Estimates obtained from equation (1). Unit of analysis is province-year. All regressions control for the second-order polynomials of precipitation. Robust standard errors clustered at the province level and Conley standard errors that allow for spatial correlation up to 150 km and serial correlation up to five lags are reported in parentheses and brackets, respectively. Province distances are computed from province geographic centroids. All regressions use sampling weights. Data from household surveys. In 2002, only hours worked for the primary job are recorded and thus data from VHLSS 2002 are dropped for consistency.

in the main results. Total hours decline by approximately 3 hours per degree day (Column 4), though this effect is not statistically significant.

Prices. For these labor productivity and hours losses to drive sectoral reallocation, prices must not fully adjust to absorb the shock. I examine this condition using the country's dominant crop: rice. In particular, I construct a province-level panel of prices prices at harvest and at sale from household surveys. Estimates using the same baseline equation (1) show that rice prices do not respond to temperature shocks: coefficients are statistically insignificant and economically negligible (Appendix Table D14). This is consistent with local agricultural prices being insensitive to local climatic shocks, as would be expected given Vietnam's integration into international rice markets and its history of government price interventions designed to insulate domestic prices from both world market volatil-

ity and local production fluctuations (Ghosh & Whalley, 2004; Government of Vietnam, 2009). With prices unaffected, productivity and hours losses translate directly into lower agricultural returns.

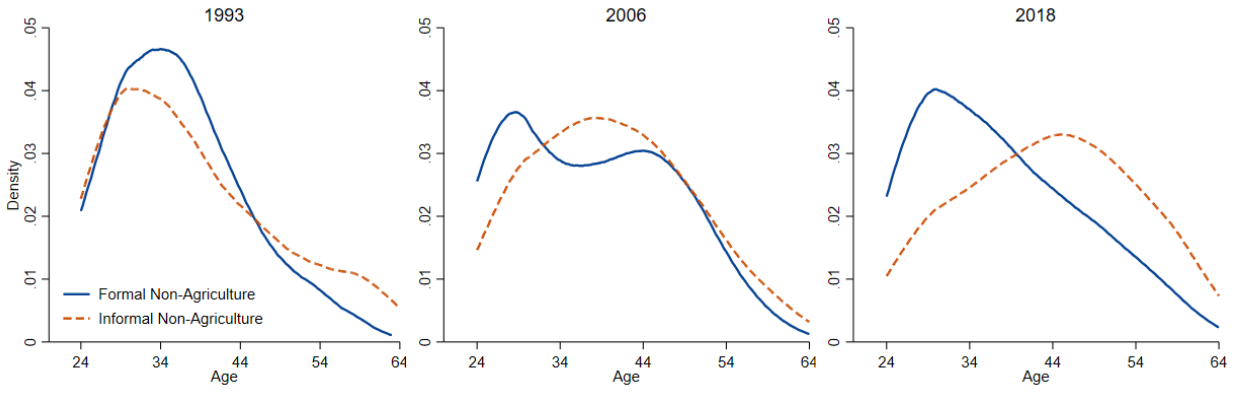
Additional evidence shows that heat also reduces crop yields by approximately 0.2 to 0.5% per degree day depending on the crop (Appendix Table D15), while cultivated area remains unaffected (Appendix Table D16). This suggests that heat's impact on agriculture operates through labor in addition to land productivity.

Together, these findings suggest that heat disproportionately reduces returns to agricultural labor, through lower productivity, fewer workable hours, and reduced yields, while non-agricultural returns remain largely stable and prices do not adjust. This shift in relative returns creates incentives for workers to move out of agriculture.

6.2 The Sorting: Age-Based Barriers to Formal Employment

The relatively larger decline in agricultural returns provides the push, but does not explain why younger workers gain access to formal employment while older workers do not. The conceptual framework identifies two reinforcing forces: accumulated agricultural attachment that raises the effective payoff of remaining in farming, and within-cohort variation in access to formal employment, with older cohorts facing a higher share of workers with limited formal access. A pattern consistent with this is visible throughout the study period. Figure 5 plots the age distribution of formal and informal non-agricultural workers across survey waves. Formal employment is consistently concentrated among younger workers relative to informal employment, and this gap does not narrow over time despite substantial cohort upgrading.

Figure 5: Age Distribution of Formal and Informal Non-Agricultural Workers Over Time



Notes: Each panel plots the kernel density of age among formal (solid) and informal (dashed) non-agricultural workers for selected VHLSS waves. Sample restricted to workers aged 24–64. Densities are estimated using survey sampling weights.

I examine the sources of age-based sorting using three complementary pieces of evidence. First, I test whether earnings patterns among sector-switchers are consistent with the model's prediction of positive selection into formal employment (Proposition A4). The earnings evidence is consistent with the framework but cannot separately identify agricultural attachment from formal barriers. Second and third, I turn to direct evidence of demand-side constraints, using age restrictions in job postings and stated employer preferences in a survey experiment.

Earnings Patterns Among Sector-Switchers. I begin by examining whether the pattern of earnings gains among sector-switchers is consistent with the model's prediction of age-based selection into formal employment. Exploiting the panel dimension of the VHLSS, I estimate within-individual log hourly earnings gaps when workers switch from agriculture into formal or informal non-agriculture. The regressions include individual and panel-by-wave fixed effects, so the earnings gaps are identified from within-individual variation in sector assignment across waves. Age groups are defined at the start of each panel to avoid contamination from workers crossing age group boundaries between waves (Table 5).

Switching into formal non-agriculture yields a substantial hourly earnings premium of approximately 0.43 log points for workers aged 24–54, rising to 0.59 for those aged 55–64 (Column 3). The joint test rejects equality across age groups ($p < 0.01$), mostly driven by the oldest group's significantly larger gain. This is consistent with Proposition A4: if older workers face higher barriers to formal entry, those who do enter must be more strongly positively selected on unobserved skill, producing a larger within-individual earnings gain. By contrast, switching into informal non-agriculture yields no significant hourly earnings premium at any age (Column 4), with point estimates indistinguishable from zero once returns to worker characteristics are allowed to vary by sector.[14]

These earnings patterns are consistent with the framework but cannot by themselves distinguish selection from barriers. The next two pieces of evidence speak directly to demand-side constraints.

Age Restrictions In Job Postings. To provide direct evidence on age-based barriers, I collected data on low-skilled job postings from a major Vietnamese job platform during October 2025–March 2026 (see Appendix E.2 for details). Out of 582 postings that do not

[14]Under uniform returns (Columns 1–2), the informal gap appears positive and significant, but this reflects compositional differences across sectors rather than a genuine premium. The estimates are broadly consistent with Hamory et al. (2021), who find approximate gains of 22% for workers moving out of agriculture in Indonesia, though their estimate pools formal and informal destinations.

Table 5: Gains in Hourly Labor Earnings among Switchers

Reference category is working in agriculture

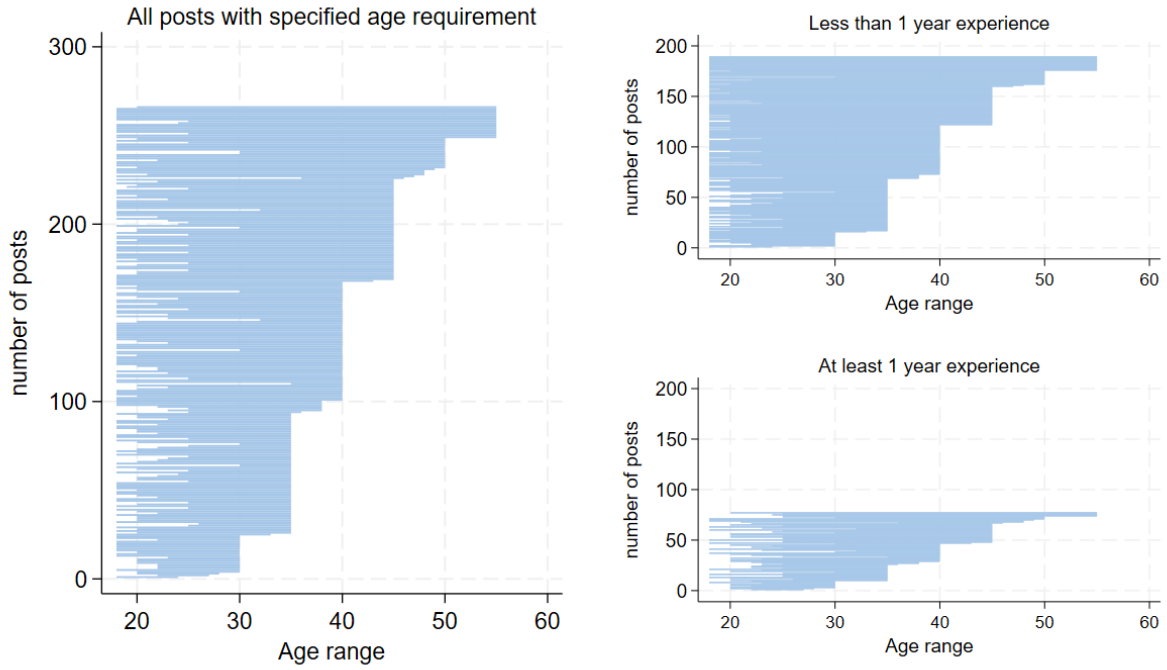
	Uniform Returns		Non-Uniform Returns	
	(1) Formal Non-Agriculture	(2) Informal Non-Agriculture	(3) Formal Non-Agriculture	(4) Informal Non-Agriculture
Age 24–39 (G1)	0.398	0.296	0.438	0.060
	(0.036)	(0.039)	(0.153)	(0.160)
Age 40–54 (G2)	0.363	0.244	0.430	0.021
	(0.035)	(0.036)	(0.157)	(0.162)
Age 55–64 (G3)	0.532	0.188	0.593	0.000
	(0.054)	(0.055)	(0.167)	(0.160)
p-value G1 = G2 = G3	0.0071	0.0289	0.0089	0.2172
p-value G1 = G2	0.2238	0.0405	0.7907	0.1350
p-value G1 = G3	0.0272	0.0258	0.0075	0.1967
p-value G2 = G3	0.0026	0.2368	0.0022	0.6567
Adj. R^2	0.407	0.407	0.420	0.420
Observations	32848	32848	32848	32848
Individuals	12630	12630	12630	12630
Panel-Wave FE	Yes	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes	Yes
Controls × Sector			Yes	Yes

Notes: Sample includes workers who switched sectors at least once in each three-wave panel. All regressions include individual and panel-by-wave fixed effects and control for age and age squared. Age group groups are defined using age at the first wave of each panel. Columns (1)–(2) impose uniform returns to worker characteristics across sectors; Columns (3)–(4) allow returns to individual controls to vary by sector. Formal and informal gaps are estimated jointly within each specification. The dependent variable is log hourly earnings, computed as total labor earnings (including wages, other benefits, and household farm/non-farm net profits) divided by annual hours worked. Household members are assumed to receive a share of household net profits proportional to their hours worked in household farm and non-farm businesses. Hourly earnings are trimmed at the top and bottom 5% within each year and sector. Individual controls include gender, ethnicity, marital status (married, single, widowed/separated), and general education qualification (no education, primary, lower secondary, upper secondary, post-secondary). Robust standard errors clustered at the province level are in parentheses. Sources: Data from VHLSS three-wave individual panel datasets 2004–2006–2008, 2010–2012–2014, 2012–2014–2016, and 2014–2016–2018.

require post-secondary education, 266 (45.7%) explicitly specified an age-related eligibility criterion. Figure 6 plots the distribution of specified age ranges among these posts.

Most postings accept candidates starting around age 18–20, but upper age limits typically fall between 35 and 45, with notably fewer opportunities extending beyond (left panel). The pattern is similar for positions requiring less than one year of experience (top-right panel)—those most accessible to agricultural workers seeking to switch sectors—where the majority of postings set upper limits below 45. Age limits persist even for

Figure 6: Distribution of Age Ranges Specified in Low-skilled Job Posts



Notes: Each horizontal bar represents a single job posting, spanning from the minimum to maximum eligible age specified. Data collected from vieclam24h.vn, one of Vietnam's leading job platforms, during October 2025–March 2026. Sample includes 582 low-skilled job postings identified using keywords "low-skilled workers" (lao động phổ thông) and "production line workers" (công nhân sản xuất) that do not require post-secondary education. Of these, 266 postings (45.7%) specified an age eligibility criterion. Left panel shows all postings with age restrictions. Right panels show breakdowns by experience requirements: less than one year (top) and at least one year (bottom). Age restrictions generally apply to both male and female applicants. See Appendix E.2 for details on data collection.

positions requiring more experience (bottom-right panel), albeit to a lesser extent. The concentration of upper age limits in the mid-thirties to mid-forties is consistent with the age-based sorting documented in the main results, and provides direct evidence of the demand-side barriers that the framework posits as one source of sorting into informality.

Stated Employer Preferences. To complement the job postings evidence, I conducted a vignette experiment with 57 manufacturing firms in Central Vietnam during September–December 2025. HR managers evaluate fictitious applicant profiles for low-skilled production positions under two scenarios: one requiring no prior experience and one requiring manufacturing experience. Profiles were identical in education, qualifications, and work background, differing only in age (mid-to-late 20s, early 40s, or mid-50s). Profiles were presented sequentially with randomized ordering to minimize comparison bias. For

a detailed description of the survey, see Appendix Section E.3.[15]

Table 6: Stated Employer Preferences

	All Scenarios				Experience Focused			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Middle (M)	-1.628	-1.627	-1.630	-1.621				
	(0.265)	(0.258)	(0.257)	(0.258)				
Old (O)	-3.196	-3.242	-3.243	-3.168				
	(0.356)	(0.354)	(0.355)	(0.359)				
Middle, Same Experience (ML)					-1.577	-1.577		
					(0.432)	(0.435)		
Middle, More Experience (MH)					-0.923	-0.923		
					(0.388)	(0.390)		
Old, Same Experience (OL)							-3.032	-2.900
							(0.507)	(0.508)
Old, More Experience (OH)							-2.452	-2.367
							(0.476)	(0.487)
p-val: M = O	0.000	0.000	0.000	0.000				
p-val: ML = MH					0.001	0.001		
p-val: OL = OH							0.054	0.083
Order FE	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Health Preference Control	No	No	Yes	Yes	Yes	Yes	Yes	Yes
Firm Size Control	No	No	No	Yes	Yes	Yes	Yes	Yes
Adjusted R2	0.237	0.329	0.331	0.321	0.152	0.185	0.252	0.240
Observations	332	332	332	326	78	78	93	90

Notes: This table presents the difference in scores of middle-aged and older applicants relative to young applicants. Respondents evaluated three profiles sequentially and rated interview likelihood on a 0–10 scale. Column (1) present the raw difference across groups. Columns (2)–(4) present the coefficients on M_i and O_i from the following model $Y_{ij} = \alpha + \beta \mathrm{M}_i + \gamma \mathrm{O}_i + X_j^{'}\delta + \varepsilon_{ij}$, where Y_{ij} is the ratings received for a profile i by manager j, and M_i and O_i are dummy variables for age of the profile (M for middle-age, and O for older). The vector X_j includes observable firm characteristics (firm size, preference for healthy candidates), the order in which each profile is presented to respondents, and scenario fixed effects. Columns (5)–(6) and Columns (7)–(8) present the results from the following models: $Y_{ij} = \alpha + \beta_L \mathrm{M}_{Li} + \beta_H \mathrm{M}_{Hi} + X_j^{'}\delta + \zeta_{ij}$ and $Y_{ij} = \alpha + \gamma_L O_{Li} + \gamma_H O_{Hi} + X_j^{'}\delta + \zeta_{ij}$, where $\mathrm{M}_{Li}/\mathrm{O}_{Li}$ denotes Middle/Old profile with similar experience as the corresponding Young profile, and $\mathrm{M}_{Hi}/\mathrm{O}_{Hi}$ denotes Middle/Old profile with more experience compared to the corresponding Young profile. Across columns, standard errors clustered at the HR manager (respondent) level are in parentheses. Source: Calculations from vignette experiment of HR managers.

The results in Table 6 reveal substantial age-based differentials. On a 0–10 scale measuring likelihood of contacting a candidate for an interview, middle-aged applicants receive ratings approximately 1.6 points lower than young applicants; older applicants re-

[15] As in standard correspondence and vignette studies (e.g., Neumark et al., 2019), the analysis entails testing for differences by age with no controls and verifying robustness to including controls. There is limited scope for specification searching in this setting, and thus I did not file a pre-analysis plan.

ceive ratings 3.2 points lower (Column 1). These gaps persist after controlling for firm characteristics and presentation order (Columns 2–4). When middle-aged and older applicants are assigned experience commensurate with their age, the gaps narrow—from 1.6 to 0.9 points for middle-aged applicants ($p < 0.01$, Columns 5–6), and from 3.0 to 2.4 points for older applicants ($p \approx 0.05$–0.08, Columns 7–8)—but remain substantial. Even with commensurate experience, middle-aged applicants still score roughly 1 point lower and older applicants roughly 2.4 points lower than their young counterparts. These residual gaps suggest that the age penalty reflects employer preferences beyond rational inference about experience or accumulated skill, consistent with age-based preferences documented in other labor markets (e.g., Carlsson & Eriksson, 2019; Lippens et al., 2023; Neumark et al., 2019; Qvist & Larsen, 2025). By holding qualifications and other characteristics constant, the experiment isolates age as the source of differential treatment, providing direct evidence that employer preferences constitute one source of the barriers channeling older workers away from formal employment.

Ruling Out Alternative Explanations. Three alternative explanations merit consideration. First, older workers may be more physiologically vulnerable to heat, potentially explaining their different labor market responses. However, differential vulnerability would predict that older workers are more likely to exit climate-exposed industries like agriculture but cannot explain why they transition into informal rather than formal employment. If anything, greater physiological sensitivity should push older workers toward climate-controlled formal work environments. I find no evidence that heat disproportionately reduces hours worked among older agricultural workers relative to younger ones. If anything, younger workers reduce hours more than older workers in response to heat (Appendix Table D17).

Second, older cohorts may have less education on average, limiting their access to formal employment. If human capital differences drive the age gradient in sectoral destinations, effects should differ across education levels. However, controlling for demographic characteristics including educational attainment does not change the main result: there are differential effects of heat on formal non-agricultural employment shares by age, but no differential effects on informal labor share (Appendix Table D18). Moreover, estimating effects separately by education group reveals similar patterns within each group (Appendix Table D5): heat increases formal employment primarily for younger workers, while informal employment increases across all age groups regardless of education. This suggests that age itself, not educational attainment, determines access to formal non-agricultural employment.

Third, formal non-farm employers may prefer younger workers because age proxies

for health rather than reflecting age-based barriers per se. Indeed, roughly 75% of survey responses from HR managers indicate a preference for healthy candidates. However, several features of the experimental design weigh against this interpretation. The older applicant's profile explicitly states "I am in good health," signaling equivalent health status across candidates. The age penalty persists after controlling for whether respondents express a preference for healthy candidates (Tables 6). While employers may still associate age with health despite these signals, additional evidence suggests this is not the binding constraint.

If health or physical capability were the barrier, older workers should face similar constraints in informal non-agricultural employment. Yet informal work is, if anything, more physically demanding: over 90% of informal workers are in manual or personal service occupations, compared to roughly 50% in the formal sector (Appendix Table D19). Moreover, heat increases employment in physically demanding occupations in both sectors (Appendix Table D20), but the patterns differ significantly by age (Appendix Table D21). In the informal sector, workers of all ages enter physically demanding jobs at similar rates: effects are statistically indistinguishable across age groups ($p = 0.51$). In the formal sector, only younger workers enter; the effect declines sharply with age and is insignificant for those aged 55–64 ($p < 0.01$ for equality across ages). If declining physical capability were the constraint, older workers would avoid demanding work in both sectors. Instead, they enter physically demanding informal jobs but are excluded from equivalent formal positions. These are pointing directly to formal hiring practices as the barrier.

Taken together, the earnings patterns among switchers are consistent with positive selection into formal employment that intensifies with age, while the age restrictions in job postings and stated employer preferences provide direct evidence of demand-side barriers to formal entry for older workers. These findings support the framework's account of age-based sorting: temperature shocks push workers out of agriculture, but formal employment is disproportionately accessible to younger workers, channeling older workers into informal non-agriculture where hourly earnings gains are negligible.

7 IMPLICATIONS FOR CLIMATE ADAPTATION UNDER AN AGING WORKFORCE

The results presented so far show that the labor reallocation response to extreme heat varies sharply with age: younger workers drive most of the exit from agriculture, while older workers largely remain. This age dependence has important implications for climate adaptation as the agricultural workforce across the developing world has been aging (HelpAge International, 2014; Ren et al., 2023; Rigg et al., 2020). In this section, I ask how the

economy's aggregate reallocation response changes under projected demographic trends, holding the climate distribution fixed.

Vietnam's agricultural workforce has aged substantially over the past quarter century. Between the 1993 and 2018 survey rounds, the share of agricultural workers in their twenties and thirties fell from 59% to 30%, while the share approaching retirement age nearly doubled, from 14% to 27%. To quantify how this aging affects the economy's capacity to adapt to heat shocks, I project the future age composition of the agricultural workforce by applying 2018 age-specific agricultural participation rates from the VHLSS to population projections from the United Nations World Population Prospects 2024 (medium variant). This approach holds participation behavior fixed and isolates the mechanical effect of population aging.

Table 7: Projected Age Composition and Aggregate Reallocation Response of the Agricultural Workforce

	2018	2030	2040	2050
Panel A: Agricultural Workforce Composition				
Share young (24–39)	0.38	0.29	0.29	0.30
Share middle (40–54)	0.39	0.44	0.41	0.36
Share older (55–64)	0.23	0.26	0.30	0.34
Panel B: Aggregate Responses				
Weighted response per degree day above 27°C	-0.0064	-0.0058	-0.0057	-0.0056
	(0.0016)	(0.0015)	(0.0015)	(0.0015)
Decline relative to 2018 (%)	–	9.4	10.9	12.5

Notes: Panel A shows the projected share of agricultural workers in each age group under the assumption that age-specific agricultural participation rates remain at their 2018 VHLSS levels. All columns, including 2018, apply age-specific agricultural participation rates from the 2018 VHLSS to the UN-projected population age structure. The 2018 projected shares differ from the observed agricultural workforce composition because the projection isolates the role of population aging and abstracts from cohort-specific differences in agricultural attachment. Panel B reports the weighted average allocation response $\bar{\beta}_t = \sum_a w_a \hat{\beta}_a$ where $\hat{\beta}_a$ are the age-specific coefficients from Figure 4 (Column 1 of Appendix Table D4). Standard errors are computed from the variance-covariance matrix of the estimated coefficients. The final row reports the percentage decline in $|\bar{\beta}(t)|$.

Panel A of Table 7 reports the results. The share of younger workers declines from 38% to 30% between 2018 and 2050, while the share of older workers (the group with the weakest reallocation response) rises from 23% to 34%. Panel B translates these compositional shifts into the economy's aggregate reallocation response per degree day $\bar{\beta}_t = \sum_a w_a \hat{\beta}_a$. The aggregate response weakens from -0.64 percentage point in 2018 to -0.56 percentage point by 2050, a decline of roughly 13%. In other words, for the same heat shock, the economy's reallocation response would be roughly 13% weaker by mid-century, purely

because of the changing age composition of the agricultural workforce. With roughly 17 million working-age adults employed in agriculture as of 2018, even a modest decline in the per-degree-day response translates into a meaningful reduction in the number of workers who reallocate away from agriculture in response to heat stress. This projection isolates the mechanical effect of population aging from other ongoing changes in Vietnam's labor market.

8 CONCLUSION

This paper examines how temperature shocks affect sectoral labor reallocation in Vietnam. Using nearly three decades of household survey data linked to high-resolution weather data, I document three main findings. First, heat exposure reduces agricultural employment and increases non-agricultural employment. Second, these effects are driven by cumulative exposure over 12–24 months, consistent with gradual labor market adjustment rather than immediate responses. Third, while workers of all ages transition into informal non-agricultural employment at similar rates, only younger workers additionally gain access to formal non-agricultural employment.

I provide evidence on the mechanisms underlying these patterns. Heat reduces relative agricultural returns through multiple channels (lower labor productivity, fewer workable hours, and reduced crop yields) while non-agricultural returns remain largely stable, providing the push out of agriculture. The differential access to formal employment reflects age-based sorting: earnings gaps among sector-switchers are consistent with positive selection into formal employment that intensifies with age, while explicit age restrictions in job postings and stated employer preferences from a vignette experiment provide direct evidence of demand-side barriers in formal hiring.

These findings have several implications. First, climate adaptation in developing countries involves significant labor market reallocation beyond adjustments within agriculture, with the informal sector serving as a key margin of adjustment that absorbs workers of all ages. Second, age-based barriers to formal hiring constrain the options available to older workers: while younger workers access formal employment and realize substantial hourly earnings gains of 0.4 to 0.6 log points, older workers are channeled into informal employment where hourly returns are no higher than in agriculture. Third, as agricultural workforces age across the developing world, the aggregate reallocation response to climate shocks is likely to weaken, since the workers who remain in agriculture are precisely those least able to access formal employment.

Several limitations warrant acknowledgment. First, while the analysis focuses on Vietnam, the core mechanisms documented here (age-based barriers to formal employment,

informal absorption of displaced agricultural workers, and the role of cumulative heat exposure) are features shared by many developing economies with large agricultural workforces and segmented labor markets. The specific magnitudes will vary with local labor market institutions and agricultural structures, but the qualitative patterns are likely to extend beyond this setting. Second, the employer-type classification captures sectoral reallocation across formal and informal production units but does not capture variation in job protection within formal firms.[16] Third, the welfare consequences of reallocation depend on the quality of jobs workers obtain, which the hourly earnings gap captures only partially. Finally, the long-run effects of sustained temperature increases may differ from the short-run adjustments documented here, as both workers and firms adapt to changing climatic conditions. That said, exploratory long-difference estimates following Burke and Emerick (2016), reported in Appendix Figure D8, suggest broadly similar patterns: younger cohorts are significantly more likely to transition into formal non-agricultural employment in response to long-term heat exposure, while older workers show little to no reallocation.

As climate change intensifies and agricultural workforce continues to age, understanding the barriers that shape worker reallocation, and the policies that might ease them, becomes increasingly important for designing effective adaptation strategies.

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[16]Recent evidence from Brazil shows that the intensive margin of informality, i.e., formal firms hiring workers without contracts, plays an important role in how labor markets absorb climate-induced migration (Imbert & Ulyssea, 2026). Understanding the interplay between the extensive and intensive margins of formality represents an important dimension for future research.

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Supplementary Materials for Online Publication

Temperature Shocks, Worker Age, and Sectoral Reallocation into Informality: Evidence from Vietnam

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A THEORETICAL MODEL

This appendix presents the formal model underlying the conceptual framework in Section 2. The model describes a three-sector economy (agriculture, informal non-agriculture, and formal non-agriculture) in which heterogeneous workers choose sectors based on comparative advantage, entry barriers, and accumulated agricultural attachment. Temperature enters through sector-specific productivity and labor performance channels.

A.1 Environment

Sectors and timing. There are three sectors: agriculture (g), informal non-agriculture (i), and formal non-agriculture (f). Workers live for two periods and supply one unit of labor inelastically each period. Period-1 employment is predetermined. At the end of period 1, workers observe temperature realizations and choose their sector of employment for period 2. The model focuses on workers initially in agriculture who consider reallocation out of farming. This structure follows the Roy selection framework (Roy, 1951), in which workers sort across sectors based on comparative advantage, as applied to developing-country labor markets by Lagakos and Waugh (2013).

Skill heterogeneity. Workers differ in non-agricultural skill ε but have identical efficiency in agriculture. The skill vector is $\{1, \varepsilon\}$.

Assumption A1 (Pareto heterogeneity)**.** *ε is Pareto-distributed with support $\varepsilon \geq 1$:*

$$F(\varepsilon) = 1 - \varepsilon^{-\theta}, \qquad \theta > 0.$$

The Pareto assumption buys tractability: closed-form expressions for employment shares, conditional means, and comparative statics. The shape parameter θ controls skill dispersion, with lower θ implying more heterogeneity.

Informal skill intensity. Informal work is less skill-intensive than formal work. A worker with non-agricultural skill ε has effective skill ε in the formal sector but a lower effective skill in informality:

Assumption A2 (Skill loading in informality)**.**

$$\varepsilon_i(\varepsilon) = (1 - b) + b\varepsilon, \qquad \varepsilon_f(\varepsilon) = \varepsilon, \qquad b \in (0, 1].$$

The parameter b controls skill intensity in informality. The formulation is a convex combination: a fraction $(1 - b)$ of informal effective skill is common to all workers, while

a fraction b depends on individual ability ε. This compresses the informal earnings distribution relative to formal. The specification normalizes effective skill to 1 at the lower bound $\varepsilon = 1$, so the least-skilled worker has the same effective productivity in both non-agricultural sectors.

Technology. Temperature affects production through two channels. An *effective labor* channel operates in all sectors: heat reduces the physiological capacity for work, captured by sector-specific shifters $\phi_s(T)$ that scale labor effectiveness (e.g., LoPalo, 2023; Somanathan et al., 2021). An *agronomic* channel is specific to agriculture: heat directly damages crop productivity through a Hicks-neutral term $Z_g(T)$ (e.g., Schlenker & Roberts, 2009).

Assumption A3 (Production technology)**.** *For each sector $s \in \{g, i, f\}$,*

$$Q_s = Z_s(T)\, F_s\big(\phi_s(T) L_s\big), \qquad F_s' > 0, \quad F_s'' < 0,$$

where $Z_g'(T) < 0$ (agronomic damage in agriculture) and $Z_i(T) = Z_i$, $Z_f(T) = Z_f$ are constant or weakly affected on average.

The effective labor shifters reflect differential climate exposure across sectors:

$$\phi_s(T) = \lambda_s \phi_{\text{out}}(T) + (1 - \lambda_s)\phi_{\text{in}}(T), \qquad \phi_{\text{out}}'(T) \ll \phi_{\text{in}}'(T) \leq 0, \qquad \lambda_g > \lambda_i > \lambda_f,$$

where λ_s is the share of work performed outdoors. Agriculture is most exposed (λ_g highest), formal non-agriculture least (λ_f lowest, reflecting the employment-weighted average across formal sub-sectors), and informality intermediate. Since outdoor labor performance declines more steeply with heat ($\phi_{\text{out}}' \ll \phi_{\text{in}}'$), agriculture suffers the largest effective labor loss.

Prices. The non-agricultural good is the numeraire. The agricultural good has relative price p.

Assumption A4 (Price insensitivity)**.** *Local agricultural commodity prices are approximately unresponsive to local productivity shocks: $p \approx \bar{p}$.*

This condition can be satisfied under several institutional arrangements. First, international market integration: because Vietnam is integrated into global rice markets, a productivity shock in any single province represents a negligible share of world supply and is therefore unlikely to move the price a local farmer receives. Second, government price stabilization: throughout much of the sample period, Vietnam directly regulated rice prices

through state marketing agencies (Ghosh & Whalley, 2004) and floor price guarantees (Government of Vietnam, 2009). Third, sufficient domestic market integration, in which local supply shortfalls are offset by inflows from other regions. The assumption is testable with price data and important to generate labor reallocation out of agriculture. Without it, reduced agricultural output could raise p enough to increase wage w_g, potentially attenuating or reversing the push out of agriculture (e.g., Liu et al., 2023).

Wages. Competitive labor markets imply wages per unit of raw efficiency equal marginal products:

$$w_s(T) = p_s\, Z_s(T)\, \phi_s(T)\, F_s'\big(\phi_s(T) L_s\big), \qquad p_g = \bar{p}, \quad p_i = p_f = 1. \tag{A1}$$

Gross earnings of a worker with skill ε in each sector are

$$y_g = w_g, \qquad y_i = w_i\,[(1-b) + b\varepsilon], \qquad y_f = w_f\,\varepsilon.$$

Entry barriers, agricultural attachment, and age. Switching from agriculture into destination $j \in \{i, f\}$ requires paying a one-time entry cost κ_j. Additionally, workers accumulate agricultural attachment $\delta(a)$ that raises the effective payoff of staying in agriculture.

The segmentation of non-agricultural employment into formal and informal sectors, with barriers governing access to formal employment, is a central feature of developing-country labor markets (La Porta & Shleifer, 2014; Ulyssea, 2020). The informal cost κ_i reflects the generally low barriers to entering informal self-employment or casual labor, which do not vary strongly by age. The formal cost κ_f captures hiring restrictions, screening, and employer preferences documented in job postings and the vignette experiment.

Agricultural attachment $\delta(a)$ captures accumulated sector-specific capital (e.g., crop knowledge, trade relationships, and social networks) that raise the value of remaining in agriculture (Foster & Rosenzweig, 1995). Importantly, $\delta(a)$ is not a cost of entering non-agriculture, but rather a (potentially non-pecuniary) benefit of staying in agriculture. Because $\delta(a)$ enters the agriculture payoff but does not appear in observed non-agricultural wages, switcher-earnings regressions are informative about age variation in entry costs such as $\kappa_i(a)$, but they do not identify $\delta(a)$. Empirically, the similar informal earnings gains across ages are consistent with an approximately age-invariant κ_i, while allowing $\delta(a)$ to rise with age and reduce agricultural exit.

Within a given age cohort, workers differ in their access to formal employment. Some face low formal barriers (e.g., workers with proximity to formal employers, connections), while others face high barriers (e.g., workers in remote areas, without networks). The

share facing high barriers is larger among older cohorts, reflecting the extensive evidence that age-based hiring restrictions limit older workers' access to formal employment (Carlsson & Eriksson, 2019; Neumark et al., 2019).

Assumption A5 (Switching costs, barriers, and attachment)**.**

(i) κ_i *is low and age-invariant.*

(ii) Within each cohort of age a*, workers draw formal entry cost* $\kappa_f \in \{\kappa_f^L, \kappa_f^H\}$ *with* $\kappa_f^L < \kappa_f^H$*, independently of* ε*. The probability of facing high barriers is* $\pi(a) \equiv \Pr(\kappa_f = \kappa_f^H \mid a)$*, with* $\pi'(a) > 0$*.*

(iii) $\delta'(a) > 0$ *(agricultural attachment increases with age).*

This binary-type structure allows me to capture within-cohort variation in formal access while preserving the closed-form expressions that follow from Pareto heterogeneity. It generates all five predictions in Section 2.2, including the informal-entry response that requires both types to be present within a cohort.

Assumption A6 (Formal has higher return to skill)**.** $w_f > b\, w_i$.

This assumption states that the formal wage premium per unit of skill exceeds the informal premium, which is empirically reasonable given the skill compression in informal work.

A.2 Sector choice and threshold structure

Consider a worker of age a initially in agriculture with formal entry cost $\kappa_f \in \{\kappa_f^L, \kappa_f^H\}$. In period 2, she chooses the sector that maximizes her net payoff:

$$\max\Big\{ \underbrace{w_g + \delta(a)}_{\text{stay in agriculture}} \;,\; \underbrace{w_i[(1-b) + b\varepsilon] - \kappa_i}_{\text{enter informal}},\; \underbrace{w_f\varepsilon - \kappa_f}_{\text{enter formal}} \Big\}. \tag{A2}$$

Each payoff is linear in ε: agriculture has slope zero, informal has slope $w_i b$, and formal has slope w_f. Under Assumption A6, these slopes are strictly ordered. The resulting partition of the skill space depends on the worker's formal entry cost, which determines whether informal employment is ever optimal.

Pairwise indifference thresholds. Setting pairs of payoffs equal yields three thresholds:

$$\varepsilon_{gi}(a) = \frac{w_g + \delta(a) + \kappa_i - w_i(1-b)}{w_i b}, \tag{A3}$$

$$\varepsilon_{gf}(a, \kappa_f) = \frac{w_g + \delta(a) + \kappa_f}{w_f}, \tag{A4}$$

$$\varepsilon_{if}(\kappa_f) = \frac{\kappa_f - \kappa_i + w_i(1-b)}{w_f - w_i b}. \tag{A5}$$

The agriculture-informal threshold $\varepsilon_{gi}(a)$ does not depend on κ_f. The informal-formal threshold $\varepsilon_{if}(\kappa_f)$ does not depend on a or $\delta(a)$.

Sector choice characterization There exists a critical formal cost $\kappa_f^*(a)$ at which all three payoff lines intersect at a single skill level. This threshold is defined by $\varepsilon_{gi}(a) = \varepsilon_{gf}(a, \kappa_f^*) = \varepsilon_{if}(\kappa_f^*)$, yielding

$$\kappa_f^*(a) = w_f\, \varepsilon_{gi}(a) - w_g - \delta(a). \tag{A6}$$

Since ε_{if} rises faster in κ_f than ε_{gf} does (as $1/(w_f - w_i b) > 1/w_f$), and both coincide with ε_{gi} at κ_f^*, the allocation shifts cleanly from one case to the other as κ_f crosses $\kappa_f^*(a)$. For κ_f just below κ_f^*, informal employment vanishes; for κ_f just above, an informal region opens up between agriculture and formal.

Proposition A1 (Two cases of sectoral allocation)**.** *Under Assumption A6:*

(a) ***Low formal barriers*** *($\kappa_f < \kappa_f^*(a)$): Formal employment is accessible enough that workers exit agriculture directly into formal jobs, bypassing informality. The skill space is divided into two regions:*

$$\textit{Agriculture: } \varepsilon < \varepsilon_{gf}(a), \qquad \textit{Formal: } \varepsilon \geq \varepsilon_{gf}(a).$$

No worker chooses informal employment.

(b) ***High formal barriers*** *($\kappa_f > \kappa_f^*(a)$): Formal barriers are high enough that agricultural exiters first encounter informality as the attractive outside option, and only the most skilled proceed to formal. The skill space is divided into three regions:*

$$\textit{Agriculture: } \varepsilon < \varepsilon_{gi}(a), \quad \textit{Informal: } \varepsilon_{gi}(a) \leq \varepsilon < \varepsilon_{if}, \quad \textit{Formal: } \varepsilon \geq \varepsilon_{if}.$$

Proof. Each payoff is linear in ε with strictly ordered slopes $0 < w_i b < w_f$, so pairwise crossing is unique. The maximum of three linear functions divides the real line into at most three consecutive intervals.

Low formal barriers. When $\kappa_f < \kappa_f^*(a)$, we have $\varepsilon_{gf}(a) < \varepsilon_{gi}(a)$: formal crosses agriculture before informal does. At the point where a worker becomes indifferent between agriculture and informal ($\varepsilon = \varepsilon_{gi}$), formal already exceeds both (since $\varepsilon_{gi} > \varepsilon_{gf}$ and formal is steeper than informal). No worker ever finds informal optimal. The partition is agriculture below ε_{gf} and formal above.

To verify the ordering: $\varepsilon_{gf} < \varepsilon_{gi}$ is equivalent to

$$\frac{w_g + \delta + \kappa_f}{w_f} < \frac{w_g + \delta + \kappa_i - w_i(1-b)}{w_i b}.$$

At $\kappa_f = \kappa_f^*$, equality holds (by definition). Since the left side increases in κ_f at rate $1/w_f$ and the right side is independent of κ_f, for $\kappa_f < \kappa_f^*$ the left side is strictly smaller.

High formal barriers. When $\kappa_f > \kappa_f^*(a)$, we have $\varepsilon_{gi}(a) < \varepsilon_{gf}(a)$ and $\varepsilon_{gi}(a) < \varepsilon_{if}$: informal crosses agriculture first. For $\varepsilon \in [\varepsilon_{gi}, \varepsilon_{if})$, informal exceeds both agriculture (by $\varepsilon > \varepsilon_{gi}$) and formal (by $\varepsilon < \varepsilon_{if}$). For $\varepsilon \geq \varepsilon_{if}$, formal exceeds informal. The ordering $\varepsilon_{gf} \leq \varepsilon_{if}$ holds because both exceed ε_{gi} and ε_{if} rises faster in κ_f than ε_{gf} does, so for $\kappa_f > \kappa_f^*$, $\varepsilon_{if} > \varepsilon_{gf}$. The binding formal threshold is therefore ε_{if}.

Transition. At $\kappa_f = \kappa_f^*(a)$, all three thresholds coincide: $\varepsilon_{gi} = \varepsilon_{gf} = \varepsilon_{if}$. The informal region collapses to a single point and the partition transitions continuously between the two cases. ■

Parameter restriction. For tractability, I assume that the two types of workers remain in their respective cases at all ages: low-barrier workers always have direct access to formal jobs, and high-barrier workers always sort through informality first. Formally, this requires

$$\kappa_f^L < \kappa_f^*(a) < \kappa_f^H \qquad \text{for all } a. \tag{A7}$$

Since $\kappa_f^*(a)$ is increasing in a (through $\delta(a)$, as $d\kappa_f^*/d\delta = (w_f - w_i b)/(w_i b) > 0$), the binding conditions are: $\kappa_f^L < \kappa_f^*(a_{\min})$ and $\kappa_f^H > \kappa_f^*(a_{\max})$.

This amounts to requiring that the difference in formal access between the two types is large relative to the variation in agricultural attachment across ages. The assumption would fail if, for example, a worker who had easy access to formal jobs at age 30 accumulated so much agricultural attachment by age 55 that she could no longer be drawn into formal employment by any decrease in the agricultural wage. In practice, this is unlikely: whether a worker has access to formal hiring networks is a more fundamental determinant of her options than how many additional years she has spent farming. Relaxing this assumption would allow some workers to move from one case to the other as they age, but the qualitative predictions are unchanged.

A.3 Employment shares

Under Assumption A1, $\Pr(\varepsilon \geq x) = x^{-\theta}$ for $x \geq 1$.

Type-specific shares. For low-barrier workers (Proposition A1a):

$$N_g^L(a) = 1 - \varepsilon_{gf}^L(a)^{-\theta}, \qquad N_f^L(a) = \varepsilon_{gf}^L(a)^{-\theta}, \qquad N_i^L(a) = 0, \tag{A8}$$

where $\varepsilon_{gf}^L(a) \equiv \varepsilon_{gf}(a, \kappa_f^L) = (w_g + \delta(a) + \kappa_f^L)/w_f$.

For high-barrier workers (Proposition A1b):

$$N_g^H(a) = 1 - \varepsilon_{gi}(a)^{-\theta}, \qquad N_i^H(a) = \varepsilon_{gi}(a)^{-\theta} - (\varepsilon_{if}^H)^{-\theta}, \qquad N_f^H(a) = (\varepsilon_{if}^H)^{-\theta}, \tag{A9}$$

where $\varepsilon_{if}^H \equiv \varepsilon_{if}(\kappa_f^H) = (\kappa_f^H - \kappa_i + w_i(1-b))/(w_f - w_i b)$ does not depend on age.

Aggregate shares. Aggregating over types within a cohort of age a:

$$N_g(a) = (1 - \pi(a))\, N_g^L(a) + \pi(a)\, N_g^H(a), \tag{A10}$$

$$N_i(a) = \pi(a)\, N_i^H(a) = \pi(a) \left[\varepsilon_{gi}(a)^{-\theta} - (\varepsilon_{if}^H)^{-\theta}\right], \tag{A11}$$

$$N_f(a) = (1 - \pi(a))\, \varepsilon_{gf}^L(a)^{-\theta} + \pi(a)\, (\varepsilon_{if}^H)^{-\theta}. \tag{A12}$$

By construction, $N_g(a) + N_i(a) + N_f(a) = 1$. Informal employment is generated exclusively by high-barrier workers, whereas low-barrier workers contribute only to agriculture and formal employment.

Under the regime-stability condition (A7), low-barrier workers satisfy $\kappa_f^L < \kappa_f^*(a)$ (Proposition A1, case (a)). Since $\varepsilon_{gf}(a, \kappa_f)$ is strictly increasing in κ_f and $\varepsilon_{gf}(a, \kappa_f^*(a)) = \varepsilon_{gi}(a)$ by construction, it follows that

$$\varepsilon_{gf}^L(a) = \varepsilon_{gf}(a, \kappa_f^L) < \varepsilon_{gi}(a).$$

Thus, holding age a fixed, low-barrier workers exit agriculture at a lower skill threshold than high-barrier workers. Intuitively, low-barrier workers have a more attractive outside option (direct access to formal jobs), making them less "stuck" in agriculture for any given level of agricultural attachment. (This is a threshold-ordering statement; the relative magnitudes of per-worker exit responses additionally depend on payoff slopes, as discussed in Corollary A1.)

A.4 Sufficient conditions for the relative-return shock

The comparative statics require that heat reduces agricultural returns relative to non-agriculture. The following assumption states this directly; below I provide sufficient conditions in primitives.

Assumption A7 (Relative-return shock (push))**.** *In the relevant range of T,*

$$\frac{d}{dT}\log w_g(T) \;<\; \min\left\{\frac{d}{dT}\log w_i(T),\; \frac{d}{dT}\log w_f(T)\right\}.$$

This assumption is directly testable. In Section 6, I examine the effect of temperature on agricultural prices, revenue per worker, and crop yields, and confirm that heat reduces agricultural returns without a compensating price increase.

A necessary condition is that agricultural commodity prices do not absorb the productivity shock. The price insensitivity assumption (Assumption A4) ensures that the productivity decline passes through to wages rather than being offset by higher output prices. If local prices were responsive to local supply conditions, reduced agricultural output would raise p, partially or fully offsetting the decline in w_g and potentially lowering or reversing the reallocation.

This is the "push" out of agriculture—a mechanism central to the structural transformation literature (Kongsamut et al., 2001; Matsuyama, 1992). Nath (2025) shows that in the absence of trade barriers, temperature-driven agricultural productivity losses generate precisely this pattern of sectoral reallocation at the global level.

Sufficient conditions. From equation (A1), the agricultural wage is

$$w_g(T) = \bar{p}\, Z_g(T)\, \phi_g(T)\, F_g'(\phi_g(T)L_g)$$

Consider first the direct effect of temperature on w_g, holding labor allocation L_g fixed. Define $x_g(T) = \phi_g(T)L_g$ and $m_g(x) \equiv xF_g'(x)$, so that $w_g = (\bar{p}/L_g)Z_g(T)m_g(x_g(T))$. If (i) $Z_g'(T) < 0$ (agronomic damage), (ii) $\phi_g'(T) < 0$ (reduced labor performance), and (iii) $m_g'(x) > 0$ (a mild regularity condition satisfied by standard decreasing-returns technologies—for example, $F_g(x) = x^\alpha$ with $\alpha \in (0,1)$ gives $m_g(x) = \alpha x^\alpha$ and $m_g'(x) = \alpha^2 x^{\alpha-1} > 0$), then $\partial w_g/\partial T < 0$ holding L_g fixed.

Equilibrium wage decline. The "holding L_g fixed" qualifier identifies the direct effect of temperature on labor demand. In equilibrium, the resulting exit from agriculture reduces L_g, which raises F_g' (diminishing returns), partially offsetting the wage decline.

However, this feedback operates through labor supply: a lower w_g induces exit (shifting labor supply left), while the diminishing-returns response is a movement along the demand curve. In the standard demand-supply diagram, the temperature shock shifts demand down, the upward-sloping supply response means equilibrium w_g falls by less than the direct effect, but the sign is unambiguous. The equilibrium wage decline is therefore weaker than the partial-equilibrium calculation but qualitatively the same.

If, additionally, $\phi_i(T)$ and $\phi_f(T)$ are less temperature-sensitive (or Z_i, Z_f are constant), then the relative-return inequality in Assumption A7 holds locally.

A.5 Exit from agriculture

Proposition A2 (Exit from agriculture)**.** *Under Assumption A7, an increase in T reduces agricultural employment: $dN_g(a)/dT < 0$.*

Proof. For low-barrier workers, $N_g^L(a) = 1 - \varepsilon_{gf}^L(a)^{-\theta}$ and

$$\frac{dN_g^L}{dT} = \theta\, \varepsilon_{gf}^L(a)^{-\theta-1} \cdot \frac{1}{w_f} \cdot \frac{dw_g}{dT} < 0,$$

since $\partial\varepsilon_{gf}^L/\partial w_g = 1/w_f > 0$ and $dw_g/dT < 0$ under Assumption A7.

For high-barrier workers, $N_g^H(a) = 1 - \varepsilon_{gi}(a)^{-\theta}$ and

$$\frac{dN_g^H}{dT} = \theta\, \varepsilon_{gi}(a)^{-\theta-1} \cdot \frac{1}{w_i b} \cdot \frac{dw_g}{dT} < 0.$$

The aggregate response is a convex combination of two negative terms:

$$\frac{dN_g(a)}{dT} = (1 - \pi(a))\frac{dN_g^L}{dT} + \pi(a)\frac{dN_g^H}{dT} < 0. \qquad \blacksquare$$

A.6 Older workers exit agriculture less

Corollary A1 (Older workers exit agriculture less)**.** *Under Assumptions A5 and A7, older cohorts are less responsive to heat shocks in agriculture. In particular, the magnitude of agricultural exit is decreasing in age if the per-worker exit responsiveness is (weakly) larger for low-barrier types than for high-barrier types:*

$$\frac{\varepsilon_{gf}^L(a)^{-\theta-1}}{w_f} \geq \frac{\varepsilon_{gi}(a)^{-\theta-1}}{w_i b} \qquad \textit{for all } a. \tag{A13}$$

Proof. Define the type-specific exit magnitudes

$$E_L(a) \equiv \theta\, \varepsilon^L_{gf}(a)^{-\theta-1} \cdot \frac{|dw_g/dT|}{w_f}, \qquad E_H(a) \equiv \theta\, \varepsilon_{gi}(a)^{-\theta-1} \cdot \frac{|dw_g/dT|}{w_i b}.$$

Then the cohort-level magnitude is a convex combination:

$$\left|\frac{dN_g(a)}{dT}\right| = (1-\pi(a))\, E_L(a) + \pi(a)\, E_H(a).$$

Attachment within types. Because $\delta'(a) > 0$, both exit thresholds $\varepsilon^L_{gf}(a)$ and $\varepsilon_{gi}(a)$ rise with age, and hence both $E_L(a)$ and $E_H(a)$ are decreasing in a: within each barrier type, fewer older workers are at the margin of exit.

Composition across types. The effect of shifting weight toward high-barrier types is

$$\frac{\partial}{\partial \pi}\left|\frac{dN_g(a)}{dT}\right| = E_H(a) - E_L(a),$$

which is non-positive under (A13). Under this sufficient condition, the increase of $\pi(a)$ with age further reduces the aggregate exit response, so $|dN_g(a)/dT|$ is decreasing in a.

When (A13) fails, this compositional channel can partially offset the within-type attenuation, but the within-type forces remain unambiguously negative. ■

A.7 Formal employment gains are concentrated among younger workers

Corollary A2 (Formal employment gains are concentrated among younger workers)**.** *The temperature-induced increase in formal employment is decreasing in age:*

$$\frac{dN_f(a)}{dT} \text{ is positive and decreasing in } a.$$

Proof. For low-barrier workers, all exit from agriculture flows to formal:

$$\frac{dN^L_f}{dT} = -\frac{dN^L_g}{dT} = \theta\, \varepsilon^L_{gf}(a)^{-\theta-1} \cdot \frac{|dw_g/dT|}{w_f} > 0.$$

For high-barrier workers, the formal threshold ε^H_{if} is independent of w_g (equation A5), so formal employment does not respond to agricultural shocks:

$$\frac{dN^H_f}{dT} = 0.$$

The main economic reason is that high-barrier workers who exit agriculture first encounter informality as the relevant outside option, and their decision between informal and formal is governed by ε_{if}^{H}, which compares informal and formal wages, which is unaffected by changes in w_g.

Aggregating:

$$\frac{dN_f(a)}{dT} = (1 - \pi(a))\, \theta\, \varepsilon_{gf}^{L}(a)^{-\theta-1} \cdot \frac{|dw_g/dT|}{w_f}. \tag{A14}$$

This is decreasing in a through two channels: $(1 - \pi(a))$ falls (fewer workers have direct formal access) and $\varepsilon_{gf}^{L}(a)$ rises through $\delta(a)$ (weaker response among those who do). ■

Intuition. For young workers with low formal barriers, the relevant comparison is agriculture versus formal: a decline in agricultural returns directly expands formal employment. For older workers facing high barriers, the relevant comparison is informal versus formal: even though more workers leave agriculture, the bar for entering formal (relative to informal) is unchanged by the agricultural shock, so the additional exiters flow into informality.

A.8 Informal entry is similar across age groups: conditions

This section formalizes the argument that the approximately equal informal response across age groups arises from offsetting forces and identifies the conditions under which this holds.

Proposition A3 (Informal employment response to heat across age groups)**.** *The temperature-induced increase in informal employment is*

$$\frac{dN_i(a)}{dT} = \pi(a)\, \theta\, \varepsilon_{gi}(a)^{-\theta-1} \cdot \frac{|dw_g/dT|}{w_i b} > 0. \tag{A15}$$

Whether this response increases or decreases with age is ambiguous: $\pi(a)$ rises with age (increasing the response), while $\varepsilon_{gi}(a)$ rises through $\delta(a)$ (decreasing the response). When these forces approximately offset, the informal response is similar across age groups.

Proof. Only high-barrier workers generate informal entry. For these workers, ε_{if}^{H} is independent of w_g, so the entire agriculture exit flows to informality:

$$\frac{dN_i^H}{dT} = -\frac{dN_g^H}{dT} = \theta\, \varepsilon_{gi}(a)^{-\theta-1} \cdot \frac{|dw_g/dT|}{w_i b}.$$

Low-barrier workers contribute nothing to informal: $dN_i^L/dT = 0$. Aggregating:

$$\frac{dN_i(a)}{dT} = \pi(a) \cdot \frac{dN_i^H}{dT} = \pi(a)\,\theta\,\varepsilon_{gi}(a)^{-\theta-1} \cdot \frac{|dw_g/dT|}{w_i b}.$$

Whether this increases or decreases with age is ambiguous because $\pi'(a) > 0$ (more high-barrier workers with age) while $\varepsilon_{gi}(a)$ is increasing in $\delta(a)$ (fewer at the exit margin among high-barrier workers). Specifically:

$$\frac{d}{da}\left[\frac{dN_i(a)}{dT}\right] = \theta\frac{|dw_g/dT|}{w_i b}\left[\pi'(a)\,\varepsilon_{gi}^{-\theta-1} - \pi(a)(\theta+1)\,\varepsilon_{gi}^{-\theta-2} \cdot \frac{\delta'(a)}{w_i b}\right],$$

which is zero when

$$\frac{\pi'(a)}{\pi(a)} = \frac{(\theta+1)\,\delta'(a)}{w_i b\,\varepsilon_{gi}(a)}. \tag{A16}$$

The left side is the growth rate of the high-barrier share. The right side captures the attenuation from agricultural attachment. The informal response is approximately similar across age groups when these are approximately equal. ■

Interpreting the similar informal response across age groups. The model provides a framework for interpreting the similar informal response across age groups. Because informal employment response to temperature is jointly determined by forces originating in agriculture (attachment $\delta(a)$) and in the formal sector (barriers $\pi(a)$), the age pattern of informal entry reflects the combined structure of both sectors.

To see this, consider each force in isolation. Under agricultural attachment alone ($\delta(a) > 0$, no within-cohort barrier variation), fewer older workers exit agriculture, and their exit is distributed across non-agricultural sectors in fixed proportions, both informal and formal responses would decline with age. Under formal barriers alone ($\pi'(a) > 0$, $\delta(a) = 0$), agriculture exit would be equal across ages, but more of the older exiters would be high-barrier types whose exit flows to informal, the informal response would be larger for older workers. That the observed informal response is instead similar across age groups is consistent with both forces operating simultaneously: agricultural attachment reduces total exit for older workers, while the rising share of high-barrier workers channels a larger fraction of that exit toward informality, and these two effects offset.

A.9 Selection and switcher gains

Proposition A4 (Selection among formal switchers)**.** *Observed earnings among formal switchers are increasing in age: older formal entrants exhibit larger average earnings gains than younger formal entrants.*

Proof. Formal workers are drawn from two pools with different entry thresholds. Under the Pareto distribution, the conditional expectation of $\log \varepsilon$ above a threshold x is:

$$\mathbb{E}[\log \varepsilon \mid \varepsilon \geq x] = \log x + \frac{1}{\theta}.$$

(This follows from $\log \varepsilon - \log x \mid \varepsilon \geq x$ is exponential with rate θ.)

Average log earnings among formal switchers of each type are therefore:

$$\mathbb{E}[\log y_f \mid f, L] = \log w_f + \log \varepsilon^L_{gf}(a) + \frac{1}{\theta},$$
$$\mathbb{E}[\log y_f \mid f, H] = \log w_f + \log \varepsilon^H_{if} + \frac{1}{\theta}.$$

Aggregating over types, with weights proportional to formal employment mass:

$$\mathbb{E}[\log y_f \mid f] = \log w_f + \frac{1}{\theta} + \lambda(a) \log \varepsilon^L_{gf}(a) + (1 - \lambda(a)) \log \varepsilon^H_{if}, \tag{A17}$$

where $\lambda(a) \equiv N^L_f(a)/[N^L_f(a)+N^H_f(a)] = (1-\pi(a))\varepsilon^L_{gf}(a)^{-\theta} \big/ \left[(1-\pi(a))\varepsilon^L_{gf}(a)^{-\theta}+\pi(a)(\varepsilon^H_{if})^{-\theta}\right]$.

As age increases, $\varepsilon^L_{gf}(a)$ rises through $\delta(a)$, directly increasing $\log \varepsilon^L_{gf}(a)$: the few low-barrier older workers who enter formal are drawn from further in the right tail. Simultaneously, $\lambda(a)$ falls (both because $\pi(a)$ rises and because $\varepsilon^L_{gf}(a)^{-\theta}$ falls), shifting weight toward the high-barrier pool with threshold ε^H_{if}.

When $\varepsilon^H_{if} > \varepsilon^L_{gf}(a)$, which holds when κ^H_f is sufficiently larger than κ^L_f (as assumed), both effects increase average formal earnings. The rising within-type selection (higher ε^L_{gf}) and the compositional shift toward a more selected pool (higher ε^H_{if}) reinforce each other.

The earnings *gain* relative to agriculture is

$$\mathbb{E}[\log y_f - \log y_g \mid f] = \log w_f - \log w_g + \frac{1}{\theta} + \lambda(a) \log \varepsilon^L_{gf}(a) + (1 - \lambda(a)) \log \varepsilon^H_{if},$$

which is also increasing in a through the same channels.

This logic parallels the classic analysis of immigrant self-selection by Borjas (1987), where higher migration barriers produce more positively selected entrants with higher observed outcomes. ■

Informal switchers: attenuated selection. Informal switchers are high-barrier workers with skill $\varepsilon \in [\varepsilon_{gi}(a), \varepsilon^H_{if})$. This interval is bounded above by the formal entry threshold: workers with skill above ε^H_{if} enter formal instead. As $\varepsilon_{gi}(a)$ rises with age, the interval shifts upward, but the most skilled workers in any cohort are always pulled into formal

employment rather than informal. This limits how much average informal skill can increase with age.

In contrast, formal switchers are drawn from $[\varepsilon_{if}^{H}, \infty)$, with no upper bound, so average formal skill can rise more freely as the entry threshold increases. The lower skill intensity of informal work ($b < 1$) reinforces this: informal earnings depend on $(1 - b) + b\varepsilon$ rather than ε, so even differences in skill translate into smaller differences in earnings. Together, these forces imply that informal earnings gains vary less with age than formal earnings gains, which is consistent with the empirical finding.

B DATA AND VARIABLE CONSTRUCTION

B.1 Employment and Labor Productivity Data

The household- and individual-level data are retrieved from the random 5% population and housing census in 1989 (Minnesota Population Center, 2015), and the household livings standard survey conducted by the General Statistics Office of Vietnam (GSO) in 1993, 1998, and every two years since 2002 (GSO, n.d.-a). The household survey is representative at the national and provincial level.

Table B1: List of Vietnamese Data Sets

Data Set	Reference Period	Sample size	Source	Data Access
Population and Housing Census 1989	12 months	5% census	IPUMS	Public
Living Standard Survey 1992/1993	12 months	4,800 households	GSO	Restricted
Living Standard Survey 1997/1998	12 months	6,000 households	GSO	Restricted
Household Living Standards Survey 2002	12 months	30,000 households	GSO	Restricted
Household Living Standards Survey 2004	12 months	45,000 households	GSO	Restricted
Household Living Standards Survey 2006	12 months	45,000 households	GSO	Restricted
Household Living Standards Survey 2008	12 months	45,000 households	GSO	Restricted
Household Living Standards Survey 2010	12 months	45,000 households	GSO	Restricted
Household Living Standards Survey 2012	12 months	45,000 households	GSO	Restricted
Household Living Standards Survey 2014	12 months	70,000 households	GSO	Restricted
Household Living Standards Survey 2016	12 months	45,000 households	GSO	Restricted
Household Living Standards Survey 2018	12 months	70,000 households	GSO	Restricted
Annual Enterprise Census 2002 to 2016	Fiscal year	All formal firms	GSO	Restricted

The key variable of interest is employment in agriculture, informal non-agriculture, and formal non-agriculture. The variable is constructed using data from the employment module of the survey, which covers hours worked, industries, as well as types of employer of the two most time-consuming jobs. I restrict the sample to 24–64 year old workers with information on industry of employment and types of employer to capture working-age individuals with completed education.

Province-level Employment Panel. For the main temperature-sectoral employment analysis, I compute properly weighted share of individuals working their principal job in agriculture, informal and formal non-agriculture, for each of the three age groups (24-39 years old, 40-54 years old, and 55-64 years old) in each of the 52 provinces for each of the 11 survey waves over the study period. The final sample includes 1,707 province-age-year observations.[1] For the temperature-sectoral hours worked analysis, in 2002, only informa-

[1] In 1993 and 1998, only 50 and 51 provinces, respectively, were surveyed.

tion of the principal job is collected. To ensure the measure of hours worked in a sector is consistent over time, I drop data from the survey wave 2002, which ends up having 1,551 province-age-year observations.

Individual-level Panel. Although the household survey is repeated cross-sectional, it contains a (random) rotating panel sub-component that tracks households and individuals over a period of up to four years, which allows me to analyze individual transition from agriculture to informal and formal non-agricultural sectors over a longer time than is usually feasible. I link individuals over time using a unique individual identification code based on household identification, and other individual information including gender, birth year, and sometimes confidential information (e.g., full name) provided by GSO in order to ensure the matching is correct.[2]

Province-level Production Panel. I assemble a province-level production dataset separately for agriculture, informal non-agriculture, and formal non-agriculture.
Agriculture and Informal Non-Agriculture: I combine multiple waves of the household survey from 2002 to 2016 to construct household-level agricultural and informal non-agricultural production datasets.

The key variable of interest is annual revenue per worker. Agricultural revenues comprise of revenues from crops, livestock, aquaculture, forestry, and farm services. Informal non-agricultural revenues include revenues from non-farm business. The number of workers are measured as the number of household members engaging in agriculture and non-agriculture as their primary job. These information are reported by the households for the 12-month reference period before the interview. Correspondingly, I restrict the sample to households that do not hire labors, because the household survey does not record information on the number of hired workers.

Household-level data are then merged with weather data in the same province using the timing of interview, similar to employment data. I then aggregate household-level to provincial-level dataset by taking a weighted average of all household producers in that province, with weight being the production size (i.e., number of workers).

Formal Non-Agriculture: The firm-level data are retrieved from the annual census conducted by the General Statistics Office of Vietnam since 2001 (VEC) (GSO, n.d.-b). While the household survey has the advantage of covering both formal and informal workers but the shortcoming that it is at best representative at the province level, VEC has the

[2]The matching codes for the survey waves 2002 to 2006, and 2010 to 2012 are graciously shared by McCaig and Pavcnik (2015) and McCaig and Pavcnik (2021), respectively.

advantage of a census and being available at yearly level, but small and informal firms are not covered.

The enterprise census collects rich information on ownership type, industry type, employment, labor compensations, as well as business performance and financial information of registered firms in the preceding fiscal year. I construct a dataset from 2002-2016 for firms whose main economic activity is non-agriculture using a unique firm identification code which comprises of tax code (available all periods), firm code (available before 2012), and branch code (available before 2014). New firms that do not have tax code yet are identified by a unique firm code assigned by the survey team.

The key variable of interest is annual revenue per worker, where revenue is calculated as the net turnover of goods and services, and the number of workers are measured at the end of fiscal year. In constructing the production data, I impose the following conditions: (i) The firm should not operate more than one branch, (ii) The firm should be in operation and report positive revenues, (iii) The firm should report positive number of workers at year end. Restriction (i) drops 0.7% of the original sample. Restrictions (ii) and (iii) mainly reflect data errors, and drop 9.6% and 0.008%, respectively, of the original sample. Firm data are then merged with weather data in the same province. I then aggregate firm-level to provincial-level dataset by taking a weighted average of all firms in that province, with weight being the firm size.

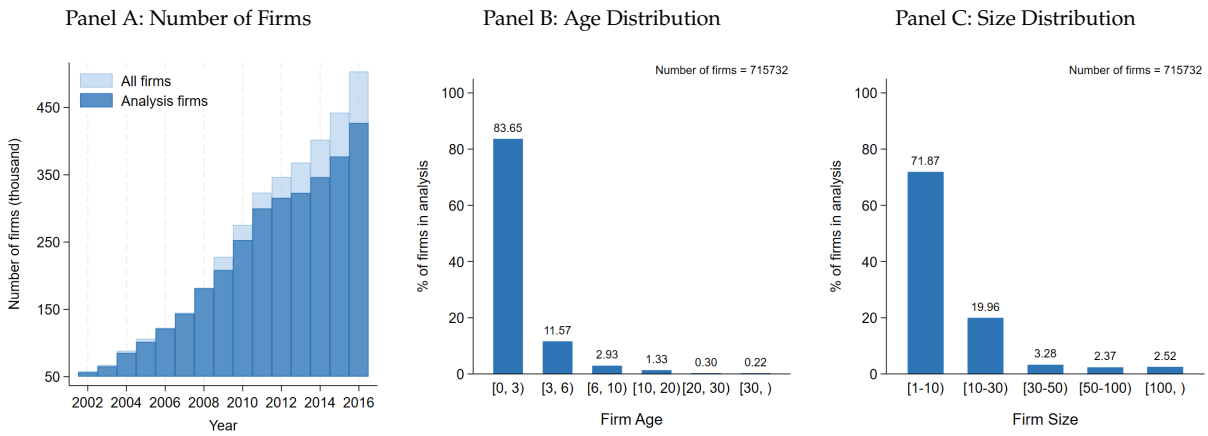
Panel A of Figure B1 shows the number of firms in the analysis sample, which reflects the increasing number of registered firms in Vietnam over the same period. As in many other low and middle income countries, a majority of these firms are young and small: more than 80% are less than three years old (Panel B), and nearly 70% have fewer than 10 workers (Panel C). Firms operating more than 10 years in the market account for less than 3% of the total sample.

B.2 Informality Measurement

Informality can broadly be defined either from the worker side or from the employer side. According to GSO and ILO (2018), informality on the worker side implies that workers do not have social security benefits and labor contract with a minimum term of three months ("informal workers"). On the employer side, informality implies that firms do not have legal status or register with the government ("informal firms"). In this paper, the notion of informality largely follows that from the employer side by assuming individuals who either are self-employed or work for household businesses or collectives are employed informally ("informal employment").

The household surveys include a question on whether a worker has benefited from

Figure B1: Non-agricultural firm-level data



social insurance since the 2010 wave, and a question on whether she has a labor contract since the 2014 wave. Although these two questions do not perfectly capture the definition of informal workers defined by GSO and ILO (2018), they allow a cross-check between the definition of informality employed in this paper and of informal workers in 2014, 2016, and 2018.

Table B2: Informality

	(1) Pearson Correlation between Informal Workers and Informal Employment	(2) Share of Formal Workers in Informal Employment	(3) Share of Informal Workers in Formal Employment
Panel A: By Year			
2014	0.8842	0.0021	0.1451
2016	0.8864	0.0017	0.1429
2018	0.8917	0.0021	0.1288
Total	0.8875	0.0020	0.1389
	(1)	(2)	(3)
24-39	0.8944	0.0026	0.1105
40-54	0.8772	0.0020	0.1630
55-64	0.8079	0.0009	0.2731

Notes: Informal employment is defined as self-employment, employment in household businesses, and collectives. Informal workers are defined as those who do not have social security benefits nor labor contract (GSO & ILO, 2018). Note that the GSO definition further requires a minimum contract term of three months. However, because the survey does not record contract duration, this criterion cannot be applied, making the classification slightly more conservative. *Source:* Data from VHLSS 2014, 2016, 2018.

Table B2 shows that the two definitions are largely similar. The Pearson correlation coefficient between informal workers and informal employment variables is nearly 0.9.

Only a small fraction (less than 0.2%) of workers with labor contract and social insurance are classified as working for informal employers (informal employment). The notion of informal employment, however, does not capture very well the intensive margin of informality: up to 14.5% of workers in formal firms does not have social security benefits and labor contract. Panel B shows that this share varies markedly by age: 11% for workers aged 24–39, compared to 27% for those aged 55–64. This suggests that even among workers who enter formal employment, older workers are substantially less likely to hold protected positions.

Table B3 provides further details by industry and highlights the differences in education, as proxied by years of schooling, between workers in the informal and formal sectors across industries. Generally, workers in the informal sector have lower educational attainment compared to their peers in the formal non-agricultural sector.

Table B3: Education level of workers by industry and informality, non-agricultural sectors

	Years of Schooling		Share of Informal Workers in Formal Firms		
	(1)	(2)	(3)	(4)	(5)
	Informal	Formal	2014	2016	2018
Manufacturing: high tech	8.945	11.086	0.037	0.015	0.031
Manufacturing: medium tech	8.192	10.686	0.159	0.117	0.125
Manufacturing: low tech	7.868	9.493	0.135	0.119	0.103
Service: knowledge intensive	9.440	13.296	0.060	0.061	0.050
Service: less knowledge-intensive	7.982	11.290	0.264	0.281	0.226
Mining and quarrying	7.405	10.831	0.131	0.140	0.147
Public utilities	8.737	11.884	0.053	0.077	0.073
Construction	7.805	10.323	0.474	0.466	0.473
Total	8.297	11.111	0.164	0.160	0.154

Notes: Informal workers are defined as those who do not have social security benefits nor labor contract. *Source:* Data from VHLSS 2014, 2016, 2018.

B.3 Weather Data and Construction of Variables

This section summarizes how weather variables are constructed from the ERA5 reanalysis data (Hersbach et al., 2020).

Wet-bulb Temperature. Wet-bulb temperature (WBT) is a nonlinear function of dry-bulb temperature (i.e., ambient air temperature) and relative humidity. It reflects the lowest temperature to which air can be cooled by the evaporation of water into the air at

a constant pressure. The measure of WBT has been increasingly used in the economics literature to study the combined effects of heat and humidity on worker productivity (Adhvaryu et al., 2020; LoPalo, 2023; Somanathan et al., 2021).

To calculate daily average WBT, I proceed in three steps. First, I calculate hourly relative humidity RH, which is defined as the ratio of vapor pressure e and saturation vapor pressure e_s, using hourly air temperature T_a (°C) and hourly dew-point temperature T_d (°C), following Bolton (1980):[3]

$$RH = 100 \times \frac{e}{e_s} = 100 \times \exp\left[\frac{17.67 \times 243.5 \times (\mathrm{T}_d - \mathrm{T}_a)}{(243.5 + \mathrm{T}_a)(243.5 + \mathrm{T}_d)}\right] \tag{B1}$$

Second, I calculate hourly WBT using hourly dry-bulb temperature T_a (°C) and hourly relative humidity RH (%), following Stull (2011):

$$\begin{aligned} \mathrm{WBT} = \quad & \mathrm{T}_a \times atan\left[0.151977 \times (RH + 8.313659)^{0.5}\right] + atan(\mathrm{T}_a + RH) - 4.686035 \\ & - atan(RH - 1.676331) + 0.00391838(RH)^{1.5} \times atan(0.023101 \times RH) \end{aligned} \tag{B2}$$

Finally, I take the mean of hourly T_w to get daily WBT.

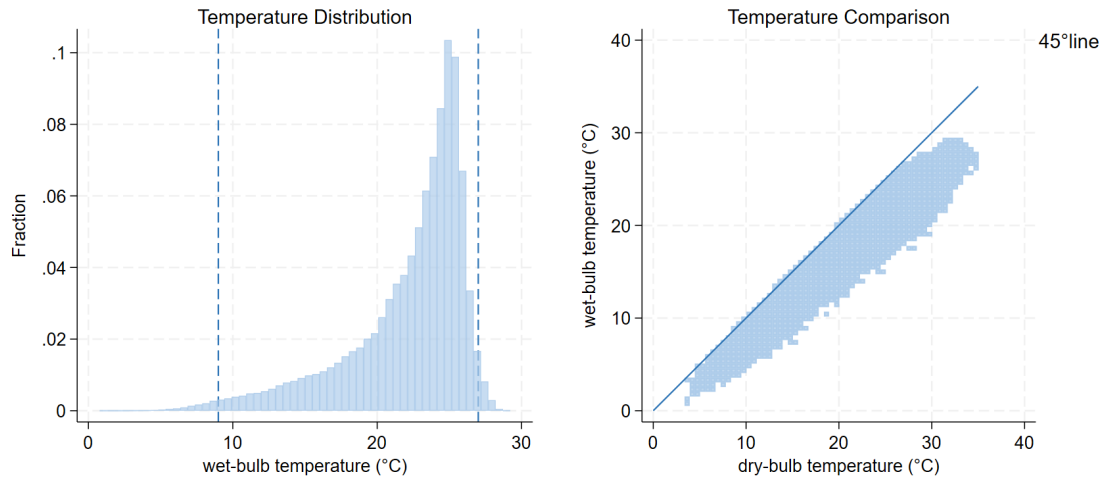
Figure B2 presents the distribution of daily wet-bulb temperatures over the study period (left panel) and a comparison between daily dry-bulb and wet-bulb temperatures (right panel). The right panel reveals substantial variation in wet-bulb temperature at any given dry-bulb temperature: for instance, when dry-bulb temperature is between 27–30°C, wet-bulb temperature can range from approximately 17°C to 28°C. This variation reflects differences in humidity across observations and has important implications for human labor capacity.

Precipitation. Daily precipitation is calculated as the sum of hourly precipitation. I then compute the second order polynomial of daily precipitation at each grid-level. This is done before the data are spatially averaged in order to accurately represent the distributions at grid level.

Aggregation of grid-level weather data to province-level weather data I transform grid-level weather data to province-level weather data using two methods. The first method is to take weighted average of four nearest grid points to province centroids, where the weight is inverse distance. The second method is to average all the points within the geographic boundary of the first administrative level–a province, except for wind speed where I use a maximum value. In both cases, nonlinear transformations of temperature and rainfall are computed at the grid level before averaging values across

[3]This equation is available as archive on University Corporation for Atmospheric Research's website.

Figure B2: Distribution of Wet-bulb Temperatures and Comparison with Dry-bulb Temperature



space, and finally summing over days during the reference period. This procedure is similar to Carleton et al. (2022).

To see how this calculation is conducted, consider the fourth-order polynomial specification for temperature. I begin with data on average temperatures for each day d at each grid point z, generating observations WBT_{zd}. These grid-level values are aggregated to the province level p for each 12-month reference period. To do this, I first raise grid-level temperature to the power n, computing $(\text{WBT}_{zd})^n$ for $n \in \{1, 2, 3, 4\}$. I then take a spatial average of these values following the two methods mentioned above. I then sum these daily polynomial terms $(\text{WBT}_{zd})^n$ over days during individual-specific reference periods, i.e., 12 full months before the survey interview. This nonlinear transformation performed prior to aggregation allows the aggregated measure of temperature to capture grid-by-day level exposure to very hot and very cold temperatures. Quadratic polynomials in precipitation are similarly calculated.

Because there has been changes in the administrative boundaries in Vietnam over the last three decades and most of the changes happen in case of splitting, I use the original administrative units in 1993, which gives a consistent sample of 52 provinces over the study period. An exception is that Ha Tay province was merged into Hanoi city in 2008 and thus I use the boundary of the new Ha Noi for consistency. This process results in the province-level vector of weather data for each 12-month interval.

C APPENDIX ROBUSTNESS CHECKS

C.1 Time-varying Demographic Controls

A potential concern is that temperature shocks may be correlated with demographic shifts that independently affect sectoral employment. For example, if hot years coincide with changes in educational attainment or labor force composition, the estimated temperature effects could reflect these demographic trends rather than direct responses to heat exposure. To address this, I augment the baseline specification with time-varying demographic controls including average age, the share of male workers, the share of ethnic minority workers, and share of workers with a high school diploma. Given that there are not much within-province change in these variables over time (Table 1), it is no surprise that the estimates remain relatively stable across these specifications (Appendix Table D7, Column 2).

C.2 Sample Restrictions

Province-year cells constructed from few underlying individual observations may introduce measurement error, and cells based on household surveys not designed for province-level representativeness may not accurately capture local labor market conditions. Both concerns are concentrated in the early survey rounds (1993 and 1998), which employed smaller sample sizes and were not designed to be representative at the province level. I address these concerns by re-estimating the model excluding province-year cells with fewer observations than the 5th percentile of the sample distribution (approximately 100 observations), which correspond to 16 cells from 1993 and 10 from 1998. The results are similar, suggesting that the findings are not driven by measurement error or representativeness issues in these early waves (Appendix Table D7, Column 3).

C.3 Leave-one-out Sensitivity

To ensure that the results are not driven by any single province, I re-estimate the model 52 times, each time excluding one province from the sample. The age-specific estimates are stable across all exclusions (Appendix Figure D1): for workers aged 24–39, coefficients remain statistically significant across all three sectors. For workers aged 40–54, coefficients are stable in magnitude, though less precisely estimated for formal non-agriculture, consistent with the smaller effect for this group. For workers aged 55–64, the agriculture and formal non-agriculture coefficients remain close to zero, confirming that heat-induced reallocation into formal employment is concentrated among younger and middle-aged workers.

C.4 Alternative Weather Measures

The construction of province-level weather variables from gridded reanalysis data involves interpolation choices that could affect the estimates. In the baseline specification, I assign weather exposure based on the inverse-distance-weighted average of the four nearest grid points to each province's centroid. As an alternative, I construct weather variables by averaging values across all grid points falling within each province's administrative boundaries. The results are not sensitive to these choices (Appendix Table D7, Column 4).

C.5 Alternative Functional Forms of Temperature

The baseline specification imposes a particular functional form on the temperature-outcome relationship through the degree day measure. To assess whether the results are sensitive to this choice, I estimate models using alternative representations of weather exposure: cumulative temperature bins, degree day bins, and fourth-order polynomials. Across these specifications, the qualitative pattern is consistent with the baseline findings: labor reallocation effects are concentrated at the upper end of the temperature distribution, above approximately 27°C (Appendix Figures D2, D3, D4). This consistency across functional forms not only provides confidence that the results reflect genuine non-linearity in the temperature-labor relationship rather than artifacts of the baseline parameterization, but also align with occupational health thresholds for heat stress and supporting the choice of 27°C as the threshold in the baseline specification.

In addition, as discussed in Section 3, I focus on wet-bulb temperature because it captures the combined effects of heat and humidity that determine thermal stress. However, much of the existing literature uses dry-bulb (ambient) temperature. To facilitate comparison with prior work, I re-estimate the main specification using dry-bulb temperature. The qualitative pattern is similar: high temperatures are associated with decreased agricultural employment and increased non-agricultural employment, but the relationships are less precisely estimated (Appendix Figure D5). This is consistent with wet-bulb temperature being the more physiologically relevant measure for outdoor labor.

C.6 Leads and Lags of Temperature

The baseline specification uses a 24-month reference window to construct weather variables, reflecting the assumption that labor reallocation is a gradual process. I test this assumption by decomposing temperature exposure into two sub-periods: 12–24 months before the interview ($t-1$) and 0–12 months before the interview (t). This decomposition reveals which period drives the reallocation effects. As a falsification test, I also include

leads (i.e., temperature exposure in the 12 months following the interview). If provinces experiencing temperature shocks were already on differential trajectories, future temperature should predict current outcomes. Insignificant lead effects would further support the causal interpretation of the estimates (Dell et al., 2012).

The results in Appendix Table D8 reveal that the lag component $t-1$ (i.e., 12–24 months before the interview) drives the reallocation effects, while contemporaneous exposure t, (i.e., 0–12 months before the interview) has no independent effect once lags are controlled for. This pattern supports the assumption underlying the baseline specification: labor reallocation is a gradual process that unfolds over one to two years as workers assess whether adverse conditions persist, search for non-agricultural opportunities, and acquire new skills. The cumulative effects remain highly significant ($p < 0.01$ for agriculture and informal non-agriculture; $p = 0.03$ for formal non-agriculture) and consistent in sign with the baseline estimates.

On the other hand, lead coefficients, which capture the association between future temperature exposure in the 12 months following the interview and current employment outcomes, are small and statistically insignificant across all sectors. Since future weather has not yet been realized, this suggests the baseline estimates reflect genuine temperature responses rather than spurious trends.

These patterns hold when examining age-specific estimates: lead and contemporaneous coefficients are insignificant across all age groups, while lagged effects drive the age-specific responses (Appendix Table D9). The lagged effects differ significantly across age groups for agriculture ($p = 0.012$) and formal non-agriculture ($p < 0.01$), but not for informal non-agriculture ($p = 0.27$), which is consistent with the baseline finding that heat-induced reallocation into informal employment is broad-based rather than concentrated among particular age groups.

C.7 Permutation Test

A valid identification strategy should not detect effects where none exist. To verify this, I conduct a Monte Carlo permutation test that randomly reassigns provincial weather series across provinces, breaking the true province-weather relationship while preserving the temporal structure within each series. Under the null hypothesis of no temperature effect, the distribution of placebo estimates should be centered at zero, with the actual estimate falling in the tails. As shown in Appendix Figure D6, the distributions of 10,000 placebo coefficients for all outcomes are centered at zero, and the baseline estimates lie far outside these distributions. The randomization inference p-values are below 0.001 for all three outcomes: none of the 10,000 placebo coefficients exceed the actual estimates in

absolute value. This confirms that the findings are not artifacts of spurious correlation or other features of the data structure.

The age-specific estimates are similarly validated by permutation test (Appendix Figure D7). For workers aged 24–39 and 40–54, the baseline estimates fall clearly in the tails of the placebo distributions across all three sectors. For workers aged 55–64, the informal non-agriculture estimates remain in the tails, while the agriculture and formal non-agriculture estimate fall within the placebo distribution, which is consistent with the smaller and less precisely estimated effect for these age-sector combinations in the baseline results.

C.8 Migration

The analysis so far has implicitly assumed that provincial labor markets are relatively bounded. Previous research, however, has documented inter-provincial migration as a response to climate change.[4] If heat-exposed provinces experience selective out-migration of agricultural workers or in-migration of non-agricultural workers, the observed sectoral shifts could reflect compositional changes rather than the within-province reallocation mechanism discussed in Section 6.

I address this concern through three exercises. First, a decomposition following McCaig and Pavcnik (2018) shows that most of the observed structural change occurs within provinces (Appendix Table D10). Second, I estimate equation (1) using province-level migration rates from statistical yearbooks over the study period. Neither in-migration, out-migration, nor net migration responds significantly to contemporaneous or lagged temperature exposure (Appendix Table D11).

Third, I use a more conservative, individual-level measure: the number of months each household member stayed at home during the reference period, available from the 2004 wave onward and aggregated to the province-year and province-age group-year level as in the main analysis.[5] This provides a stricter test than the province-level analysis, which classifies anyone remaining within the same province as geographically immobile, including individuals who may have moved across districts or between rural and urban

[4] See Cattaneo et al. (2019) for a review.

[5] The surveys use households as sampling units and define household membership on the basis of physical presence. Individuals must stay and eat in the household for at least six months during the 12-month reference period, and contribute to collective income and expenses to be considered members. This requirement means that individuals who have moved away for work or school (e.g., migrants) are not considered household members. Considering an individual as a seasonal migrant if they left the household for work during the year but are still considered as a household member (as in de Brauw and Harigaya (2007)), then more than 96% of household members staying in households during the last 12 months also suggests a low seasonal migration rate of 4%.

areas within the province. In contrast, months stayed at home captures whether individuals remained in their own household, ruling out even within-province mobility as a driver of the results. If the age differential findings were driven by selective in-migration of young formal workers, one would expect younger workers to spend fewer months at home in response to heat. Appendix Table D12 shows the opposite: heat exposure increases months at home for younger workers ($p < 0.01$), with no effect on middle-aged or older groups. The point estimate is small relative to the mean (0.009 additional months against a mean of 11.68), indicating that younger workers are, if anything, marginally more likely to stay home when exposed to heat. These are consistent with in-situ adaptation rather than migration-driven compositional change.

A related concern is that the repeated cross-sectional design means different workers populate each province-age group cell across waves, and the observed sectoral shifts could reflect compositional turnover rather than behavioral responses. Estimating the specification on the longitudinal panel component (2004–2018), which tracks the same workers across waves with weather measured at the district level, yields consistent average and age-specific results (Appendix Table D6), ruling out both compositional change and attenuation from coarse weather measurement.

Together, these results indicate that inter-provincial migration is not a first-order response to temperature shocks in this setting, and that the observed sectoral reallocation occurs among local, geographically immobile workers.

D APPENDIX TABLES AND FIGURES

D.1 Appendix Tables

Table D1: Residual Variation in Temperature Variables

	Total SD	Residual SD	Residual SD/Total SD (%)	% | Residual | > 1 DD
	(1)	(2)	(3)	(4)
DD9 (Cold Exposure)	12.84	4.58	35.7	34.3
DD27 (Heat Exposure)	8.70	1.33	15.3	23.0

Notes: Total SD is the weighted standard deviation of the raw variable. Residual SD is the standard deviation after removing province and region-by-year fixed effects. Residual/Total reports the residual SD as a percentage of the total SD. The final column reports the percentage of observations with absolute residuals exceeding one degree day. Unit of analysis is province-year. Sample weights applied throughout. For comparison, Fisher et al. (2012) report that only 1.3% of observations in Deschênes and Greenstone (2007) exceed one degree Fahrenheit after removing county and state-by-year fixed effects with corrected US temperature data.

Table D2: Wet-bulb Temperature and Sectoral Labor Share

	Agriculture	Formal Non-Agriculture	Informal Non-Agriculture	Inactive and Unemployed
	(1)	(2)	(3)	(4)
DD9	0.0061	-0.0012	-0.0045	-0.0004
	(0.0033)	(0.0009)	(0.0028)	(0.0004)
	[0.0029]	[0.0012]	[0.0025]	[0.0004]
DD27	-0.0430	0.0177	0.0252	0.0002
	(0.0108)	(0.0078)	(0.0076)	(0.0035)
	[0.0104]	[0.0057]	[0.0078]	[0.0026]
Precipitation	-0.0355	0.0125	0.0197	0.0033
	(0.0112)	(0.0057)	(0.0104)	(0.0036)
	[0.0118]	[0.0055]	[0.0094]	[0.0029]
Precipitation Squared	0.0147	-0.0032	-0.0097	-0.0018
	(0.0069)	(0.0037)	(0.0063)	(0.0019)
	[0.0071]	[0.0031]	[0.0056]	[0.0017]
Mean Outcome	0.457	0.174	0.281	0.088
Observations	569	569	569	569
Province FE	Yes	Yes	Yes	Yes
Region-Year FE	Yes	Yes	Yes	Yes

Notes: Each column presents the effects of one within-province standard deviation increase in each weather variable on sectoral labor shares. Estimates obtained from equation (1). Unit of analysis is province-year. All regressions control for the second-order polynomials of precipitation. Robust standard errors clustered at the province level and Conley standard errors that allow for spatial correlation up to 150 km and serial correlation up to five lags are reported in parentheses and brackets, respectively. Province distances are computed from province geographic centroids. All regressions use sampling weights.

Table D3: Wet-bulb Temperature and Sectoral Labor Share by Industry

	Agriculture	Less Skill-Intensive Non-Agriculture	Medium Skill-Intensive Non-Agriculture	High Skill-Intensive Non-Agriculture
	(1)	(2)	(3)	(4)
DD9	0.0007	-0.0005	0.0000	-0.0002
	(0.0004)	(0.0003)	(0.0001)	(0.0001)
	[0.0003]	[0.0003]	[0.0000]	[0.0001]
DD27	-0.0064	0.0042	0.0014	0.0008
	(0.0016)	(0.0011)	(0.0002)	(0.0006)
	[0.0015]	[0.0012]	[0.0003]	[0.0004]
Mean Outcome	0.457	0.332	0.032	0.091
Observations	569	569	569	569
Province FE	Yes	Yes	Yes	Yes
Region-Year FE	Yes	Yes	Yes	Yes

Notes: Each column presents the effects of one additional degree day below 9 °C (DD9) or above 27°C (DD27) on sectoral labor shares. Estimates obtained from equation (1). Unit of analysis is province-year. Less skill-intensive industries include those classified as low-tech manufacturing, less knowledge-intensive services, construction and mining; medium skill-intensive industries include those classified as medium-tech manufacturing and public utilities; high skill-intensive industries include those classified as high-tech manufacturing and knowledge-intensive services according to the Statistical Classification of Economic Activities in the European Community. The mean educational attainment of workers in each industry is presented in Appendix Table B3. All regressions control for the second-order polynomials of precipitation. Robust standard errors clustered at the province level and Conley standard errors that allow for spatial correlation up to 150 km and serial correlation up to five lags are reported in parentheses and brackets, respectively. Province distances are computed from province geographic centroids. All regressions use sampling weights.

Table D4: Wet-bulb Temperature and Sectoral Labor Share: Heterogeneity by Age Group

	Agriculture	Formal Non-Agriculture	Informal Non-Agriculture	Inactive and Unemployed
	(1)	(2)	(3)	(4)
DD9 × Age 24–39 (DD9G1)	0.0005	-0.0002	-0.0003	-0.0000
	(0.0005)	(0.0002)	(0.0004)	(0.0000)
	[0.0004]	[0.0002]	[0.0003]	[0.0000]
DD9 × Age 40–54 (DD9G2)	0.0009	-0.0002	-0.0006	0.0000
	(0.0004)	(0.0001)	(0.0003)	(0.0001)
	[0.0004]	[0.0001]	[0.0003]	[0.0000]
DD9 × Age 55–64 (DD9G3)	0.0010	0.0001	-0.0010	-0.0002
	(0.0006)	(0.0001)	(0.0005)	(0.0003)
	[0.0005]	[0.0001]	[0.0004]	[0.0002]
p-val: DD9G1 = DD9G2	0.4157	0.8042	0.1829	0.6255
p-val: DD9G2 = DD9G3	0.7425	0.0655	0.4798	0.4679
p-val: DD9G1 = DD9G3	0.4995	0.2707	0.2857	0.5137
p-val: DD9G1 = DD9G2 = DD9G3	0.7130	0.1784	0.4060	0.7423
DD27 × Age 24–39 (DD27G1)	-0.0096	0.0053	0.0043	-0.0000
	(0.0021)	(0.0017)	(0.0016)	(0.0004)
	[0.0021]	[0.0015]	[0.0016]	[0.0004]
DD27 × Age 40–54 (DD27G2)	-0.0060	0.0020	0.0038	0.0002
	(0.0015)	(0.0010)	(0.0013)	(0.0004)
	[0.0015]	[0.0008]	[0.0013]	[0.0002]
DD27 × Age 55–64 (DD27G3)	-0.0017	-0.0005	0.0029	-0.0007
	(0.0018)	(0.0005)	(0.0012)	(0.0016)
	[0.0015]	[0.0003]	[0.0011]	[0.0013]
p-val: DD27G1 = DD27G2	0.0173	0.0053	0.7407	0.6538
p-val: DD27G2 = DD27G3	0.0027	0.0030	0.4696	0.5319
p-val: DD27G1 = DD27G3	0.0005	0.0005	0.5062	0.6123
p-val: DD27G1 = DD27G2 = DD27G3	0.0017	0.0020	0.7453	0.8025
Mean Outcome, Age 24–39	0.431	0.225	0.288	0.056
Mean Outcome, Age 40–54	0.473	0.156	0.305	0.066
Mean Outcome, Age 55–64	0.495	0.063	0.200	0.242
Observations	1707	1707	1707	1707
Province-Age Group FE	Yes	Yes	Yes	Yes
Region-Age Group-Year FE	Yes	Yes	Yes	Yes

Notes: Each column presents the effects of one additional degree day below 9 °C (DD9) or above 27°C (DD27) on sectoral labor shares of different age groups. Unit of analysis is province-age group-year. Estimates obtained from equation (2). All regressions control for cold exposure (DD9) and the second-order polynomials of precipitation and their interactions with group dummies. Robust standard errors clustered at the province level and Conley standard errors that allow for spatial correlation up to 150 km and serial correlation up to five lags are reported in parentheses and brackets, respectively. Province distances are computed from province geographic centroids. p-values from the F-test of significant age cohort differences using standard errors clustered at the province level are reported. The results are qualitatively similar when using Conley standard errors.

Table D5: Wet-bulb Temperature and Sectoral Labor Share: Heterogeneity by Education and Age

	Without High School Diploma		With High School Diploma	
	Formal Non-Agriculture (1)	Informal Non-Agriculture (2)	Formal Non-Agriculture (3)	Informal Non-Agriculture (4)
DD27 × Age 24–39 (DD27G1)	0.0041	0.0068	0.0056	-0.0000
	(0.0014)	(0.0019)	(0.0017)	(0.0022)
	[0.0014]	[0.0018]	[0.0013]	[0.0015]
DD27 × Age 40–54 (DD27G2)	0.0023	0.0040	0.0017	0.0021
	(0.0009)	(0.0013)	(0.0018)	(0.0013)
	[0.0007]	[0.0014]	[0.0013]	[0.0012]
DD27 × Age 55–64 (DD27G3)	-0.0001	0.0032	-0.0003	0.0024
	(0.0003)	(0.0012)	(0.0016)	(0.0014)
	[0.0003]	[0.0013]	[0.0014]	[0.0016]
p-val: DD27G1 = DD27G2	0.0808	0.0859	0.0273	0.3553
p-val: DD27G2 = DD27G3	0.0025	0.5405	0.2513	0.8420
p-val: DD27G1 = DD27G3	0.0020	0.0837	0.0098	0.3775
p-val: DD27G1 = DD27G2 = DD27G3	0.0032	0.1832	0.0277	0.6352
Mean Outcome, Age 24–39	0.100	0.311	0.548	0.231
Mean Outcome, Age 40–54	0.073	0.311	0.476	0.283
Mean Outcome, Age 55–64	0.032	0.200	0.213	0.204
Observations	1707	1707	1634	1634
Province-Age Group FE	Yes	Yes	Yes	Yes
Region-Age Group-Year FE	Yes	Yes	Yes	Yes

Notes: Each column presents the effects of one additional degree day above 27°C (DD27) on sectoral labor shares of different age groups. Unit of analysis is province-age group-education-year. Columns (1)–(2) and (3)–(4) are estimated separately for each sub-sample. Estimates obtained from equation (2). All regressions control for cold exposure (DD9) and the second-order polynomials of precipitation and their interactions with group dummies, as well as controlling for demographics including the share of male workers, the share of ethnic minority workers, and share of workers with a high school diploma. Robust standard errors clustered at the province level and Conley standard errors that allow for spatial correlation up to 150 km and serial correlation up to five lags are reported in parentheses and brackets, respectively. Province distances are computed from province geographic centroids. p-values from the F-test of significant age cohort differences using standard errors clustered at the province level are reported. The results are qualitatively similar when using Conley standard errors.

Table D6: Wet-bulb Temperature and Sectoral Labor Share: Individual-level Panel

	Agriculture	Formal Non-Agriculture	Informal Non-Agriculture	Inactive and Unemployed
	(1)	(2)	(3)	(4)
Panel A: Average Effects				
DD27	-0.0049	0.0022	0.0019	0.0008
	(0.0015)	(0.0010)	(0.0009)	(0.0004)
Mean Outcome	0.450	0.169	0.295	0.085
Observations	206576	206576	206576	206576
Province FE	Yes	Yes	Yes	Yes
Region-Wave FE	Yes	Yes	Yes	Yes
Panel FE	Yes	Yes	Yes	Yes
Panel B: Age-Heterogeneity Effects				
DD27 × Age 24–39 (G1)	-0.0071	0.0048	0.0021	0.0003
	(0.0021)	(0.0016)	(0.0012)	(0.0003)
DD27 × Age 40–54 (G2)	-0.0054	0.0022	0.0027	0.0004
	(0.0014)	(0.0010)	(0.0008)	(0.0004)
DD27 × Age 55–64 (G3)	-0.0027	0.0005	0.0013	0.0009
	(0.0019)	(0.0004)	(0.0019)	(0.0013)
p-val: G1 = G2 = G3	0.109	0.016	0.515	0.879
Province-Age Group FE	Yes	Yes	Yes	Yes
Region-Wave-Age Group FE	Yes	Yes	Yes	Yes
Panel-Age Group FE	Yes	Yes	Yes	Yes

Notes: This table replicates the main results using the longitudinal panel component of the VHLSS, which tracks the same individuals across up to three consecutive survey waves. The sample comprises four stacked panels (2004–2006–2008, 2010–2012–2014, 2012–2014–2016, 2014–2016–2018) covering 63 provinces. Weather variables are constructed at the district level using inverse-distance-weighted averages of the four nearest ERA5-Land grid points to each district centroid, and aggregated over the 24 months preceding each individual's interview date. Panel A reports average effects. Panel B reports age-specific effects, with p-values from joint tests of equality across age groups. All regressions in Panel A include province, region-by-wave, and panel cohort fixed effects. All regressions in Panel B interact these fixed effects with age group. Standard errors clustered at the province level are reported in parentheses.

Table D7: Wet-bulb Temperature and Sectoral Labor Share: Robustness Checks

	Baseline	Controlling for Demographics	Sampling Restrictions	Alternative Weather Construction Method
	(1)	(2)	(3)	(4)
Agriculture	-0.0064	-0.0047	-0.0062	-0.0051
	(0.0016)	(0.0012)	(0.0016)	(0.0019)
	[0.0015]	[0.0011]	[0.0015]	[0.0015]
Formal Non-Agriculture	0.0026	0.0022	0.0023	0.0026
	(0.0012)	(0.0009)	(0.0012)	(0.0010)
	[0.0008]	[0.0007]	[0.0008]	[0.0008]
Informal Non-Agriculture	0.0037	0.0027	0.0038	0.0028
	(0.0011)	(0.0009)	(0.0011)	(0.0012)
	[0.0012]	[0.0010]	[0.0012]	[0.0011]

Notes: Each cell presents the effects of one additional degree day above 27°C on sectoral labor shares. Estimates obtained from equation (1). Unit of analysis is province-year. All regressions control for the second-order polynomials of precipitation. Robust standard errors clustered at the province level and Conley standard errors that allow for spatial correlation up to 150 km and serial correlation up to five lags are reported in parentheses and brackets, respectively. Province distances are computed from province geographic centroids. All regressions use sampling weights. Column (2) controls for demographic characteristics: the share of male individuals, share of ethnic majority (Kinh), average age, and share with a high school diploma. Column (3) excludes province-year cells with fewer observations than the 5th percentile of the sample distribution (approximately 100 observations). Column (4) uses weather variables constructed as simple averages across all grid points within each province's geographic boundary, rather than inverse-distance weighted averages.

Table D8: Wet-bulb Temperature and Sectoral Labor Share: Lead and Lag of Weather Variable

	Agriculture	Formal Non-Agriculture	Informal Non-Agriculture
	(1)	(2)	(3)
DD27 (t + 1)	-0.0009	0.0009	0.0011
	(0.0020)	(0.0013)	(0.0022)
	[0.0016]	[0.0011]	[0.0016]
DD27 (t)	-0.0005	0.0008	0.0012
	(0.0018)	(0.0014)	(0.0022)
	[0.0021]	[0.0011]	[0.0022]
DD27 (t − 1)	-0.0131	0.0045	0.0063
	(0.0049)	(0.0021)	(0.0028)
	[0.0037]	[0.0019]	[0.0027]
p-value: DD27 (t − 1) + (t) = 0	0.001	0.032	0.005
Mean Outcome	0.457	0.174	0.281
Observations	569	569	569
Province FE	Yes	Yes	Yes
Region-Year FE	Yes	Yes	Yes

Notes: Each column presents the effects of one additional degree day above 27°C (DD27) on sectoral labor shares. Estimates obtained from equation (1). Unit of analysis is province-year. All regressions control for the lag, current, and lead of cold exposure (DD9), as well as of second-order polynomials of precipitation. Robust standard errors clustered at the province level and Conley standard errors that allow for spatial correlation up to 150 km and serial correlation up to five lags are reported in parentheses and brackets, respectively. Province distances are computed from province geographic centroids. All regressions use sampling weights.

Table D9: Wet-bulb Temperature and Sectoral Labor Share: Heterogeneity by Age Group

Including Lead and Lag of Weather Variable

	Agriculture (1)	Formal Non-Agriculture (2)	Informal Non-Agriculture (3)
DD27 (t + 1) × Age 24–39 (G1)	-0.0006	0.0012	-0.0005
	(0.0024)	(0.0022)	(0.0029)
	[0.0019]	[0.0017]	[0.0022]
DD27 (t + 1) × Age 40–54 (G2)	-0.0026	0.0011	0.0018
	(0.0023)	(0.0011)	(0.0017)
	[0.0018]	[0.0009]	[0.0013]
DD27 (t + 1) × Age 55–64 (G3)	-0.0015	0.0011	0.0050
	(0.0024)	(0.0004)	(0.0025)
	[0.0014]	[0.0003]	[0.0018]
p-val: (t + 1) G1 = G2 = G3	0.2769	0.9997	0.0423
DD27 (t) × Age 24–39 (G1)	-0.0016	0.0009	0.0016
	(0.0023)	(0.0020)	(0.0027)
	[0.0027]	[0.0019]	[0.0025]
DD27 (t) × Age 40–54 (G2)	-0.0011	0.0015	0.0003
	(0.0023)	(0.0014)	(0.0021)
	[0.0024]	[0.0009]	[0.0021]
DD27 (t) × Age 55–64 (G3)	0.0019	-0.0000	0.0029
	(0.0033)	(0.0009)	(0.0022)
	[0.0030]	[0.0007]	[0.0021]
p-val: (t) G1 = G2 = G3	0.6737	0.5111	0.2844
DD27 (t − 1) × Age 24–39 (G1)	-0.0203	0.0118	0.0069
	(0.0059)	(0.0029)	(0.0036)
	[0.0043]	[0.0027]	[0.0033]
DD27 (t − 1) × Age 40–54 (G2)	-0.0114	0.0021	0.0075
	(0.0055)	(0.0023)	(0.0037)
	[0.0040]	[0.0019]	[0.0031]
DD27 (t − 1) × Age 55–64 (G3)	-0.0058	-0.0014	0.0034
	(0.0034)	(0.0009)	(0.0024)
	[0.0036]	[0.0008]	[0.0025]
p-val:(t − 1) G1 = G2 = G3	0.0140	0.0000	0.2782
Mean Outcome, Age 24–39	0.431	0.225	0.288
Mean Outcome, Age 40–54	0.473	0.156	0.305
Mean Outcome, Age 55–64	0.495	0.063	0.200
Observations	1707	1707	1707
Province-Age Group FE	Yes	Yes	Yes
Region-Age Group-Year FE	Yes	Yes	Yes

Notes: Each column presents the effects of one additional degree day above 27°C (DD27) on sectoral labor shares of different age groups. Estimates obtained from equation (2). Unit of analysis is province-age group-year. All regressions control for the lag, current, and lead of cold exposure (DD9), as well as of second-order polynomials of precipitation, and their interaction with age group dummies. p-values from the F-test of significant age cohort differences using standard errors clustered at the province level are reported. The results are qualitatively similar when using Conley standard errors.

Table D10: Decomposing Changes in Sectoral Labor Share Shares: 1992–2018

	Agriculture	Informal Non-Agriculture	Formal Non-Agriculture
	(1)	(2)	(3)
Total Change	-0.314	0.111	0.203
Within Province	-0.312	0.121	0.200
Between Province	-0.002	-0.009	0.002

Notes: This table presents the results from the decomposition exercise following McCaig and Pavcnik (2018). I decompose the change in the share of workers in each sector in total employment between 1992 and 2018, denoted by ΔS, into within and between province shifts, respectively: $\Delta S_t = S_t - S_{t-1} = \sum_p \Delta s_{pt} h_p + \sum_p \Delta h_{pt} s_p$ where h_{pt} is the share of province p's employment in total employment at time t, s_{pt} is the share of workers in sector s in total employment in province p, $h_p = 0.5(h_{pt} + h_{pt-1})$, and $s_p = 0.5(s_{pt} + s_{pt-1})$. The first summation term captures the importance of mobility of workers across sectors within a province, and the second summation captures the prevalence of mobility of workers across provinces as sources of changes in aggregate sectoral employment shares. Estimates based on VLSS 1992/1993 and VHLSS 2018. Sample includes workers aged 24-64 inclusive. Survey sampling weights included. The results are qualitatively similar when using other household survey rounds as the start point.

Table D11: Wet-bulb Temperature and Migration Rates

	In-Migration (1)	Out-Migration (2)	Net-Migration (3)
DD27 (t)	0.0215	-0.0149	0.0396
	(0.0735)	(0.1017)	(0.1355)
	[0.0690]	[0.0819]	[0.1097]
DD27 (t − 1)	0.0799	-0.0731	0.1517
	(0.0892)	(0.1058)	(0.1315)
	[0.0854]	[0.0773]	[0.1184]
p-value: DD27 (t − 1) + (t) = 0	0.198	0.280	0.054
Mean Outcome	5.060	6.712	-1.653
Observations	312	312	312
Province FE	Yes	Yes	Yes
Region-Year FE	Yes	Yes	Yes

Notes: Each column presents the effects of one additional degree day below 9 or above 27°C (DD27) on migration rates at the province level. Estimates obtained from equation (1). Unit of analysis is province-year. All regressions control for second-order polynomials of precipitation. Robust standard errors clustered at the province level and Conley standard errors that allow for spatial correlation up to 150 km and serial correlation up to five lags are reported in parentheses and brackets, respectively. Province distances are computed from province geographic centroids. "In-migration" rate is the number of people from different territorial units (out-migration province) in-migrate to a territorial unit in the reference period on an average per 1000 population of that territorial unit (in-migration province). Out-migration rate is the number out-migrants of a territorial unit in the reference period on an average per 1000 population of that territorial unit. Net-migration rate is the difference between number of in-migrants and number of out-migrants of a territorial unit in the reference period on an average per 1000 population of that territorial unit.

Table D12: Wet-bulb Temperature and Number of Months at Home

	Average	Age 24–39	Age 40–54	Age 55–64
	(1)	(2)	(3)	(4)
DD27	0.0044	0.0093	-0.0006	-0.0006
	(0.0017)	(0.0030)	(0.0014)	(0.0014)
	[0.0014]	[0.0025]	[0.0011]	[0.0011]
Mean Outcome	11.794	11.678	11.870	11.916
Observations	1248	1248	1248	1248
Province FE	Yes			
Region-Year FE	Yes			
Province-Age Group FE		Yes	Yes	Yes
Region-Age Group-Year FE		Yes	Yes	Yes

Notes: Each column presents the effects of one additional degree day above 27°C (DD27) on number of months staying at home during the 12-month reference period. Estimates obtained from equation (1). Unit of analysis is province-year for Column (1) and province-age group-year for Columns (2)–(4). Columns (2)–(4) are jointly estimated. All regressions control for second-order polynomials of precipitation. Robust standard errors clustered at the province level and Conley standard errors that allow for spatial correlation up to 150 km and serial correlation up to five lags are reported in parentheses and brackets, respectively. Province distances are computed from province geographic centroids. Data available for VHLSS 2004–2018.

Table D13: Wet-bulb Temperature and Labor Productivity: Firm-Level Estimates

Sample of mining and quarrying firms

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Firm Size	Average Firm	[1, 10)	[10, 30)	[30, 50)	[50-100)	[100,)
DD27	-0.0040	-0.0118	-0.0076	-0.0009	-0.0002	-0.0029
	(0.0039)	(0.0047)	(0.0040)	(0.0040)	(0.0050)	(0.0073)
Panel B: Firm Age	Average Worker	[0, 3)	[3, 6)	[6, 10)	[10-20)	[20,)
DD27	-0.0149	0.0117	-0.0014	0.0037	-0.0105	-0.0126
	(0.0053)	(0.0115)	(0.0108)	(0.0090)	(0.0076)	(0.0059)

Notes: This table presents the results from the following regression: $y_{iprt} = f(\mathrm{WBT}_{pt}) + g(\mathrm{R}_{pt}) + \gamma_i + \gamma_{rt} + \varepsilon_{prt}$ where y_{iprt} represents log of annual revenue per worker (2010 million VND) of firm i in province p of region r in year t, and γ_i represents firm-specific fixed effects. Unit of analysis is firm-year. Column (1) reports the average effect from a separate regression, unweighted in Panel A, and weighted by average firm employment in Panel B. Columns (2)–(6) in Panel A examine heterogeneity by firm size; estimates are unweighted, reflecting effects on the average firm within each size category. Columns (2)–(6) in Panel B examine heterogeneity by firm age; estimates are weighted by average firm employment, reflecting effects on the average worker within each age category. Note that the effect on the average worker exceeds the effect on the average firm, consistent with employment being concentrated in older establishments. All regressions control for cold exposure (DD9) and second-order polynomials of precipitation (and their interactions with either firm size or firm age dummies for the heterogeneity analysis). Robust standard errors clustered at the province level are reported.

Table D14: Wet-bulb Temperature and Rice Prices

	Harvest Price		Sold Price	
	Mean	Median	Mean	Median
	(1)	(2)	(3)	(4)
DD9	0.0011	0.0012	0.0016	0.0041
	(0.0010)	(0.0016)	(0.0009)	(0.0010)
	[0.0010]	[0.0015]	[0.0011]	[0.0012]
DD27	0.0003	-0.0029	-0.0034	-0.0092
	(0.0040)	(0.0047)	(0.0060)	(0.0067)
	[0.0025]	[0.0029]	[0.0051]	[0.0050]
Mean Outcome	3.996	3.959	4.029	3.974
Observations	416	416	415	415
Province FE	Yes	Yes	Yes	Yes
Region-Year FE	Yes	Yes	Yes	Yes

Notes: Each column presents the effects of one additional degree day below 9 or above 27°C (DD27) on rice prices. In Columns (1)-(2), dependent variables are price of rice at harvest, measured as the province-level mean/median of household-level revenue at harvest divided by output (thousand VND/kg). In Columns (3)-(4), dependent variables are price of rice sold, measured as the province-level mean/median of household-level revenue sold divided by output sold (thousand VND/kg). Estimates obtained from equation (1). Unit of analysis is province-year. All regressions control for the second-order polynomials of precipitation. Robust standard errors clustered at the province level and Conley standard errors that allow for spatial correlation up to 150 km and serial correlation up to five lags are reported in parentheses and brackets, respectively. Province distances are computed from province geographic centroids. All regressions are weighted by household survey sampling weights interacted with cultivated areas.

Table D15: Wet-bulb Temperature and Crop Yields

	Rice Aggregate	Rice Winter-Spring	Maize	Grains
	(1)	(2)	(3)	(4)
DD27	-0.0018	-0.0047	-0.0023	-0.0021
	(0.0011)	(0.0026)	(0.0018)	(0.0011)
	[0.0009]	[0.0023]	[0.0014]	[0.0009]
Mean Outcome	5.140	6.064	3.910	5.023
Observations	520	510	510	520
Province FE	Yes	Yes	Yes	Yes
Region-Year FE	Yes	Yes	Yes	Yes

Notes: Each column presents the effects of one additional degree day above 27°C (DD27) on rice prices. Dependent variables are crop yields, measured in tonnes per hectare, at the province level, entered in logs for the estimation. Mean outcomes are in levels. Estimates obtained from equation (1). Unit of analysis is province-year. All regressions control for cold exposure (DD9) and the second-order polynomials of precipitation. Robust standard errors clustered at the province level and Conley standard errors that allow for spatial correlation up to 150 km and serial correlation up to five lags are reported in parentheses and brackets, respectively. Province distances are computed from province geographic centroids. All regressions are weighted by province-level cultivated areas in 1995.

Table D16: Wet-bulb Temperature and Planting Areas

	Rice Aggregate	Rice Winter-Spring	Maize	Grains
	(1)	(2)	(3)	(4)
DD27	-0.0019	-0.0005	0.0010	-0.0020
	(0.0013)	(0.0011)	(0.0037)	(0.0013)
	[0.0013]	[0.0017]	[0.0036]	[0.0018]
Mean Outcome	145.066	58.387	19.763	164.906
Observations	520	520	516	520
Province FE	Yes	Yes	Yes	Yes
Region-Year FE	Yes	Yes	Yes	Yes

Notes: Each column presents the effects of one additional degree day above 27°C (DD27) on rice prices. Dependent variables are planting area, measured in thousand hectares, at the province level (logs). Estimates obtained from equation (1). Unit of analysis is province-year. All regressions control for cold exposure (DD9) and the second-order polynomials of precipitation. Robust standard errors clustered at the province level and Conley standard errors that allow for spatial correlation up to 150 km and serial correlation up to five lags are reported in parentheses and brackets, respectively. Province distances are computed from province geographic centroids.

Table D17: Wet-bulb Temperature and Conditional Hours of Work: Heterogeneity by Age Group

	Agriculture	Formal Non-Agriculture	Informal Non-Agriculture
	(1)	(2)	(3)
DD27 × Age 24–39 (DD27G1)	-24.7056	11.3670	3.0642
	(7.5843)	(5.8289)	(4.4537)
	[6.7863]	[5.0403]	[3.2026]
DD27 × Age 40–54 (DD27G2)	-17.4367	7.1301	1.3900
	(6.1400)	(3.9630)	(3.9145)
	[5.7113]	[3.1800]	[4.0356]
DD27 × Age 55–64 (DD27G3)	-15.3957	-3.7714	5.3019
	(4.9465)	(9.2608)	(3.4110)
	[6.0791]	[6.8301]	[3.3797]
p-val: DD27G1 = DD27G2	0.0565	0.4527	0.6626
p-val: DD27G2 = DD27G3	0.4999	0.2161	0.4961
p-val: DD27G1 = DD27G3	0.0095	0.1272	0.6944
p-val: DD27G1 = DD27G2 = DD27G3	0.0325	0.3086	0.7734
Mean Outcome, Age 24–39	1333	1839	1809
Mean Outcome, Age 40–54	1306	1868	1832
Mean Outcome, Age 55–64	1168	1661	1753
Observations	1551	1499	1514
Province-Age Group FE	Yes	Yes	Yes
Region-Age Group-Year FE	Yes	Yes	Yes

Notes: Each column presents the effects of one additional degree day above 27°C (DD27) on sectoral hours, conditional on working in a sector (intensive margin). Unit of analysis is province-age group-year. Estimates obtained from equation (2). All regressions control for cold exposure (DD9) and the second-order polynomials of precipitation and their interactions with group dummies. Robust standard errors clustered at the province level and Conley standard errors that allow for spatial correlation up to 150 km and serial correlation up to five lags are reported in parentheses and brackets, respectively. Province distances are computed from province geographic centroids. p-values from the F-test of significant age cohort differences using standard errors clustered at the province level are reported. The results are qualitatively similar when using Conley standard errors.

Table D18: Wet-bulb Temperature and Sectoral Labor Share: Heterogeneity by Age Group

Controlling for demographic characteristics

	Agriculture	Formal Non-Agriculture	Informal Non-Agriculture	Inactive and Unemployed
	(1)	(2)	(3)	(4)
DD27 × Age 24–39 (DD27G1)	-0.0076	0.0038	0.0040	-0.0002
	(0.0016)	(0.0012)	(0.0015)	(0.0004)
	[0.0017]	[0.0011]	[0.0015]	[0.0004]
DD27 × Age 40–54 (DD27G2)	-0.0055	0.0019	0.0034	0.0002
	(0.0014)	(0.0010)	(0.0012)	(0.0004)
	[0.0014]	[0.0007]	[0.0012]	[0.0002]
DD27 × Age 55–64 (DD27G3)	-0.0018	0.0002	0.0023	-0.0007
	(0.0018)	(0.0004)	(0.0011)	(0.0016)
	[0.0014]	[0.0004]	[0.0011]	[0.0013]
p-val: DD27G1 = DD27G2	0.2323	0.0514	0.7100	0.4418
p-val: DD27G2 = DD27G3	0.0082	0.0680	0.3692	0.5730
p-val: DD27G1 = DD27G3	0.0157	0.0031	0.4199	0.7548
p-val: DD27G1 = DD27G2 = DD27G3	0.0207	0.0120	0.6370	0.7117
Mean Outcome, Age 24–39	0.431	0.225	0.288	0.056
Mean Outcome, Age 40–54	0.473	0.156	0.305	0.066
Mean Outcome, Age 55–64	0.495	0.063	0.200	0.242
Observations	1707	1707	1707	1707
Province-Age Group FE	Yes	Yes	Yes	Yes
Region-Age Group-Year FE	Yes	Yes	Yes	Yes

Notes: Each column presents the effects of one additional degree day above 27°C (DD27) on sectoral labor shares of different age groups. Unit of analysis is province-age group-year. Estimates obtained from equation (2). All regressions control for cold exposure (DD9) and the second-order polynomials of precipitation and their interactions with group dummies, as well as controlling for demographics including the share of male workers, the share of ethnic minority workers, and share of workers with a high school diploma. Robust standard errors clustered at the province level and Conley standard errors that allow for spatial correlation up to 150 km and serial correlation up to five lags are reported in parentheses and brackets, respectively. Province distances are computed from province geographic centroids. p-values from the F-test of significant age cohort differences using standard errors clustered at the province level are reported. The results are qualitatively similar when using Conley standard errors.

Table D19: Occupational Distribution in Formal and Informal Non-Agriculture

	Formal Non-Agriculture (1)	Informal Non-Agriculture (2)	p-value (1) = (2) (3)
Distribution by Occupation			
Leaders	0.062	0.001	0.000
High-level professionals	0.225	0.008	0.000
Mid-level professionals	0.152	0.011	0.000
Low-level professionals	0.078	0.005	0.000
Personal services and sales	0.074	0.287	0.000
Skilled manual workers	0.152	0.285	0.000
Assemblers and operators	0.136	0.063	0.000
Unskilled labor	0.120	0.339	0.000

Notes: This table presents the distribution of workers by occupation in formal and informal non-agricultural sectors. Sample includes workers aged 24–64 from VHLSS 2004–2018, during which occupation codes are consistent over time. Columns (1) and (2) report shares for formal and informal non-agriculture, respectively. Column (3) reports p-values from regressions of each occupation indicator on a formal non-agriculture dummy, controlling for province and year fixed effects, with standard errors clustered at the province level.

Table D20: Wet-bulb Temperature and Sectoral Labor Share by Occupation Group

	Formal Non-Agriculture		Informal Non-Agriculture	
	Physically Demanding (1)	Others (2)	Physically Demanding (3)	Others (4)
DD9	0.0001	-0.0002	-0.0005	-0.0000
	(0.0001)	(0.0001)	(0.0003)	(0.0000)
	[0.0001]	[0.0001]	[0.0003]	[0.0000]
DD27	0.0024	0.0002	0.0037	0.0000
	(0.0008)	(0.0006)	(0.0011)	(0.0001)
	[0.0007]	[0.0004]	[0.0011]	[0.0001]
Mean Outcome	0.080	0.090	0.273	0.007
Observations	569	569	569	569
Province FE	Yes	Yes	Yes	Yes
Region-Year FE	Yes	Yes	Yes	Yes

Dependent variables are employment shares in physically demanding and non-physically demanding occupations within each sector. "Physically demanding" includes skilled manual workers, assemblers and operators, unskilled labor, and personal services and sales. "Others" includes leaders, high- and mid-level professionals, and office staff. Estimates obtained from equation (1). Unit of analysis is province-year. All regressions control for the second-order polynomials of precipitation. Robust standard errors clustered at the province level and Conley standard errors that allow for spatial correlation up to 150 km and serial correlation up to five lags are reported in parentheses and brackets, respectively. Province distances are computed from province geographic centroids. All regressions use sampling weights.

Table D21: Wet-bulb Temperature and Sectoral Labor Share by Occupation Group:

Heterogeneity by Age Group

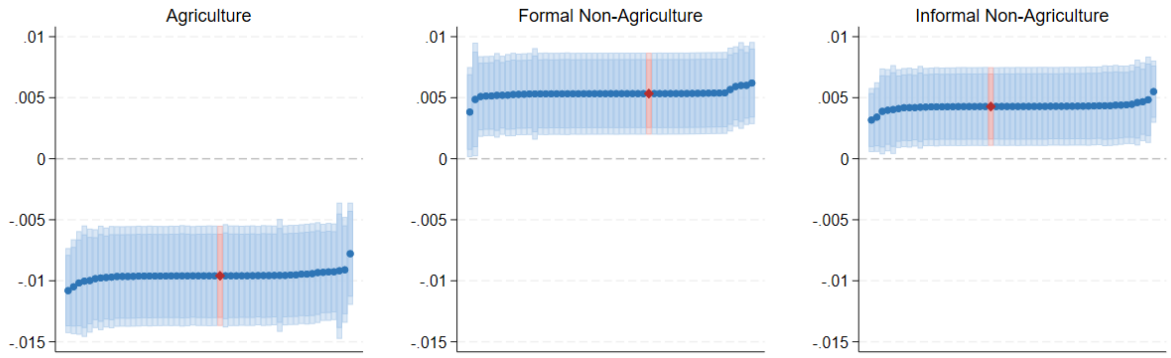
	Formal Non-Agriculture		Informal Non-Agriculture	
	Physically Demanding	Others	Physically Demanding	Others
	(1)	(2)	(3)	(4)
DD27 × Age 24–39 (DD27G1)	0.0048	0.0005	0.0042	0.0001
	(0.0012)	(0.0008)	(0.0016)	(0.0001)
	[0.0012]	[0.0006]	[0.0016]	[0.0001]
DD27 × Age 40–54 (DD27G2)	0.0013	0.0007	0.0039	-0.0001
	(0.0007)	(0.0005)	(0.0013)	(0.0001)
	[0.0006]	[0.0005]	[0.0013]	[0.0001]
DD27 × Age 55–64 (DD27G3)	0.0001	-0.0006	0.0027	0.0002
	(0.0003)	(0.0004)	(0.0011)	(0.0002)
	[0.0002]	[0.0003]	[0.0010]	[0.0002]
p-val: DD27G1 = DD27G2	0.0002	0.7812	0.8146	0.1092
p-val: DD27G2 = DD27G3	0.0478	0.0002	0.2527	0.2718
p-val: DD27G1 = DD27G3	0.0003	0.0946	0.4310	0.6864
p-val: DD27G1 = DD27G2 = DD27G3	0.0008	0.0008	0.5152	0.2558
Mean Outcome, Age 24–39	0.111	0.109	0.280	0.007
Mean Outcome, Age 40–54	0.064	0.087	0.297	0.007
Mean Outcome, Age 55–64	0.023	0.038	0.191	0.008
Observations	1707	1707	1707	1707
Province-Age Group FE	Yes	Yes	Yes	Yes
Region-Age Group-Year FE	Yes	Yes	Yes	Yes

Dependent variables are employment shares in physically demanding and non-physically demanding occupations within each sector. "Physically demanding" includes skilled manual workers, assemblers and operators, unskilled labor, and personal services and sales. "Others" includes leaders, high- and mid-level professionals, and office staff. Unit of analysis is province-age group-year. Estimates obtained from equation (2). Robust standard errors clustered at the province level and Conley standard errors that allow for spatial correlation up to 150 km and serial correlation up to five lags are reported in parentheses and brackets, respectively. Province distances are computed from province geographic centroids. All regressions use sampling weights. p-values from the F-test of significant age cohort differences using standard errors clustered at the province level are reported.

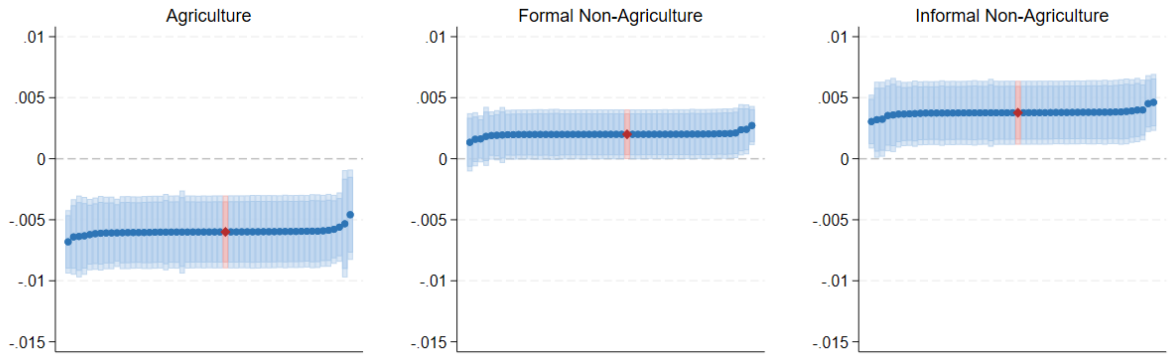
D.2 Appendix Figures

Figure D1: Leave-one-out Sensitivity: Heterogeneity by Age Group

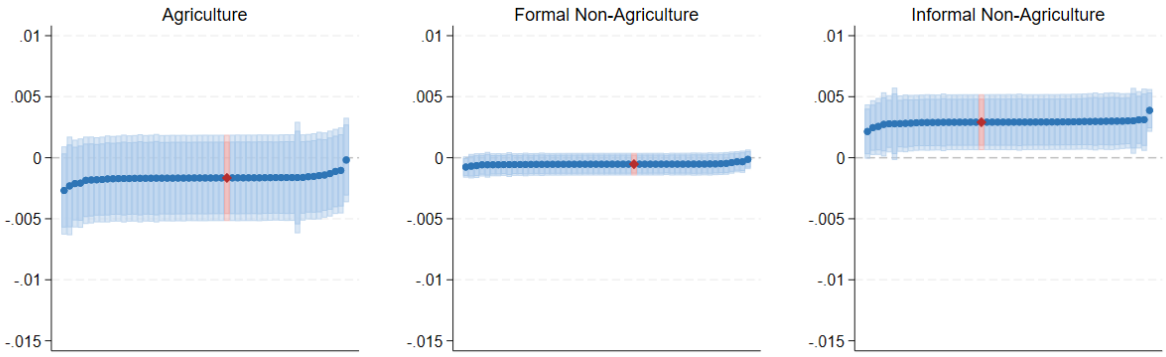
Panel A: Age 24–39



Panel B: Age 40–54

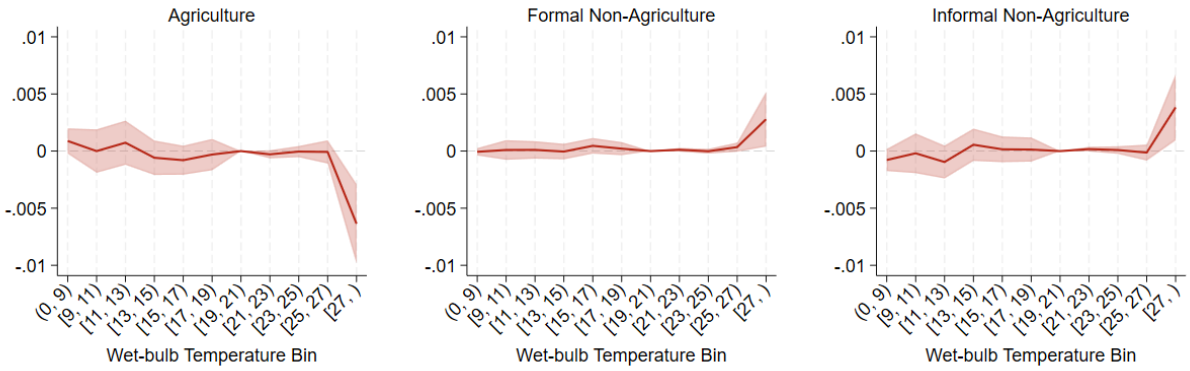


Panel C: Age 55–64



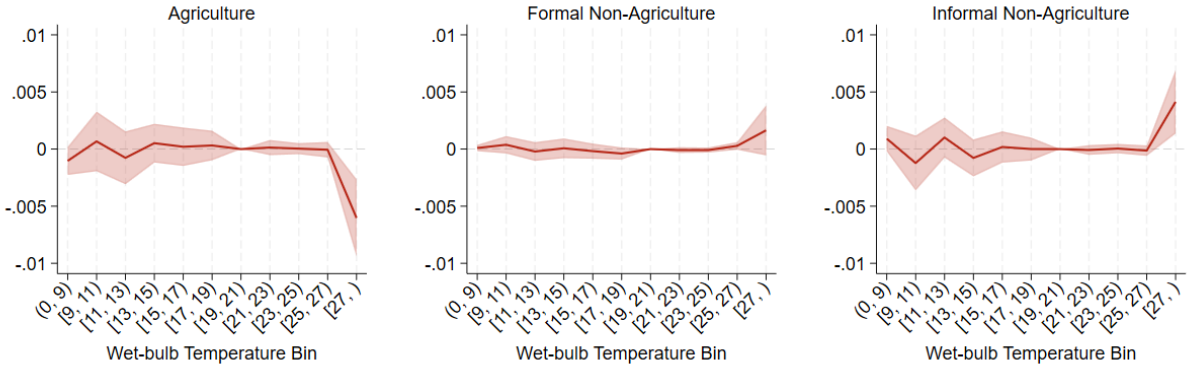
Notes: This figure presents the coefficients and 90/95% confidence intervals obtained from estimating the heterogeneous model (equation 2) 52 times, each time excluding one province from the sample. Results are shown separately by age group. Red diamond and bars represent the baseline estimates and confidence intervals.

Figure D2: Wet-bulb Temperature and Sectoral Labor Share: Cumulative Temperature Bins



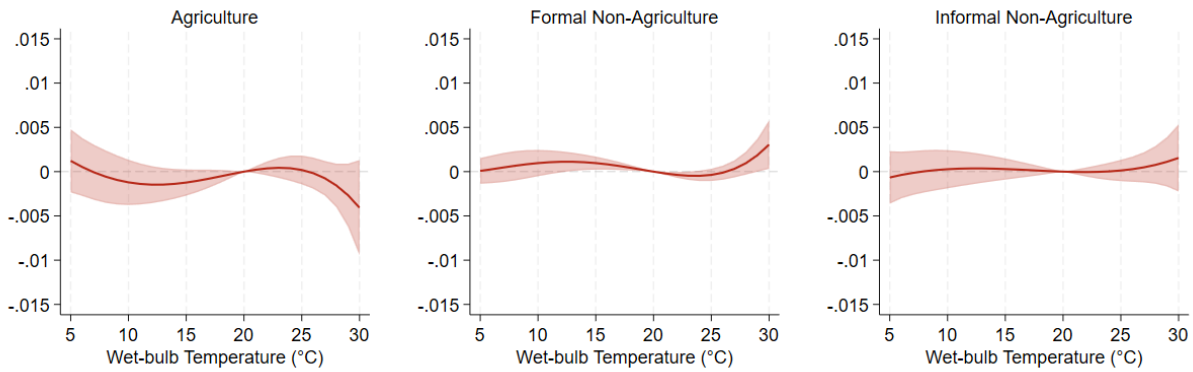
Notes: This figure plots the estimated effect of temperature on sectoral labor shares using cumulative temperature bins. Daily wet-bulb temperatures are allocated to eleven bins: below 9°C, nine two-degree bins from 9–27°C, and above 27°C. For temperatures below 19°C, each day contributes (bin upper bound - WBT) degree days; for temperatures above 19°C, each day contributes (WBT - bin lower bound) degree days. Coefficients represent the effect of one additional degree day in each bin, with 19–21°C as the omitted category. All specifications include province fixed effects, region-by-year fixed effects, and controls for precipitation. Shaded areas represent 95% confidence intervals based on standard errors clustered at the province level.

Figure D3: Wet-bulb Temperature and Sectoral Labor Share: Degree Day Bins



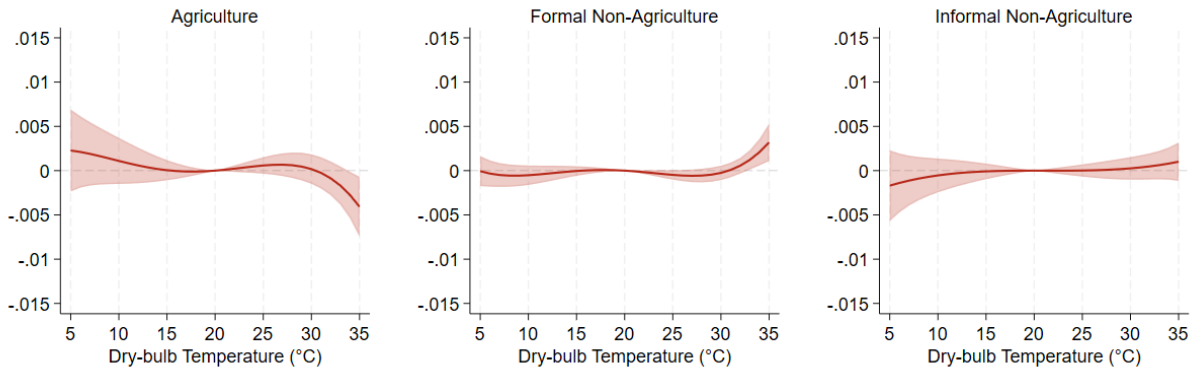
Notes: This figure plots the estimated effect of temperature on sectoral labor shares using degree day bins following Somanathan et al. (2021). Daily wet-bulb temperatures are allocated to eleven bins: below 9°C, nine two-degree bins from 9–27°C, and above 27°C. Each day contributes degree days to all bins below its temperature: if WBT exceeds the bin's upper bound, the day contributes the full bin width; if WBT falls within the bin, it contributes (WBT - bin lower bound). Coefficients represent the effect of one additional degree day in each bin, with 19–21°C as the omitted category. All specifications include province fixed effects, region-by-year fixed effects, and controls for precipitation. Shaded areas represent 95% confidence intervals based on standard errors clustered at the province level.

Figure D4: Wet-bulb Temperature and Sectoral Labor Share: Polynomials



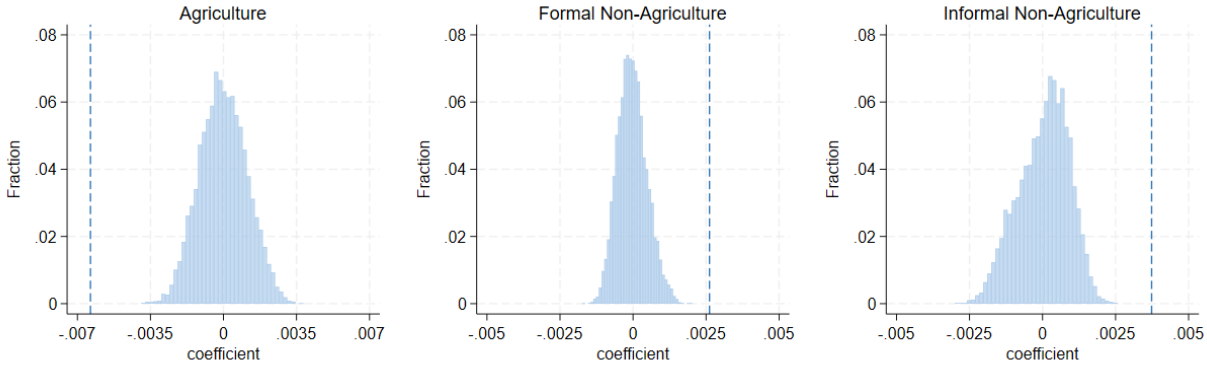
Notes: This figure plots the estimated effect of wet-bulb temperature on sectoral labor shares using a fourth-order polynomial specification. Following Carleton et al. (2022), polynomial terms are constructed by first raising grid-level daily temperatures to powers 1 through 4, then aggregating to the province level using inverse-distance weighting from the four nearest grid points to each province's geographic centroid, and summing over days in the reference period. All specifications include province fixed effects, region-by-year fixed effects, and controls for precipitation. Effects are normalized to zero at 20°C. Shaded areas represent 95% confidence intervals based on standard errors clustered at the province level.

Figure D5: Dry-bulb Temperature and Sectoral Labor Share



Notes: This figure plots the estimated effect of dry-bulb temperature on sectoral labor shares using a fourth-order polynomial specification. Following Carleton et al. (2022), polynomial terms are constructed by first raising grid-level daily temperatures to powers 1 through 4, then aggregating to the province level using inverse-distance weighting from the four nearest grid points to each province's geographic centroid, and summing over days in the reference period. All specifications include province fixed effects, region-by-year fixed effects, and controls for precipitation polynomials. Effects are normalized to zero at 20°C. Shaded areas represent 95% confidence intervals based on standard errors clustered at the province level.

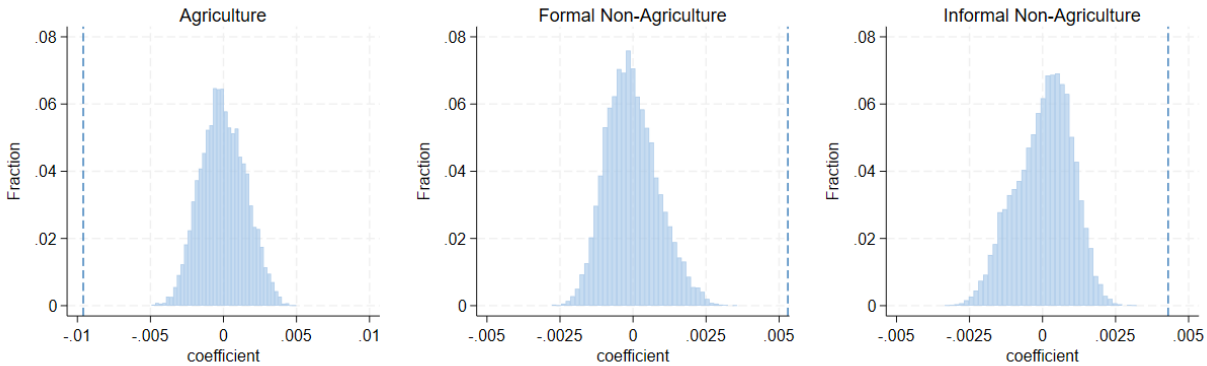
Figure D6: Monte-Carlo Permutation-Based Placebo Test



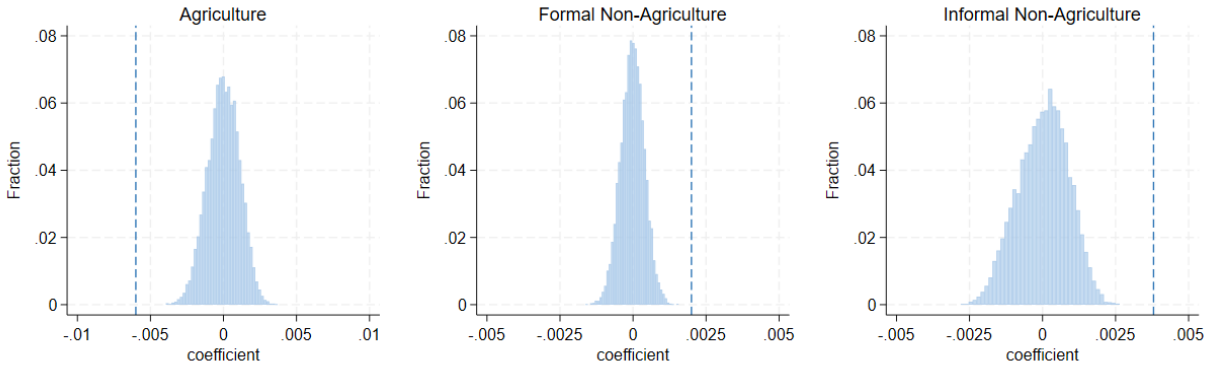
Notes: Each iteration randomly reassigns province weather series to different provinces, breaking the true province-weather relationship while preserving temporal structure. The model is re-estimated 10,000 times. Histograms show the the distribution of placebo coefficients, and vertical dashed lines indicate baseline estimates.

Figure D7: Monte-Carlo Permutation-Based Placebo Test: Heterogeneity by Age Group

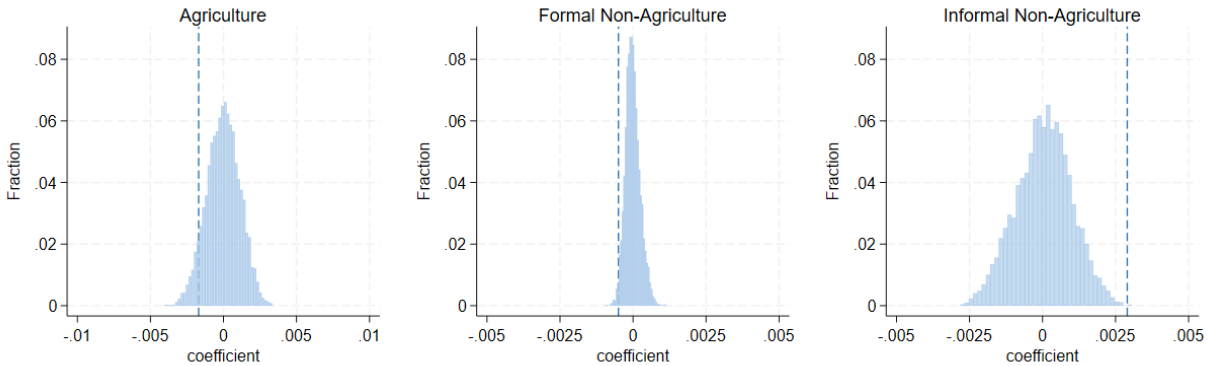
Panel A: Age 24–39



Panel B: Age 40–54

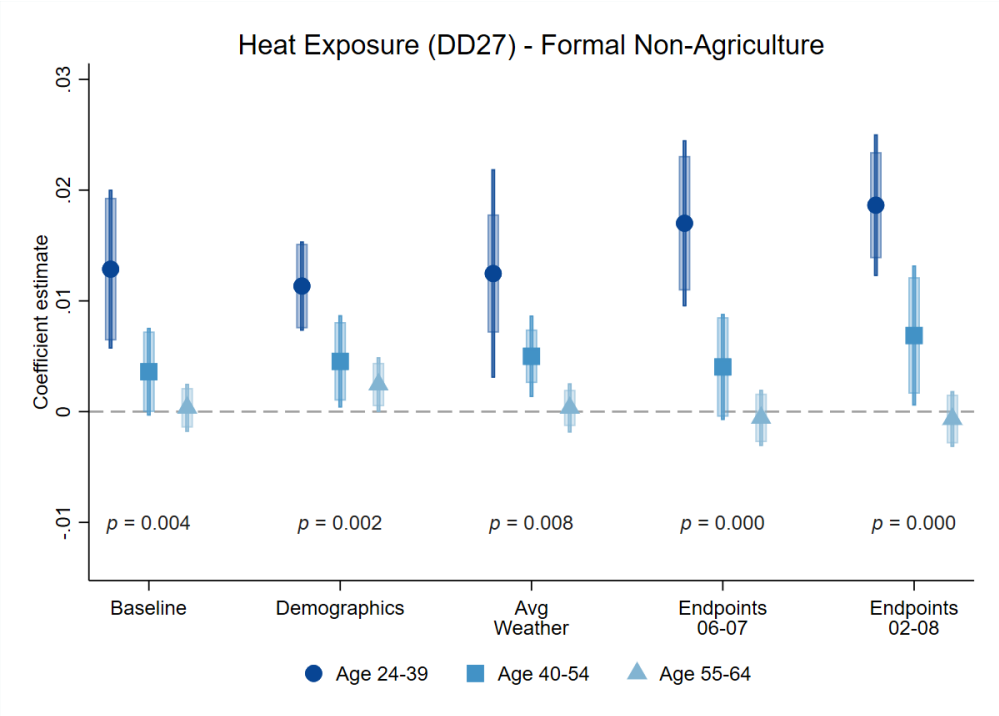


Panel C: Age 55–64



Notes: Each iteration randomly reassigns province weather series to different provinces, breaking the true province-weather relationship while preserving temporal and age-group structure. The model is re-estimated 1,000 times. Histograms show the distribution of placebo coefficients for each age group, and vertical dashed lines indicate baseline estimates.

Figure D8: Long-Difference Estimates: Heterogeneity by Age Group



Notes: This figure reports long-difference estimates of the effect of heat exposure on formal non-agricultural employment by age group. Following Burke and Emerick (2016), I regress changes in sectoral employment shares between two sub-periods on changes in weather variables:

$$\Delta y_{apr} = f_a(\Delta \mathrm{WBT}_p) + g_a(\Delta \mathrm{R}_p) + \gamma_{ar} + \varepsilon_{apr}$$

where Δy_{apr} is the change in employment share of age group $a \in \{24\text{–}39,\ 40\text{–}54,\ 55\text{–}64\}$ in province p and region r, $\Delta \mathrm{WBT}_p$ denotes changes in wet-bulb temperature distribution (degree days above 27°C and below 9°C), $\Delta \mathrm{R}_p$ includes changes in precipitation and its square, and γ_{ar} are region-by-age group fixed effects. The baseline specification compares 1992–2008 to 2009–2018, with shares in each sub-period calculated as the average across survey waves. "Demographics" adds controls for changes in gender composition, ethnic composition, and high school completion rates. "Avg Weather" uses simple averaging instead of inverse-distance weighting for weather variable construction. "Endpoints 06-07" and "Endpoints 02-08" use alternative period definitions (1992–2006 vs. 2007–2018 and 1992–2002 vs. 2008–2018, respectively). Point estimates are shown with 95% confidence intervals based on province-clustered standard errors (thin bars) and Conley spatial standard errors with a 150km bandwidth (thick bars). The p-values report joint tests of equality of coefficients across the three age groups.

E APPENDIX MECHANISM: THE SORTING

E.1 Earnings Patterns Among Sector-Switchers

To estimate the earnings gains among sector-switchers, I rely on the sample of sector-switchers from a pool of three-consecutive-survey-wave individual panel data sets from VHLSS over the period 2004–2018. I estimate the following model:

$$\ln w_{jt} = \sum_{a} \sum_{s \in \{f,i\}} \kappa_{a,s} \mathbb{I}_a \mathbb{I}_s + \psi U_{jt} + \gamma_j + \gamma_t + \epsilon_{jt} \tag{E1}$$

where $\ln w_{jt}$ is the log real earnings of individual j per hour at panel-wave t. Earnings are computed as the sum of labor wages, benefits, and household farm or non-farm net profits.[6] The term $\mathbb{I}_s$ takes the value of one if the individual works in sector s for their main job and zero otherwise. There are three sectors: agriculture (g), formal non-agriculture (f) and informal non-agriculture (i). The term $\mathbb{I}_a$ denotes whether the individual j belongs to age group $a \in \{24-39, 40-54, 55-64\}$ that corresponds to the three age groups in the main empirical analysis, measured at the first wave of each panel. Other time-variant controls such as age and age squared are included in vector U. The vectors γ_j and γ_t denote individual and panel-wave fixed effects, respectively.

The coefficients of interest are the differences $\kappa_{a,f} - \kappa_{a,g}$ and $\kappa_{a,i} - \kappa_{a,g}$, which capture earnings gains for workers who switch from agriculture into formal and informal non-agriculture, respectively. Individual fixed effects absorb time-invariant unobservables such as ability, so these gains are identified from within-person sector switches. Under the framework, earnings gains among switchers reflect selection: higher barriers to entry raise the skill threshold for switching, so observed gains are increasing in the severity of barriers. I also estimate a variant in which time-invariant individual characteristics (including education level, gender, and ethnic minority status) are interacted with sector dummies, allowing returns to these observables to vary across sectors.[7]

E.2 Age Restrictions In Job Postings

To investigate age-based restrictions in access to formal employment, I conducted a systematic collection of current job vacancies using `https://vieclam24h.vn`, one of Vietnam's leading and most widely used job posting platforms. This website caters to a na-

[6] Household members are assumed to receive a share of household net profits that is proportional to their hours worked in household farm and non-farm business.

[7] I do not interact time-varying characteristics with sector dummies to reflect the idea that it is purely the change in relative returns to observables, not any change in individual characteristics or their behaviors, that induces workers to switch sectors.

tional audience and features both free and paid listings for a wide range of occupations, from low-skilled to high-skilled jobs, across various industries and geographic areas.

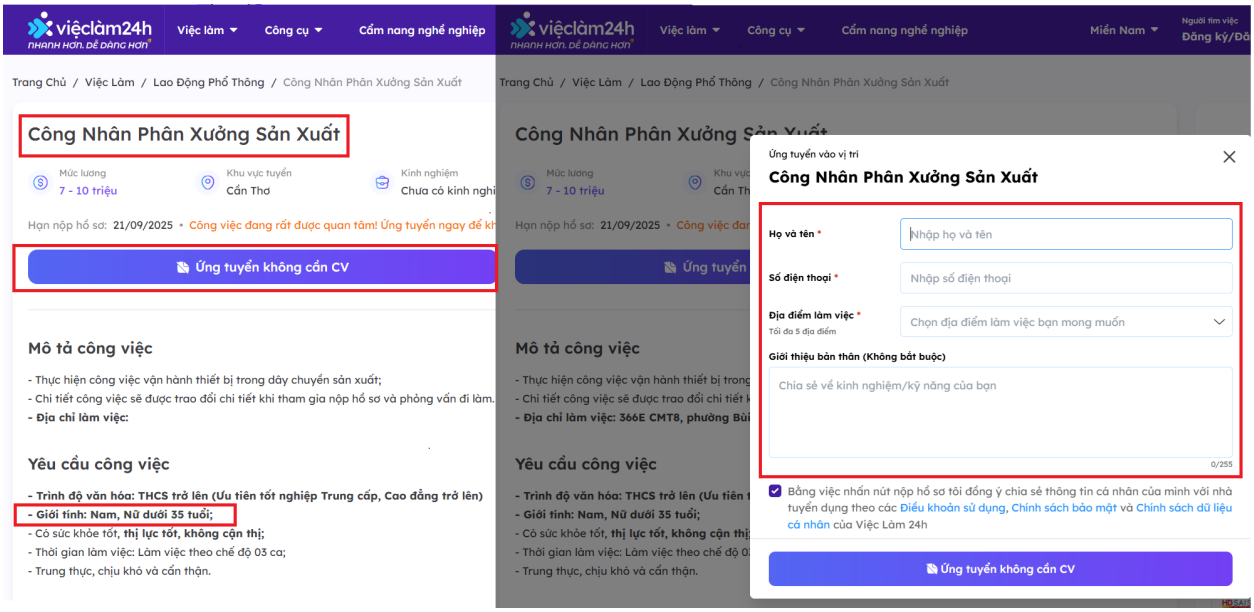
The search focused on two keywords most relevant for low-skilled employment:

- "low-skilled workers" (lao động phổ thông)
- "production line workers" (công nhân sản xuất)

For each keyword, I reviewed the first 10–20 pages of search results during October 2025–March 2026 to ensure a comprehensive snapshot of available postings. Duplicate listings, irrelevant posts or those that required a post-secondary school degrees (e.g., supervisory or skilled positions) were excluded.

Figure E1 provides an example of a typical job posting for a production line position with no experience requirements, offered by a medium-sized manufacturing firm (150–300 employees). Several features stand out. First, age limits are explicitly stated, restricting applicants (regardless of gender) to those under 35 years old. Second, job posts commonly do not require submission of a detailed CV, instead asking for basic details such as name, contact information, preferred work location, and a short summary of previous experience or skills. This streamlined process is characteristic of entry-level job recruitment in Vietnam and reflects the intent to attract a broad pool of candidates while maintaining firm age preferences.

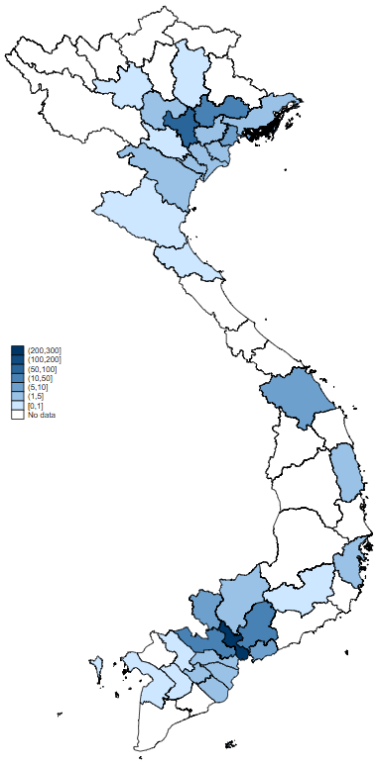
Figure E1: Example of a job post for production line worker with no experience requirement



The search yielded 1,026 unique job posts. I exclude 444 posts that required higher education (college or university degree) or advertised managerial positions, as these are not relevant to the low-skilled agricultural-to-non-agricultural transitions studied here. The final sample comprises 582 low-skilled job posts.

Figure E2 illustrates the geographic spread of job postings collected from the platform, highlighting the website's broad coverage across regions, with concentrations of posts in the country's important industrial hubs including Ho Chi Minh City, Binh Duong, Long An, Ha Noi, Dong Nai, Bac Ninh, and Ba Ria-Vung Tau.

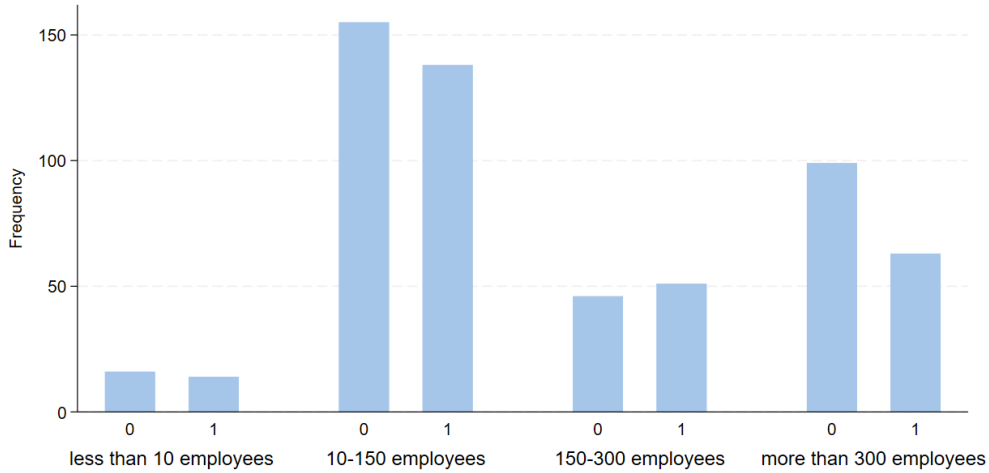
Figure E2: Spatial Distribution of Job Posts



Darker color denotes higher number of job posts.

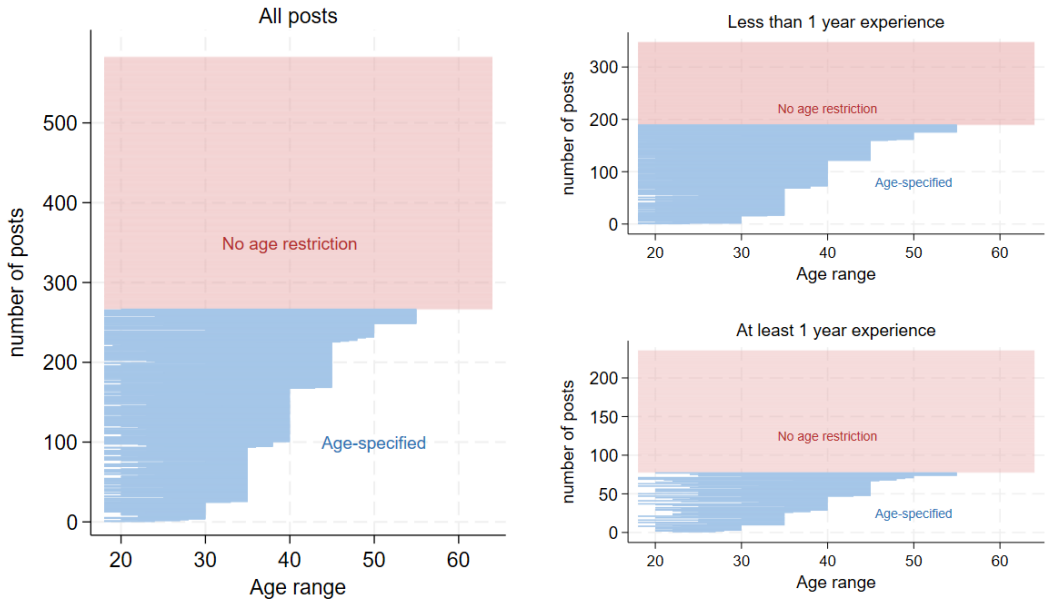
There is no noticeable systematic difference in firm size between job postings that specify age eligibility criteria and those that do not, as illustrated in Figure E3. Across the sample, nearly half of all postings (45.7%) specified age eligibility criteria, most narrowing opportunities to job seekers in their twenties or thirties. This restriction applies to both male and female applicants and is especially pronounced for positions requiring little or no prior experience, which are likely the main destination for workers transitioning out of agriculture (Figure E4).

Figure E3: Firm Size Breakdown of Job Vacancy Postings



"0" denotes job posts without age restrictions, "1" indicates posts that specify an age eligibility criterion.

Figure E4: Distribution of Age Ranges Specified in Low-skilled Job Posts



E.3 Stated Employer Preferences

A vignette experiment was conducted with a sample of 57 manufacturing firms in the Central Region of Vietnam. While the sample is geographically limited to one region, manufacturing firms in Central Vietnam share similar hiring practices with those in other industrial areas. Vignette experiments present respondents with carefully constructed hy-

pothetical scenarios and applicant profiles that systematically vary key characteristics—here, candidate age and experience—while holding others constant. This method allows for estimation of the impact of these traits on stated employer preferences.

Design and Implementation Respondents, who are HR managers responsible for hiring decisions, evaluate two hiring scenarios representing different experience requirements:

- Scenario 1: No Experience Required

  This scenario simulates hiring for positions open to candidates without prior manufacturing experience, typical of agricultural workers transitioning to manufacturing jobs.

- Scenario 2: Experience Required

  This scenario represents hiring for positions requiring prior manufacturing experience, such as agricultural workers who have some manufacturing experience as a secondary job and seek full transition.

A common approach in audit and correspondence studies involves sending applicants' CVs to open job positions. However, in Vietnam and many other developing countries, formal CV submissions are rarely required, especially for low-skilled jobs (Figure E1). Therefore, to align with common local recruitment practices for this type of job, only concise applicant summaries (instead of formal CVs) are presented to respondents.

Applicant profiles were constructed using data from the 2017 Labor Force Survey on formal manufacturing workers. Three age groups were selected (mid-to-late 20s, early 40s, and mid-50s), corresponding roughly to the 25th, 75th, and 95th percentiles of the age distribution in formal manufacturing. Experience levels reflect observed patterns: younger workers typically have 1–5 years in the position, while approximately half of workers aged 55 have more than 10 years of experience.

Within each scenario, profiles are identical in qualifications, education, and demographics, differing only in age. Table E1 illustrates an example for Scenario 1, in which no prior experience is required for a production line worker position. Each profile includes name, year of birth, gender, education, work experience, availability, and a candidate statement. To ensure that specific wording does not influence ratings, these items are randomized across profiles while preserving substantive equivalence.

Scenario 2 addresses a potential confound: older workers with the same experience as younger workers may signal lower ability: why have they not accumulated more experience? Following Neumark et al. (2019), I include two conditions: experience equal

Table E1: Recruitment of Production Line Workers with No Experience Required

Suppose your company is currently recruiting new production workers to work at the factory. This position involves assisting with tasks on the assembly line, operating simple machinery, and complying with safety regulations under supervision, and it does not require prior work experience. You have reviewed many candidates with similar qualifications. Below is the profile of one candidate: *[Three profiles are presented sequentially in randomized order. Respondents rate each on a 0–10 scale: "To what extent would you consider contacting this candidate for a job interview?"]*

Items (* denotes randomized)	Young	Middle	Old
Full Name*	Nguyễn Minh Tuấn	Trần Văn Hùng	Lê Văn Minh
Year of Birth (Age)*	1999 (26 years old)	1984 (41 years old)	1970 (55 years old)
Gender	Male	Male	Male
Education	High school graduate	High school graduate	High school graduate
Work Experience*	No direct experience in a production line worker position.	Has never worked on a production line.	No experience in the industrial manufacturing sector.
	Has experience with family farming work, familiar with manual labor and work environments that require perseverance	Has many years of experience in family agricultural work, and is familiar with physically demanding work and environments that require endurance.	Many years of experience in agricultural labor, familiar with manual work.
Start Date*	Available to start work immediately.	Ready to start work immediately when needed.	Ready to start work immediately.
Candidate's Note*	Although I do not yet have experience working on a production line, I am hardworking, eager to learn, and have a strong sense of responsibility. I sincerely hope to have the opportunity to work and build a long-term commitment with the company.	I am hardworking, diligent, and eager to learn more in a production environment. I hope to have the opportunity to contribute and to build a stable, long-term commitment with the company.	I am in good health, patient in my work, and responsible. Although I have never worked as a production line worker, I am eager to learn and will make every effort to complete assigned tasks well.

to younger workers, or experience commensurate with age. Middle-aged candidates are assigned 3 or 7 years; older candidates are assigned 5 or 15 years. If age discrimination drives employer preferences, gaps should persist even with commensurate experience. Table E2 illustrates the profiles for middle-aged candidates.

Table E2: Recruitment of Production Line Workers Requiring Experience

Suppose your company is currently recruiting production workers to work at the factory. This position involves directly participating in the assembly line, operating and monitoring production machinery, and strictly complying with occupational safety regulations. Candidates are required to have experience working in a manufacturing environment or operating industrial machinery in order to quickly adapt and ensure work efficiency. You have reviewed many candidates with similar qualifications. Below is the profile of one candidate: *[Three profiles are presented sequentially in randomized order. Respondents rate each on a 0–10 scale: "To what extent would you consider contacting this candidate for a job interview?"]*

Items (* denotes randomized)	Young	Middle (Same Experience)	Middle (More Experience)
Full Name*	Nguyễn Thanh Tùng	Trần Quang Huy	Lê Văn Bảo
Year of Birth (Age)*	1997 (28 years old)	1983 (42 years old)	1983 (42 years old)
Gender	Male	Male	Male
Education	High school graduate	High school graduate	High school graduate
Work Experience*	Has 3 years of experience working in a manufacturing environment, familiar with assembly lines and operating simple machinery.	Has 3 years of experience working in a similar position.	Has more than 7 years of experience in the manufacturing sector, proficient in assembly line tasks and operating standard machinery.
	Has previously engaged in work requiring perseverance, physical endurance, and strict compliance with occupational safety regulations.	Works well in a team, is proactive in learning, and quickly adapts to new job requirements.	Has accumulated extensive experience in high-intensity work environments, consistently ensuring production schedules and product quality.
	Has good teamwork skills and is quick to learn new work processes.	Familiar with high-intensity labor, has good endurance, and always strictly complies with safety regulations.	Has thorough knowledge of and strictly complies with safety regulations, and supports colleagues during work processes.
			Has strong teamwork skills and is willing to share experience with new employees.
Start Date*	Available to start work immediately.	Ready to start work immediately when needed.	Ready to start work immediately.
Candidate's Note*	I am hardworking, eager to learn, and have a strong sense of responsibility. I hope to apply my experience and build a long-term commitment with the company.	I am hardworking, motivated, and have a strong sense of responsibility at work. I hope to apply my existing experience, continue learning, and build a long-term commitment with the company.	With more than 7 years of experience in the manufacturing industry, I am confident in my endurance, sense of responsibility, and discipline. I wish to continue applying my experience, contribute tangible value, and build a long-term commitment with the company.

Respondents were asked to evaluate each applicant profile of a total three profiles sequentially, indicating the likelihood (on a 0–10 scale, where 0 = "very unlikely" and 10 = "very likely") that they would invite the applicant to interview . Presenting profiles individually minimizes direct comparison bias and reflects realistic sequential screening practices, despite multiple applicants typically competing for a single job.

To ensure unbiased estimates of age and experience effects on interview likelihood, the experiment implemented two randomizations:

- Profile order: Randomized within each scenario to control for order effects such as fatigue or primacy/recency bias.
- Scenario presentation order: Randomized so respondents receive either the "no experience" or "experience required" scenario first, avoiding contrast effects.

Estimation and Results. Table E3 presents mean interview ratings by applicant age group, for the pooled sample (combining both scenarios) and separately by scenario. For each case, I report pairwise tests of equality in mean ratings across applicant age groups based on linear regressions with standard errors clustered at the HR manager level. These tests assess whether employers' evaluations differ systematically by applicant age.

Table E3: Summary of Scores Across Profiles

	Young	Middle	Old	(1) = (2)	(2) = (3)	(1) = (3)	(1) = (2) = (3)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
All Scenarios	7.885	6.257	4.689	0.00	0.00	0.00	0.00
	[1.731]	[2.279]	[2.773]				
Scenario 1: No Experience Required	7.404	5.386	3.298	0.00	0.00	0.00	0.00
	[1.879]	[2.336]	[2.375]				
Scenario 2: Experience Required							
Experience Not Varies With Age	8.596	6.885	5.677	0.00	0.04	0.00	0.00
	[1.266]	[1.774]	[2.613]				
Experience Varies With Age	8.596	7.538	6.258	0.01	0.02	0.00	0.00
	[1.266]	[1.794]	[2.380]				

Notes: Applicants differ only by age and sometimes, experience. Experience assignments reflect observed age-experience patterns. Respondents evaluated three profiles sequentially and rated interview likelihood on a 0–10 scale. Columns (1)–(3) present the mean scores for each profile type, with standard deviations in brackets. Columns (4)–(6) present the p-value of equality specified in each column, obtained from regressions of scores on type dummies, with standard errors clustered at the HR manager (respondent) level. Source: Calculations from survey of HR managers.

Pooling the two scenarios, I find evidence of age-based preferences consistent with

discrimination documented in other contexts: the average rating is approximately 1.6 points lower (on the 0–10 scale) for middle-aged applicants and about 3.2 points lower for older applicants, relative to young applicants. In the second scenario, where experience is required for the job post, these gaps narrow slightly when experience is allowed to vary proportionally with age; yet the raw differences remain substantial: 1.06 and 1.34 points lower for middle-aged and older applicants, respectively. For both combined and scenario-specific analyses, tests of equality strongly reject the null hypothesis that document screening ratings are unrelated to applicant age (Columns 4–7).

Next, I present results from a richer model with other covariates:

$$Y_{ij} = \alpha + \beta \mathrm{M}_i + \gamma O_i + X_j^{'}\delta + \varepsilon_{ij} \tag{E2}$$

where Y_i is the ratings received for a profile i by manager j, and M_i and O_i are dummy variables for age (M for middle-age, and O for older). The vector X_j includes observable firm characteristics (firm size, preference for healthy candidates), the order in which each profile is presented to respondents, and scenario fixed effects. Standard errors are clustered at the manager level. Because profile group assignment is randomized within the experimental design, these controls are not expected to materially affect the estimated differences in ratings across age groups. However, as noted by Neumark et al. (2019), this logic does not always hold in age discrimination studies, since experience and age are often correlated, and conditioning on experience can influence the measured effect of age. Therefore, reporting results with and without such controls helps demonstrate the robustness of the core findings.

The results are summarized in Table E4. Even controlling for a rich set of variables including profile order fixed effects, scenario fixed effects, as well as other firms' characteristics such as firm size and whether they specify preference for healthy applicants in their evaluations, the results across all three columns consistently indicate that HR managers give substantially lower ratings to middle-aged and older applicants.

Finally, I turn to models incorporating all types of profiles to compare the findings for older versus younger applicants depending on whether the profiles for the older applicants show the same low experience as the younger profiles or instead experience that is commensurate with age. I consider two sub-scenarios in Scenario 2:

- Applicant $Y1$ is 28 years old with 3 years of experience. Applicant M_{Li} is 42 years old with 3 years of experience. Applicant M_{Hi} is 42 years old with 7 years of experience.
- Applicant $Y2$ is 28 years old with 5 years of experience. Applicant O_{Li} is 50 years old with 5 years of experience. Applicant O_{Hi} is 50 years old with 15 years of experience.

Table E4: Stated Employer Preferences: All Scenarios

	(1)	(2)	(3)
Middle	-1.627	-1.630	-1.621
	(0.258)	(0.257)	(0.258)
Old	-3.242	-3.243	-3.168
	(0.354)	(0.355)	(0.359)
p-val: M = O	0.000	0.000	0.000
Order FE	Yes	Yes	Yes
Scenario FE	Yes	Yes	Yes
Health Preference Control	No	Yes	Yes
Firm Size Control	No	No	Yes
Adjusted R2	0.329	0.331	0.321
Observations	332	332	326

Notes: This table presents the difference in scores of middle-aged and older applicants relative to young applicants. Respondents evaluated three profiles sequentially and rated interview likelihood on a 0–10 scale. Columns (1)–(3) increase the number of control variables. Standard errors clustered at the HR manager (respondent) level are in parentheses. Source: Calculations from survey of HR managers.

and estimate the following corresponding models:

$$\begin{aligned} Y_{ij1} &= \alpha_1 + \beta_L \mathrm{M}_{Li} + \beta_H \mathrm{M}_{Hi} + X_j^{'}\delta_1 + \zeta_{ij1} \\ Y_{ij2} &= \alpha_2 + \gamma_L O_{Li} + \gamma_H O_{Hi} + X_j^{'}\delta_2 + \zeta_{ij2} \end{aligned} \tag{E3}$$

where the vector X_j includes observable firm characteristics (firm size, preference for healthy candidates) and the order in which each profile is presented to respondents. I am interested in testing whether $\beta_H = \beta_L$ and $\gamma_H = \gamma_L$.

The results in Table E5 confirm the existence of age-based preferences in hiring. For middle-aged applicants, assigning experience commensurate with age (7 years vs. 3 years) significantly reduces the score gap relative to young applicants, from approximately 1.6 to 0.9 points ($p < 0.01$ for the test $\beta_H = \beta_L$ across all specifications). For older applicants, commensurate experience (15 years vs. 5 years) also narrows the gap, from approximately 3.0 to 2.4 points, though this difference is only marginally significant ($p \approx$ 0.05–0.08). Importantly, even with experience commensurate with age, substantial gaps remain: middle-aged applicants still score roughly 1 point lower, and older applicants roughly 2.4 points lower, on a 10-point scale. These residual gaps are consistent with age-based preferences that cannot be rationalized by differences in experience or inferred ability (Neumark et al., 2019).

This experimental evidence complements observational findings, offering insight into employer preferences that may contribute to heterogeneous labor market frictions shap-

Table E5: Stated Employer Preferences: The Role of Experience

	(1)	(2)	(3)	(4)	(5)	(6)
Middle, Same Experience (ML)	-1.577	-1.577	-1.577			
	(0.429)	(0.432)	(0.435)			
Middle, More Experience (MH)	-0.923	-0.923	-0.923			
	(0.385)	(0.388)	(0.390)			
Old, Same Experience (OL)				-3.032	-3.032	-2.900
				(0.504)	(0.507)	(0.508)
Old, More Experience (OH)				-2.452	-2.452	-2.367
				(0.473)	(0.476)	(0.487)
p-val: ML = MH	0.001	0.001	0.001			
p-val: OL = OH				0.053	0.054	0.083
Order FE	Yes	Yes	Yes	Yes	Yes	Yes
Health Preference Control	No	Yes	Yes	No	Yes	Yes
Firm Size Control	No	No	Yes	No	No	Yes
Adjusted R2	0.119	0.152	0.185	0.258	0.252	0.240
Observations	78	78	78	93	93	90

Notes: This table presents the difference in scores of middle-aged and older applicants, with higher or similar experience levels, relative to young applicants. Respondents evaluated three profiles sequentially and rated interview likelihood on a 0–10 scale. Standard errors clustered at the HR manager (respondent) level are in parentheses. Source: Calculations from survey of HR managers.

ing sectoral labor reallocation. Yet, two caveats apply. First, the experiment captures stated preferences, which may differ from actual hiring behavior. Second, the sample is limited to 57 firms in Central Vietnam, and results may not generalize to other regions or sectors. Nevertheless, the consistency of findings across specifications and their alignment with job postings evidence suggest these patterns reflect broader labor market realities.

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